

Co-ordination Compounds

Question1

Given below are two statements :

Statement I : Among $[\text{Cu}(\text{NH}_3)_4]^{2+}$, $[\text{Ni}(\text{en})_3]^{2+}$, $[\text{Ni}(\text{NH}_3)_6]^{2+}$ and $[\text{Mn}(\text{H}_2\text{O})_6]^{2+}$, $[\text{Mn}(\text{H}_2\text{O})_6]^{2+}$ has the maximum number of unpaired electrons.

Statement II : The number of pairs among $\{[\text{NiCl}_4]^{2-}, [\text{Ni}(\text{CO})_4]\}$, $\{[\text{NiCl}_4]^{2-}, [\text{Ni}(\text{CN})_4]^{2-}\}$ and $\{[\text{Ni}(\text{CO})_4], [\text{Ni}(\text{CN})_4]^{2-}\}$ that contain only diamagnetic species is two.

In the light of the above statements, choose the correct answer from the options given below :

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Options:

A.

Statement I is false but Statement II is true

B.

Both Statement I and Statement II are false

C.

Both Statement I and Statement II are true

D.

Statement I is true but Statement II is false

Answer: D

Solution:

$[\text{Cu}(\text{NH}_3)_4]^{2+} \Rightarrow d^9, dsp^2$ one unpaired electron

$[\text{Ni}(\text{en})_3]^{2+} \Rightarrow d^8, sp^3 d^2$ two unpaired electrons

$[\text{Ni}(\text{NH}_3)_6]^{2+} \Rightarrow d^8, sp^3 d^2$ two unpaired electrons

$[\text{Mn}(\text{H}_2\text{O})_6]^{2+} \Rightarrow d^5, sp^3 d^2$ five unpaired electrons

$[\text{Ni}(\text{CO})_4]$ (diamagnetic)

$[\text{NiCl}_4]^{2-}$ (paramagnetic)

$[\text{Ni}(\text{CN})_4]^{2-}$ (diamagnetic)

Question2

Given below are two statements :

Statement I : Crystal Field Stabilization Energy (CFSE) of $[\text{Cr}(\text{H}_2\text{O})_6]^{2+}$ is greater than that of $[\text{Mn}(\text{H}_2\text{O})_6]^{2+}$.

Statement II : Potassium ferricyanide has a greater spin-only magnetic moment than sodium ferrocyanide.

In the light of the above statements, choose the correct answer from the options given below :

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Options:

A.

Statement I is true but Statement II is false

B.

Both Statement I and Statement II are true

C.

Both Statement I and Statement II are false

D.

Statement I is false but Statement II is true

Answer: B

Solution:

Statement I:

In $[\text{Cr}(\text{H}_2\text{O})_6]^{2+}$, Cr^{2+} is d^4 and with H_2O (weak field) it is **high spin**: $t_{2g}^3 e_g^1$.

$$\text{CFSE} = 3(-0.4\Delta_o) + 1(+0.6\Delta_o) = -0.6\Delta_o$$

In $[\text{Mn}(\text{H}_2\text{O})_6]^{2+}$, Mn^{2+} is d^5 high spin: $t_{2g}^3 e_g^2$.

$$\text{CFSE} = 3(-0.4\Delta_o) + 2(+0.6\Delta_o) = 0$$

So CFSE of the chromium complex is greater (more stabilizing). **Statement I is true.**

Statement II:

- $K_3[\text{Fe}(\text{CN})_6]$: Fe^{3+} is d^5 ; CN^- is strong field $\rightarrow$ **low spin** t_{2g}^5 , **1 unpaired electron** $\rightarrow \mu_{so} = \sqrt{1(1+2)} = \sqrt{3}$ BM.
- $\text{Na}_4[\text{Fe}(\text{CN})_6]$: Fe^{2+} is d^6 ; strong field $\rightarrow$ **low spin** t_{2g}^6 , **0 unpaired electrons** $\rightarrow \mu_{so} = 0$.

Thus ferricyanide has a greater spin-only magnetic moment. **Statement II is true.**

Correct option: B (Both Statement I and Statement II are true).

Question3

A first row transition metal (M) does not liberate H_2 gas from dilute HCl . 1 mol of aqueous solution of MSO_4 is treated with excess of aqueous KCN and then H_2S (g) is passed through the solution. The amount of MS (metal sulphide) formed from the above reaction is ____ mol.

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Options:

A.

1

B.

2

C.

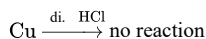
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D.

3

Answer: C

Solution:

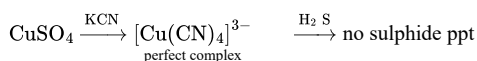


Step 1: Identify the metal M

A first row transition metal that does not liberate H_2 gas with dilute HCl is Cu (copper). So, $M = \text{Cu}$.

Step 2: Reaction of MSO_4 with excess KCN

So MSO_4 is CuSO_4 . When CuSO_4 is treated with excess aqueous KCN, copper forms a very stable (perfect) cyanide complex.



Step 3: Passing H_2S through the solution

Because copper is present as the stable complex $[\text{Cu}(\text{CN})_4]^{3-}$, free Cu^{2+} ions are not available in solution. Therefore, H_2S cannot precipitate copper sulphide.

Final Answer

No metal sulphide MS is formed. Hence, the amount of MS formed is 0 mol.

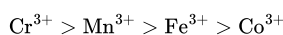
Question4

Consider the transition metal ions Mn^{3+} , Cr^{3+} , Fe^{3+} and Co^{3+} and all form low spin octahedral complexes. The correct decreasing order of unpaired electrons in their respective d-orbitals of the complexes is

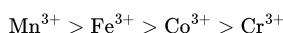
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Options:

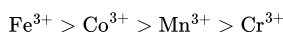
A.



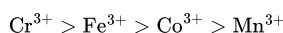
B.



C.



D.

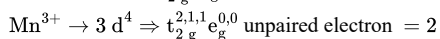
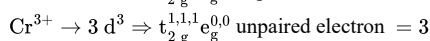
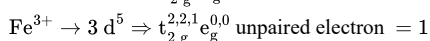
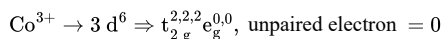


Answer: A

Solution:

For each ion, first find its 3d electron count after losing electrons to form the 3+ charge.

Since all are **low spin octahedral** complexes, electrons will first pair up in the lower-energy t_{2g} orbitals before entering the higher-energy e_g orbitals.



Now compare the number of unpaired electrons: Cr^{3+} has 3, Mn^{3+} has 2, Fe^{3+} has 1, and Co^{3+} has 0.

So, the decreasing order of unpaired electrons is: $\text{Cr}^{3+} > \text{Mn}^{3+} > \text{Fe}^{3+} > \text{Co}^{3+}$

Question 5

$[\text{Ni}(\text{PPh}_3)_2\text{Cl}_2]$ is a paramagnetic complex. Identify the **INCORRECT** statements about this complex.

A. The complex exhibits geometrical isomerism.

B. The complex is white in colour.

C. The calculated spin-only magnetic moment of the complex is 2.84 BM .

D. The calculated CFSE (Crystal Field Stabilization Energy) of Ni in this complex is $-0.8\Delta_o$.

E. The geometrical arrangement of ligands in this complex is similar to that in $\text{Ni}(\text{CO})_4$.

Choose the correct answer from the options given below :

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Options:

A.

C and D Only

B.

A, B and D Only

C.

C, D and E Only

D.

A and B Only

Answer: B

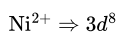
Solution:

For $[\text{Ni}(\text{PPh}_3)_2\text{Cl}_2]$:

1) Oxidation state and d -electron count

PPh_3 is neutral and each Cl is -1 .

So oxidation state of Ni is $+2$.



2) Geometry using given information (paramagnetic)

For Ni^{2+} (d^8):

- **Square planar** complexes are usually **diamagnetic** (all electrons paired).
- Since the complex is **paramagnetic**, it must be **tetrahedral** (in NCERT level treatment).

So geometry is **tetrahedral** and it has **2 unpaired electrons**.

Check each statement

A. The complex exhibits geometrical isomerism.

In a **tetrahedral** complex, all four positions are equivalent, so an MA_2B_2 type complex **does not show** cis–trans isomerism.

So **A is incorrect**.

B. The complex is white in colour.

Tetrahedral Ni^{2+} complexes are generally **coloured** due to $d-d$ transitions; “white” is not expected.

So **B is incorrect**.

C. Spin-only magnetic moment is 2.84 BM.

Number of unpaired electrons $n = 2$.

$$\mu_{\text{so}} = \sqrt{n(n+2)} = \sqrt{2(2+2)} = \sqrt{8} = 2.83 \text{ BM} \approx 2.84 \text{ BM}$$

So **C is correct**.

D. CFSE is $-0.8\Delta_o$.

For **tetrahedral** splitting, we must use Δ_t , not Δ_o .

For d^8 tetrahedral: $e^4t_2^4$

$$\text{CFSE} = 4(-0.6\Delta_t) + 4(+0.4\Delta_t) = (-2.4 + 1.6)\Delta_t = -0.8\Delta_t$$

So writing $-0.8\Delta_o$ is **wrong**.

So **D is incorrect**.

E. Geometry similar to $\text{Ni}(\text{CO})_4$.

$\text{Ni}(\text{CO})_4$ is **tetrahedral**, and this complex is also **tetrahedral** (as concluded from paramagnetism).

So **E is correct**.

Incorrect statements: A, B and D

☒ Correct option: Option B (A, B and D Only)

Question6

Given below are two statements :

Statement I : $[\text{CoBr}_4]^{2-}$ ion will absorb light of lower energy than $[\text{CoCl}_4]^{2-}$ ion.

Statement II : In $[\text{CoI}_4]^{2-}$ ion, the energy separation between the two set of d-orbitals is more than $[\text{CoCl}_4]^{2-}$ ion.

In the light of the above statements, choose the correct answer from the options given below :

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Options:

A.

Both Statement I and Statement II are false

B.

Both Statement I and Statement II are true

C.

Statement I is false but Statement II is true

D.

Statement I is true but Statement II is false

Answer: D

Solution:

Statement I (True):

The strength of a ligand means how strong it can split the d-orbitals of the metal ion.

Among Cl^- and Br^- , chloride (Cl^-) is a stronger ligand than bromide (Br^-).

This means for $[\text{CoCl}_4]^{2-}$, the energy gap between the two sets of d-orbitals (Δ_t) will be larger than in $[\text{CoBr}_4]^{2-}$.

So, $[\text{CoCl}_4]^{2-}$ absorbs light of higher energy, and $[\text{CoBr}_4]^{2-}$ absorbs light of lower energy.

$$\Delta_t : [\text{CoCl}_4]^{2-} > [\text{CoBr}_4]^{2-}$$

$$\text{Energy absorbed} : [\text{CoCl}_4]^{2-} > [\text{CoBr}_4]^{2-}$$

Statement II (False):

Here, we compare I^- and Cl^- as ligands. Iodide (I^-) is a weaker ligand than chloride (Cl^-).

This means that in $[\text{CoI}_4]^{2-}$, the energy difference between the two sets of d-orbitals (Δ_t) is smaller than in $[\text{CoCl}_4]^{2-}$.

$$\Delta_t : [\text{CoI}_4]^{2-} < [\text{CoCl}_4]^{2-}$$

Question 7

The statements that are incorrect about the nickel(II) complex of dimethylglyoxime are :

A. It is red in colour.

B. It has a high solubility in water at $\text{pH} = 9$.

C. The Ni ion has two unpaired d-electrons.

D. The $\text{N} - \text{Ni} - \text{N}$ bond angle is almost close to 90° .

E. The complex contains four five-membered metallacycles (metal containing rings).

Choose the correct answer from the options given below :

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Options:

A.

C and E Only

B.

A, D and B Only

C.

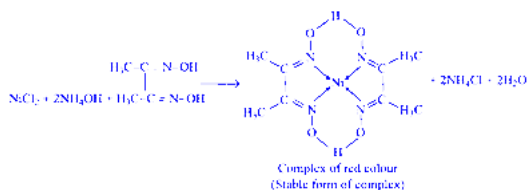
B, C and E Only

D.

C and D Only

Answer: C

Solution:



In this complex, Ni is in the +2 oxidation state (that is, it is present as Ni^{2+}).

A) The nickel dimethylglyoxime complex is a rosy red coloured precipitate, so the statement that it is red in colour is correct.

B) This complex is formed (precipitated) in a basic medium (around $\text{pH} = 9$), but it is sparingly soluble in water, not highly soluble. So the statement about high solubility in water at $\text{pH} = 9$ is incorrect.

C) $\text{Ni}^{+2} : 3d^8$ means that Ni^{2+} has 8 electrons in the d-subshell. In this complex, it forms a square planar dsp^2 complex in which all these electrons get paired. Therefore, the statement that Ni has two unpaired d-electrons is incorrect.

Hybridisation : dsp^2 , which leads to a square planar arrangement of the ligands around the Ni ion.

Unpaired $e^- = 0$, so the complex is diamagnetic (no unpaired electrons are present).

Geometry : Square planar, which is typical for dsp^2 hybridisation in such Ni^{2+} complexes.

D) $\text{N} - \text{Ni} - \text{N}$ Bond angle is close to 90° in a square planar complex because adjacent bond angles in square planar geometry are approximately 90° . So statement D is correct.

E) Each dimethylglyoxime ligand forms one five-membered chelate ring with the Ni ion. Since there are only two such ligands, 2 five-membered metal-containing rings are formed, not four. So statement E is incorrect.

Question8

Identify the **CORRECT** set of details from the following :

A. $[\text{Co}(\text{NH}_3)_6]^{3+}$: Inner orbital complex; d^2sp^3 hybridized

B. $[\text{MnCl}_6]^{3-}$: Outer orbital complex; $\text{sp}^3 \text{d}^2$ hybridized

C. $[\text{CoF}_6]^{3-}$: Outer orbital complex; d^2sp^3 hybridized

D. $[\text{FeF}_6]^{3-}$: Outer orbital complex; $\text{sp}^3 \text{d}^2$ hybridized

E. $[\text{Ni}(\text{CN})_4]^{2-}$: Inner orbital complex; sp^3 hybridized

Choose the correct answer from the options given below :

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Options:

A.

C & D Only

B.

A, C & E Only

C.

A, B, C, D & E

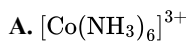
D.

A, B & D Only

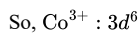
Answer: D

Solution:

Check each complex using NCERT (VBT + field strength):



Oxidation state of Co: +3



NH_3 is a stronger ligand than F^- , Cl^- and with Co^{3+} (high oxidation state) it causes **pairing**.

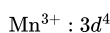
So it forms an **inner orbital (low spin) octahedral complex**:

- Hybridisation: d^2sp^3

✔ A is correct



Oxidation state of Mn: +3



Cl^- is a **weak field ligand**, so **no pairing** (high spin). Hence it uses outer d -orbitals.

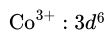
So it is an **outer orbital octahedral complex**:

- Hybridisation: sp^3d^2

✔ B is correct



Oxidation state of Co: +3



F^- is a **weak field ligand**, so complex will be **high spin** and thus **outer orbital**.

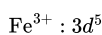
Therefore hybridisation should be:

- sp^3d^2 (not d^2sp^3)

✗ C is incorrect



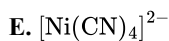
Oxidation state of Fe: +3



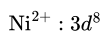
F^- is weak field $\rightarrow$ **no pairing** $\rightarrow$ **outer orbital octahedral**:

- Hybridisation: sp^3d^2

✔ D is correct



Oxidation state of Ni: +2



CN^- is a **strong field ligand**, so it forms **square planar** complex (pairing occurs).

Square planar hybridisation is:

- dsp^2 (not sp^3)

✗ E is incorrect

Correct set: A, B and D only $\rightarrow$ Option D

Question9

Consider a mixture ' X ' which is made by dissolving 0.4 mol of $[\text{Co}(\text{NH}_3)_5\text{SO}_4]\text{Br}$ and 0.4 mol of $[\text{Co}(\text{NH}_3)_5\text{Br}]\text{SO}_4$ in water to make 4 L of solution. When 2 L of mixture ' X ' is allowed to react with

excess of AgNO_3 , it forms precipitate 'Y'. The rest 2 L of mixture 'X' reacts with excess BaCl_2 to form precipitate 'Z'. Which of the following statements is CORRECT?

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Options:

A.

0.1 mol of 'Y' is formed.

B.

0.4 mol of 'Z' is formed.

C.

0.2 mol of 'Z' is formed.

D.

'Y' is BaSO_4 and 'Z' is AgBr .

Answer: C

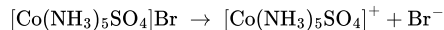
Solution:

For coordination compounds, only the **ions outside the coordination sphere** dissociate in water and give precipitates.

1) Identify free ions in solution

(i) $[\text{Co}(\text{NH}_3)_5\text{SO}_4]\text{Br}$

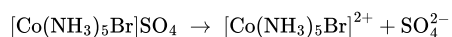
Here Br^- is **outside** the bracket, so in water:



So it gives 1 mol Br^- per mol of salt.

(ii) $[\text{Co}(\text{NH}_3)_5\text{Br}]\text{SO}_4$

Here SO_4^{2-} is **outside** the bracket, so:



So it gives 1 mol SO_4^{2-} per mol of salt.

2) Moles present in 2 L of mixture X

Total solution volume = 4 L, and we take 2 L, i.e. **half**.

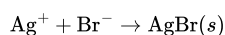
So in 2 L:

- From 0.4 mol of $[\text{Co}(\text{NH}_3)_5\text{SO}_4]\text{Br}$, we have 0.2 mol of this salt $\Rightarrow$ 0.2 mol Br^- .
- From 0.4 mol of $[\text{Co}(\text{NH}_3)_5\text{Br}]\text{SO}_4$, we have 0.2 mol of this salt $\Rightarrow$ 0.2 mol SO_4^{2-} .

3) Precipitates formed

With excess AgNO_3 (in 2 L):

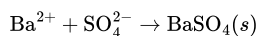
Ag^+ precipitates Br^- as AgBr :



So moles of precipitate Y = 0.2 mol (as AgBr).

With excess BaCl_2 (in remaining 2 L):

Ba^{2+} precipitates SO_4^{2-} as BaSO_4 :



So moles of precipitate Z = 0.2 mol (as BaSO_4).

Correct option

Option C: 0.2 mol of Z is formed.

(Also, $Y = \text{AgBr}$ and $Z = \text{BaSO}_4$.)

Question 10

Given below are two statements:

Statement I : The number of paramagnetic species among $[\text{CoF}_6]^{3-}$, $[\text{TiF}_6]^{3-}$, V_2O_5 and $[\text{Fe}(\text{CN})_6]^{3-}$ is 3.

Statement II :

$\text{K}_4[\text{Fe}(\text{CN})_6] < \text{K}_3[\text{Fe}(\text{CN})_6] < [\text{Fe}(\text{H}_2\text{O})_6]\text{SO}_4 \cdot \text{H}_2\text{O} < [\text{Fe}(\text{H}_2\text{O})_6]\text{Cl}_3$ is the correct order in terms of number of unpaired electron(s) present in the complexes.

In the light of the above statements, choose the correct answer from the options given below

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Options:

A.

Statement I is true but Statement II is false

B.

Both Statement I and Statement II are false

C.

Statement I is false but Statement II is true

D.

Both Statement I and Statement II are true

Answer: D

Solution:

Paramagnetic species are those which have one or more unpaired electrons.

Let us check each species one by one.

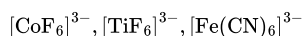
1. In $[\text{CoF}_6]^{3-}$, oxidation state of Co is +3 (because $6 \times (-1) = -6$ and overall charge is -3). So Co is Co^{3+} i.e. d^6 . Since F^- is a weak field ligand, it forms a high spin octahedral complex, so unpaired electrons are present. Hence, $[\text{CoF}_6]^{3-}$ is **paramagnetic**.

2. In $[\text{TiF}_6]^{3-}$, oxidation state of Ti is +3. So Ti is Ti^{3+} i.e. d^1 . One electron will remain unpaired, so it is **paramagnetic**.

3. In $[\text{Fe}(\text{CN})_6]^{3-}$, oxidation state of Fe is +3. So Fe is Fe^{3+} i.e. d^5 . Since CN^- is a strong field ligand, it gives a low spin octahedral complex, so only **one** electron remains unpaired. Hence, it is **paramagnetic**.

4. In V_2O_5 , V is in +5 oxidation state, so it is V^{5+} i.e. d^0 . There are no d -electrons, so there cannot be any unpaired electron. Hence, V_2O_5 is **diamagnetic**.

So, the **paramagnetic species** are :



And the **diamagnetic species** is : V_2O_5

Now, compare the complexes in Statement II by counting unpaired electrons.

In $\text{K}_4[\text{Fe}(\text{CN})_6] \Rightarrow \text{Fe}$ is +2 (because complex anion is $4-$), so Fe^{2+} is d^6 . With CN^- (strong field), it is low spin, so all electrons pair up. Therefore, No. of unpaired electron = 0

$\text{K}_3[\text{Fe}(\text{CN})_6] \Rightarrow \text{Fe}$ is +3 (complex anion is $3-$), so Fe^{3+} is d^5 . With CN^- (strong field), it is low spin, so No. of unpaired electron = 1

$[\text{Fe}(\text{H}_2\text{O})_6]\text{SO}_4 \cdot \text{H}_2\text{O} \Rightarrow$ Here the cation is $[\text{Fe}(\text{H}_2\text{O})_6]^{2+}$, so Fe is +2 i.e. d^6 . Since H_2O is a weak field ligand, it is high spin, so No. of unpaired electron = 4

$[\text{Fe}(\text{H}_2\text{O})_6]\text{Cl}_3 \Rightarrow$ Here the cation is $[\text{Fe}(\text{H}_2\text{O})_6]^{3+}$, so Fe is +3 i.e. d^5 . With H_2O (weak field), it is high spin, so No. of unpaired electron = 5

Question11

Given below are two statements :

Statement I : Hybridisation, shape and spin only magnetic moment of $\text{K}_3 [\text{Co}(\text{CO}_3)_3]$ is $\text{sp}^3 d^2$, octahedral and 4.9 BM respectively.

Statement II : Geometry, hybridisation and spin only magnetic moment values (BM) of the ions $[\text{Ni}(\text{CN})_4]^{2-}$, $[\text{MnBr}_4]^{2-}$ and $[\text{CoF}_6]^{3-}$ respectively are square planar, tetrahedral, octahedral; dsp^2 , sp^3 , $\text{sp}^3 d^2$ and 0, 5.9, 4.9.

In the light of the above statements, choose the correct answer from the options given below

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Options:

A.

Both Statement I and Statement II are false

B.

Statement I is true but Statement II is false

C.

Both Statement I and Statement II are true

D.

Statement I is false but Statement II is true

Answer: C

Solution:

To decide hybridisation, geometry and magnetic moment, we mainly look at (i) the type of ligands (strong field or weak field) and (ii) the number of unpaired electrons. The spin-only magnetic moment depends on the number of unpaired electrons n .

The given results for each complex are summarised below (hybridisation $\Rightarrow$ geometry $\Rightarrow$ number of unpaired electrons $\Rightarrow$ magnetic moment in B.M.).

In $\text{K}_3 [\text{Co}(\text{CO}_3)_3] \Rightarrow \text{sp}^3 d^2$ hybridized, octahedral
 $\Rightarrow$ 4 unpaired electron
 $\Rightarrow$ 4.9 B.M.
 $[\text{Ni}(\text{CN})_4]^{2-} \Rightarrow \text{dsp}^2$ hybridized, square planar
 $\Rightarrow$ 0 unpaired electron
 $\Rightarrow$ 0 B.M.
 $[\text{MnBr}_4]^{2-} \Rightarrow \text{sp}^3$ hybridized, tetrahedral
 $\Rightarrow$ 5 unpaired electron
 $\Rightarrow$ 5.9 B.M.
 $[\text{CoF}_6]^{3-} \Rightarrow \text{sp}^3 d^2$ hybridized, octahedral
 $\Rightarrow$ 4 unpaired electron
 $\Rightarrow$ 4.9 B.M.

So, the hybridisation/geometry and magnetic moments mentioned in the statements match these standard results (based on unpaired electrons in each case).

Question12

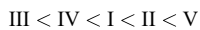
The wavelength of light absorbed for the following complexes are in the order



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Options:

A.



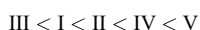
B.



C.



D.



Answer: C

Solution:

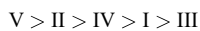
Wavelength of light absorbed increases as C.F.S.E of complex decreases.

$[\text{Co}(\text{CN})_6]^{3-}$ has maximum CFSE

$[\text{CoF}_6]^{3-}$ has least CFSE

Ligand field strength $\uparrow$; C.F.S.E $\uparrow$

Correct wavelength order.



Question13

The correct statement among the following is:

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Options:

A.

$[\text{Ni}(\text{CN})_4]^{2-}$ and $[\text{NiCl}_4]^{2-}$ are diamagnetic and $\text{Ni}(\text{CO})_4$ is paramagnetic.

B.

$\text{Ni}(\text{CO})_4$ is diamagnetic and $[\text{NiCl}_4]^{2-}$ and $[\text{Ni}(\text{CN})_4]^{2-}$ are paramagnetic.

C.

$\text{Ni}(\text{CO})_4$ and $[\text{NiCl}_4]^{2-}$ are diamagnetic and $[\text{Ni}(\text{CN})_4]^{2-}$ is paramagnetic.

D.

$\text{Ni}(\text{CO})_4$ and $[\text{Ni}(\text{CN})_4]^{2-}$ are diamagnetic and $[\text{NiCl}_4]^{2-}$ is paramagnetic.

Answer: D

Solution:

Option **D** is correct.

1) $\text{Ni}(\text{CO})_4$

- Oxidation state of Ni: CO is a neutral ligand, so Ni is in 0 state.
- Electronic configuration: $\text{Ni}(0) \Rightarrow 3d^{10}4s^0$
- Since it is d^{10} , all electrons are paired.
- Hence, $\text{Ni}(\text{CO})_4$ is **diamagnetic**.

2) $[\text{Ni}(\text{CN})_4]^{2-}$

- Oxidation state of Ni:
$$x + 4(-1) = -2 \Rightarrow x = +2$$
- So, Ni^{2+} is $3d^8$.
- CN^- is a **strong field ligand**, causes pairing of electrons.
- For $\text{Ni}^{2+}(d^8)$ with strong field ligands, complex becomes **square planar** (inner orbital), and all electrons get paired.
- Hence, $[\text{Ni}(\text{CN})_4]^{2-}$ is **diamagnetic**.

3) $[\text{NiCl}_4]^{2-}$

- Oxidation state of Ni:
$$x + 4(-1) = -2 \Rightarrow x = +2$$
- So, again Ni^{2+} is $3d^8$.
- Cl^- is a **weak field ligand**, so it does **not** cause pairing.
- Such complexes are generally **tetrahedral** (outer orbital) and remain **high spin**, leaving **2 unpaired electrons**.
- Hence, $[\text{NiCl}_4]^{2-}$ is **paramagnetic**.

Therefore:

- $\text{Ni}(\text{CO})_4 \rightarrow$ diamagnetic
- $[\text{Ni}(\text{CN})_4]^{2-} \rightarrow$ diamagnetic
- $[\text{NiCl}_4]^{2-} \rightarrow$ paramagnetic

So, the correct statement is **Option D**.

Question 14

The correct increasing order of spin-only magnetic moment values of the complex ions

$[\text{MnBr}_4]^{2-}$ (A), $[\text{Cu}(\text{H}_2\text{O})_6]^{2+}$ (B), $[\text{Ni}(\text{CN})_4]^{2-}$ (C) and $[\text{Ni}(\text{H}_2\text{O})_6]^{2+}$ (D) is :

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Options:

A.

$A = B < D < C$

B.

$C = D < B < A$

C.

$C < B < D < A$

D.

$A = B < C < D$

Answer: C

Solution:

$\text{Mn}^{2+} : 3d^5$

In $[\text{MnBr}_4]^{2-}$, Mn is in +2 oxidation state, so configuration is $3d^5$.

Br^- is a weak field ligand and the complex is tetrahedral, so it is high spin.

Hence number of unpaired electrons (n) = 5.

$\text{Cu}^{2+} : 3d^9$

In $[\text{Cu}(\text{H}_2\text{O})_6]^{2+}$, Cu is in +2 oxidation state, so configuration is $3d^9$.

This is an octahedral complex (six water ligands). In d^9 , there is always one unpaired electron.

So $t_{2g}^{2,2,2} e_g^{2,1}$ and (n) = 1.

$\text{Ni}^{2+} : 3d^8$

In $[\text{Ni}(\text{CN})_4]^{2-}$, Ni is in +2 oxidation state, so configuration is $3d^8$.

CN^- is a strong field ligand, so electrons pair up and the complex becomes square planar.

In square planar d^8 , all electrons are paired, so (n) = 0.

$\text{Ni}^{2+} : 3d^8$ (tetrahedral)

In $[\text{Ni}(\text{H}_2\text{O})_6]^{2+}$, Ni is in +2 oxidation state, so configuration is $3d^8$.

With H_2O (weak field), the complex is high spin; the distribution gives two unpaired electrons.

So $e_g^{2,2} t_{2g}^{2,1,1}$ and (n) = 2.

Question 15

Match List - I with List - II.

	List - I Chromium (III) Complexes (en = ethylene diamine)	List - II Δ_o (cm^{-1})
A	$[\text{Cr}(\text{CN})_6]^{3-}$	I. 15,060
B	$[\text{CrF}_6]^{3-}$	II. 17,400
C	$[\text{Cr}(\text{H}_2\text{O})_6]^{3+}$	III. 22,300
D	$[\text{Cr}(\text{en})_3]^{3+}$	IV. 26,600

Choose the correct answer from the options given below:

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Options:

A.

A-I, B-II, C-III, D-IV

B.

A-II, B-III, C-IV, D-I

C.

A-III, B-IV, C-I, D-II

D.

A-IV, B-I, C-II, D-III

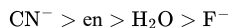
Answer: D

Solution:

The crystal field splitting energy (CFSE), represented by Δ_o , depends on how strong the ligands are and also on chelation (when a ligand forms a ring with the metal ion).

The stronger the ligand, the higher the value of Δ_o . Chelating ligands like ethylenediamine (en) also increase stability and splitting energy.

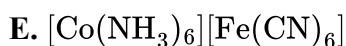
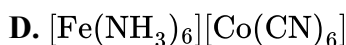
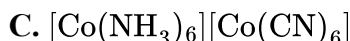
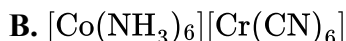
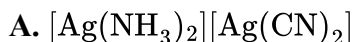
Among the given ligands, their strength follows this order:



This order means complexes with CN^- have the largest Δ_o , and those with F^- have the smallest Δ_o .

Question 16

Which of the following complexes will show coordination isomerism?



Choose the correct answer from the options given below:

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Options:

A.

B, C and D Only

B.

B, D and E Only

C.

A, C and D Only

D.

C, D and E Only

Answer: B

Solution:

Coordination Isomerism arises from the interchange of ligands between cationic and anionic entities of **different metal ions** present in a complex. When the complete interchange of ligands takes place, it must result in a new, chemically distinct isomer.

Let us evaluate each of the given choices under this concept:

- A.** $[\text{Ag}(\text{NH}_3)_2][\text{Ag}(\text{CN})_2]$: Both the cationic and anionic entities contain the same metal ion (Ag). If we fully interchange the ligands, we get $[\text{Ag}(\text{CN})_2][\text{Ag}(\text{NH}_3)_2]^+$, which is exactly the same compound written in reverse order. Further, a partial interchange leads to neutral molecules $[\text{Ag}(\text{NH}_3)(\text{CN})]$, which destroys the cation-anion pair. Hence, it **does not** show coordination isomerism.
- B.** $[\text{Co}(\text{NH}_3)_6][\text{Cr}(\text{CN})_6]$: This complex contains two different metal ions (Co and Cr). By interchanging the ligands between the cationic and anionic coordination spheres, we get a completely new distinct complex: $[\text{Cr}(\text{NH}_3)_6][\text{Co}(\text{CN})_6]$. Hence, it **shows** coordination isomerism. (This is also the classic example mentioned in NCERT).
- C.** $[\text{Co}(\text{NH}_3)_6][\text{Co}(\text{CN})_6]$: Both entities contain the same metal ion (Co). A complete interchange of all ligands gives $[\text{Co}(\text{CN})_6]^{3-}[\text{Co}(\text{NH}_3)_6]^{3+}$, which is exactly the same coordination compound. According to the strict NCERT definition requiring different metal ions for complete interchange, it **does not** exhibit coordination isomerism.
- D.** $[\text{Fe}(\text{NH}_3)_6][\text{Co}(\text{CN})_6]$: This complex has two different metal ions (Fe and Co). Exchanging the primary coordination spheres yields a distinct coordination isomer: $[\text{Co}(\text{NH}_3)_6][\text{Fe}(\text{CN})_6]$. Hence, it **shows** coordination isomerism.

- E. $[\text{Co}(\text{NH}_3)_6][\text{Fe}(\text{CN})_6]$: This complex is exactly the coordination isomer of the complex described in option D. Since it consists of different metal ions (Co and Fe), exchanging its ligands will yield the complex in D. Thus, it also **shows** coordination isomerism.

Conclusion:

Complexes **B, D, and E** exhibit coordination isomerism because they possess differently named metal ions in their cationic and anionic parts, allowing the exchange of ligands to produce distinctly new isomers.

Correct Answer:

Option B (B, D and E Only)

Question 17

Given below are two statements :

Statement I : Among Zn, Mn, Sc and Cu, the energy required to remove the third valence electron is highest for Zn and lowest for Sc.

Statement II : The correct order of the following complexes in terms of CFSE is $[\text{Co}(\text{H}_2\text{O})_6]^{2+} < [\text{Co}(\text{H}_2\text{O})_6]^{3+} < [\text{Co}(\text{en})_3]^{3+}$.

In the light of the above statements, choose the correct answer from the options given below :

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Options:

A.

Both **Statement I** and **Statement II** are true

B.

Both **Statement I** and **Statement II** are false

C.

Statement I is true but **Statement II** is false

D.

Statement I is false but **Statement II** is true

Answer: A

Solution:

Statement I

We need to compare the energy required to remove the **third valence electron**, that is, the **third ionization enthalpy**.

First write the electronic configurations:

- $\text{Zn} : [\text{Ar}] 3d^{10} 4s^2$
- $\text{Mn} : [\text{Ar}] 3d^5 4s^2$
- $\text{Sc} : [\text{Ar}] 3d^1 4s^2$
- $\text{Cu} : [\text{Ar}] 3d^{10} 4s^1$

Now, after removal of two electrons:

- $\text{Zn}^{2+} : [\text{Ar}] 3d^{10}$
- $\text{Mn}^{2+} : [\text{Ar}] 3d^5$
- $\text{Sc}^{2+} : [\text{Ar}] 3d^1$
- $\text{Cu}^{2+} : [\text{Ar}] 3d^9$

The third electron is now removed from these ions.

Which one will have highest third ionization enthalpy?

For Zn^{2+} , the configuration is $3d^{10}$, which is a **completely filled and very stable** configuration. Removing one electron from this is difficult.

So, third ionization enthalpy is **highest for Zn**.

Which one will have lowest third ionization enthalpy?

For Sc^{2+} , configuration is $3d^1$. Removing this one electron gives $Sc^{3+} : [Ar]$, which is a **noble gas configuration**, hence this removal is comparatively easier.

So, third ionization enthalpy is **lowest for Sc**.

Therefore, **Statement I is true**.

Statement II

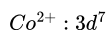
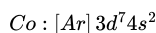
We need the order of complexes in terms of CFSE.

Given complexes:

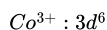
1. $[Co(H_2O)_6]^{2+}$
2. $[Co(H_2O)_6]^{3+}$
3. $[Co(en)_3]^{3+}$

Step 1: Find oxidation state and d -configuration

- In $[Co(H_2O)_6]^{2+}$, cobalt is Co^{2+}



- In $[Co(H_2O)_6]^{3+}$, cobalt is Co^{3+}



- In $[Co(en)_3]^{3+}$, cobalt is also Co^{3+}

so it is also $3d^6$

Step 2: Nature of ligand and spin state

- H_2O is a **weak field ligand**
- en is a **stronger field ligand** than H_2O

So:

- $[Co(H_2O)_6]^{2+}$ is octahedral, weak field, d^7
generally **high spin**
- $[Co(H_2O)_6]^{3+}$ is octahedral, d^6
because of higher charge on Co^{3+} , splitting is larger; this complex is effectively **low spin**
- $[Co(en)_3]^{3+}$ is octahedral, d^6 , strong field
definitely **low spin**

Step 3: Calculate CFSE

1. $[Co(H_2O)_6]^{2+}$: **high-spin** d^7

Configuration: $t_{2g}^5 e_g^2$

$$CFSE = 5(-0.4\Delta_o) + 2(+0.6\Delta_o) = -2.0\Delta_o + 1.2\Delta_o = -0.8\Delta_o$$

$$\text{Magnitude of CFSE} = 0.8\Delta_o$$

2. $[Co(H_2O)_6]^{3+}$: **low-spin** d^6

Configuration: $t_{2g}^6 e_g^0$

$$CFSE = 6(-0.4\Delta_o) = -2.4\Delta_o$$

$$\text{Magnitude} = 2.4\Delta_o$$

3. $[Co(en)_3]^{3+}$: **low-spin** d^6

Also $t_{2g}^6 e_g^0$, so

$$CFSE = -2.4\Delta_o$$

But here en is stronger field than H_2O , so Δ_o is larger.

Hence CFSE magnitude is greater for $[Co(en)_3]^{3+}$ than for $[Co(H_2O)_6]^{3+}$.

So the order is:

$$[Co(H_2O)_6]^{2+} < [Co(H_2O)_6]^{3+} < [Co(en)_3]^{3+}$$

Therefore, **Statement II is also true.**

Final Answer

Option A: Both Statement I and Statement II are true

Question18

Match the LIST-I with LIST-II

List-I Complex ion		List-II Calculated spin only magnetic moment (BM)	
A.	$[Cr(H_2O)_6]^{2+}$	I.	3.87
B.	$[Co(H_2O)_6]^{2+}$	II.	5.92
C.	$[Cu(H_2O)_6]^{2+}$	III.	4.90
D.	$[Mn(H_2O)_6]^{2+}$	IV.	1.73

Choose the correct answer from the options given below :

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Options:

A.

A-I, B-III, C-IV, D-II

B.

A-II, B-I, C-III, D-IV

C.

A-IV, B-II, C-I, D-III

D.

A-III, B-I, C-IV, D-II

Answer: D

Solution:

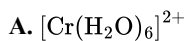
For matching, we use the spin-only magnetic moment formula:

$$\mu = \sqrt{n(n+2)} \text{ BM}$$

where n = number of unpaired electrons.

Since H_2O is a weak field ligand, all these octahedral complexes will be high spin.

Now find oxidation state, electronic configuration, number of unpaired electrons, and then magnetic moment.



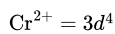
Here, chromium is in +2 oxidation state.

Atomic number of Cr = 24

Neutral Cr:



So,



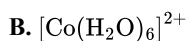
For high spin d^4 , number of unpaired electrons:

$$n = 4$$

Thus,

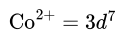
$$\mu = \sqrt{4(4+2)} = \sqrt{24} \approx 4.90 \text{ BM}$$

So, A $\rightarrow$ III



Cobalt is in +2 oxidation state.

Atomic number of Co = 27



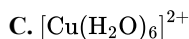
For high spin d^7 , number of unpaired electrons:

$$n = 3$$

Thus,

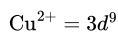
$$\mu = \sqrt{3(3+2)} = \sqrt{15} \approx 3.87 \text{ BM}$$

So, B $\rightarrow$ I



Copper is in +2 oxidation state.

Atomic number of Cu = 29



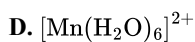
For d^9 , number of unpaired electrons:

$$n = 1$$

Thus,

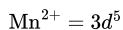
$$\mu = \sqrt{1(1+2)} = \sqrt{3} \approx 1.73 \text{ BM}$$

So, C $\rightarrow$ IV



Manganese is in +2 oxidation state.

Atomic number of Mn = 25



For high spin d^5 , number of unpaired electrons:

$$n = 5$$

Thus,

$$\mu = \sqrt{5(5+2)} = \sqrt{35} \approx 5.92 \text{ BM}$$

So, D $\rightarrow$ II

Final matching

A-III, B-I, C-IV, D-II

So the correct option is:

Option D

Question 19

Consider the metal complexes $[\text{Ni}(\text{en})_3]^{2+}$ (A), $[\text{NiCl}_4]^{2-}$ (B) and $[\text{Ni}(\text{NH}_3)_6]^{2+}$ (C). Choose the CORRECT option by considering the number of unpaired electrons present in (A), (B) and (C) respectively and the order of frequency of absorption.

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Options:

A.

2, 2, 2 and $(A) > (C) > (B)$

B.

0, 2, 0 and $(A) > (C) > (B)$

C.

2, 2, 0 and $(B) > (C) > (A)$

D.

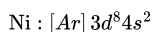
2, 2, 2 and $(C) > (A) > (B)$

Answer: A

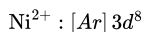
Solution:

For all the three complexes, the metal ion is Ni^{2+} .

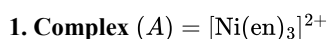
The electronic configuration of Ni is



So,



Now let us examine each complex.



Here, en is ethane-1,2-diamine, a neutral bidentate ligand.

- Coordination number = 6
- Geometry = octahedral

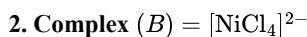
For an octahedral d^8 configuration, the arrangement is:



This gives **2 unpaired electrons**.

So, for (A):

Number of unpaired electrons = 2



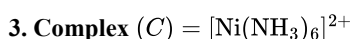
- Coordination number = 4
- Since Cl^- is a weak ligand, this complex is **tetrahedral**.

For tetrahedral d^8 , the splitting is small, and the electrons remain high spin.

Hence, it has **2 unpaired electrons**.

So, for (B):

Number of unpaired electrons = 2



- Coordination number = 6
- Geometry = octahedral

Again, for octahedral d^8 :



So it also has **2 unpaired electrons**.

Thus, for (C):

Number of unpaired electrons = 2

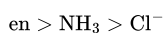
Frequency of absorption

The frequency of absorbed light depends on crystal field splitting:

$$\Delta \propto \nu$$

Greater the splitting, greater the frequency of absorption.

Now compare ligand strengths:

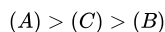


Also, tetrahedral complexes have smaller splitting than octahedral complexes.

So,

- $[\text{Ni}(\text{en})_3]^{2+}$ has the highest Δ
- $[\text{Ni}(\text{NH}_3)_6]^{2+}$ has next
- $[\text{NiCl}_4]^{2-}$ has the least

Therefore, order of frequency of absorption is:



Hence the correct option is:

Option A

Question 20

Given below are two statements:

Statement I: Each electron in e_g orbitals destabilizes the orbitals by $+0.6\Delta_o$ and each electron in the t_{2g} orbitals stabilizes the orbitals by $-0.4\Delta_o$ in an octahedral field on the basis of crystal field theory.

Statement II: All the d - orbitals of the transition metals have the same energy in their free atomic state but when a complex is formed the ligands destroy the degeneracy of these orbitals on the basis of crystal field theory.

In the light of the above statements, choose the correct answer from the options given below

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Options:

A.

Both Statement I and Statement II are correct

B.

Both Statement I and Statement II are incorrect

C.

Statement I is correct but Statement II is incorrect

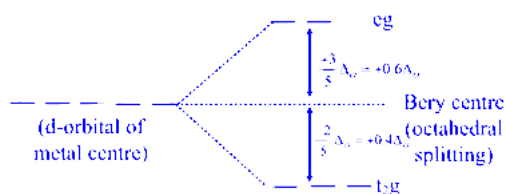
D.

Statement I is incorrect but Statement II is correct

Answer: A

Solution:

Statement I :



In an octahedral field, the five d -orbitals split into two sets: lower-energy t_{2g} and higher-energy e_g .

Each electron placed in the higher-energy e_g orbitals increases (destabilizes) the energy by $+0.6\Delta_o$.

Each electron placed in the lower-energy t_{2g} orbitals decreases (stabilizes) the energy by $-0.4\Delta_o$.

So, when electrons go into e_g , the crystal field stabilisation becomes less (energy goes up), and when electrons go into t_{2g} , the stabilisation becomes more (energy goes down).

Statement II : In free atomic state of transition metal, degeneracy of d -orbital is maintained.

In the free (isolated) metal ion, all five d -orbitals have the same energy, so they are degenerate.

When ligands approach to form a complex, the metal–ligand interactions are not the same for all d -orbitals.

Because of this, the five d -orbitals split into two sets (like t_{2g} and e_g in octahedral complexes), and the degeneracy is removed.

Question21

The correct statements about metal carbonyls are

A. The metal-carbon bonds in metal carbonyls possess both σ and π character.

B. Due to synergic bonding interactions between metal and CO ligand, the metal-carbon bond becomes weak.

C. The metal-carbon σ bond is formed by the donation of lone pair of electrons on the carbonyl carbon into a vacant orbital of metal.

D. The metal-carbon π bond is formed by the donation of electrons from filled d -orbital of metal into vacant π^* orbital of CO .

Choose the correct answer from the options given below :

JEE Main 2026 (Online) 5th April Morning Shift

Options:

A.

A and B Only

B.

A, C and D Only

C.

B and C Only

D.

A and D Only

Answer: B

Solution:

In metal carbonyls, bonding between the metal and CO ligand is explained by **synergic bonding**.

Statement A

The metal-carbon bond in metal carbonyls has both σ and π character.

This is **correct**.

- CO donates a lone pair from carbon to the metal, forming a σ bond.
- The metal donates electron density back to CO into its vacant π^* orbital, forming a π bond.

So, the bond has **both σ and π character**.

Statement B

Due to synergic bonding interactions between metal and CO ligand, the metal-carbon bond becomes weak.

This is **incorrect**.

Synergic bonding actually **strengthens the metal-carbon bond** because:

- CO donates electrons to metal by σ donation.
- Metal donates electrons back to CO by π back bonding.

These two processes reinforce each other, so the **M–C bond becomes stronger**, not weaker.

Statement C

The metal-carbon σ bond is formed by donation of lone pair of electrons on the carbonyl carbon into a vacant orbital of metal.

This is **correct**.

This is exactly the σ donation from CO to the metal.

Statement D

The metal-carbon π bond is formed by donation of electrons from filled d -orbital of metal into vacant π^* orbital of CO.

This is **correct**.

This is called **π back bonding** or **back donation**.

So the correct statements are:

A, C, and D

Hence, the correct answer is:

Option B

Question22

Identify the correct statements from the following

A. $[\text{Fe}(\text{C}_2\text{O}_4)_3]^{3-}$ is the most stable complex among $[\text{Fe}(\text{OH})_6]^{3-}$, $[\text{Fe}(\text{C}_2\text{O}_4)_3]^{3-}$ and $[\text{Fe}(\text{SCN})_6]^{3-}$

B. The stability of $[\text{Cu}(\text{NH}_3)_4]^{2+}$ is greater than that of $[\text{Cu}(\text{en})_2]^{2+}$

C. The hybridization of Fe in $\text{K}_4[\text{Fe}(\text{CN})_6]$ is d^2sp^3

D. $[\text{Fe}(\text{NO}_2)_3\text{Cl}_3]^{3-}$ exhibits linkage isomerism

E. NO_2^- and SCN^- ligands are NOT ambidentate ligands

Choose the correct answer from the options given below :

JEE Main 2026 (Online) 5th April Evening Shift

Options:

A.

A, B, C, D and E

B.

B, C and D only

C.

A, C and D only

D.

A, C and E only

Answer: C

Solution:

(A) The stability of a complex increases when the ligand is stronger (i.e., stronger field/stronger bonding ligand gives more stable complex).

The ligand strength order given is:



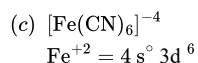
Also, $\text{C}_2\text{O}_4^{2-}$ (oxalate) is a **bidentate** ligand. Because of the **chelate effect**, complexes with bidentate ligands are generally more stable.

So, $[\text{Fe}(\text{C}_2\text{O}_4)_3]^{3-}$ is the most stable among the given complexes. Hence, statement (A) is true.

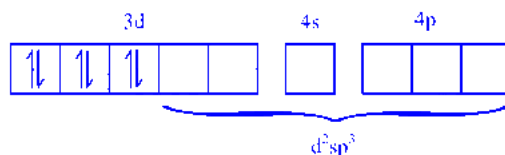
(B) en (ethylenediamine) is also a **chelating (bidentate)** ligand, while NH_3 is **monodentate** (non-chelating).

Due to the chelate effect, $[\text{Cu}(\text{en})_2]^{2+}$ is more stable than $[\text{Cu}(\text{NH}_3)_4]^{2+}$.

So, statement (B) is false.

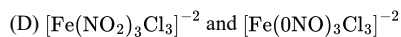


CN^- = strong field ligand, so it forms a **low spin** complex.

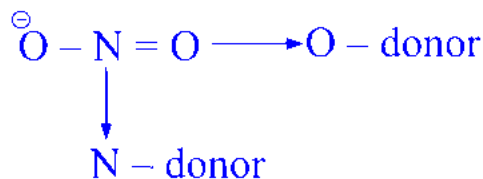


Because CN^- is a strong field ligand, electrons pair up in the $3d$ orbitals. This leads to an **inner orbital complex** with d^2sp^3 hybridisation for Fe.

So, statement (C) is true.



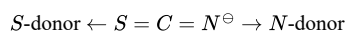
Linkage isomerism is shown when an **ambidentate** ligand can donate electrons through two different atoms.

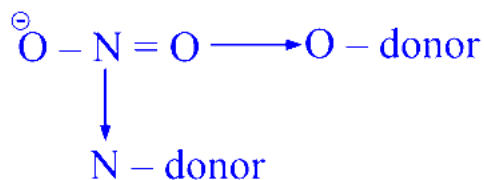


Here, NO_2^- can coordinate through N (nitro) or through O (nitrito). Therefore, the complex shows linkage isomerism.

So, statement (D) is true.

(E) NO_2^- and SCN^- are **ambidentate** ligands (they can attach through different atoms).





So, the statement “ NO_2^- and SCN^- ligands are NOT ambidentate ligands” is incorrect.

Hence, statement (E) is false.

Question 23

Given below are two statements :

Statement I : Presence of large number of unpaired electrons in transition metal atoms results in higher enthalpies of their atomisation.

Statement II : $d_{xy} = d_{xz} = d_{yz} < d_{x^2-y^2} = d_{z^2}$ and $d_{x^2-y^2} = d_{z^2} < d_{xy} = d_{xz} = d_{yz}$ are the d-orbital splittings in $[\text{Fe}(\text{H}_2\text{O})_6]^{3+}$ and $[\text{Ni}(\text{Cl})_4]^{2-}$ complex ions respectively.

In the light of the above statements, choose the correct answer from the options given below :

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Options:

A.

Both Statement I and Statement II are correct

B.

Both Statement I and Statement II are incorrect

C.

Statement I is correct but Statement II is incorrect

D.

Statement I is incorrect but Statement II is correct

Answer: A

Solution:

Enthalpy of atomisation is the energy needed to break 1 mole of a metal (in solid state) into its separate atoms (in gaseous state).

In transition metals, metallic bonding becomes stronger when there are more unpaired electrons. This is because unpaired electrons can take part more effectively in metal-metal bonding.

So, if a transition metal atom has a large number of unpaired electrons, its metallic bonds are stronger, and more energy is required to convert the metal into atoms.

Therefore, more unpaired electrons $\propto$ stronger metallic bonding $\propto$ higher enthalpy of atomisation. Hence, **Statement I is correct.**

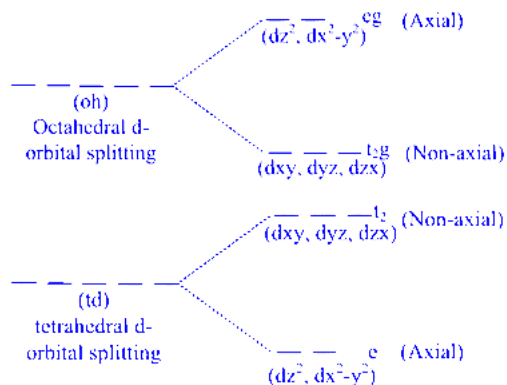
$[\text{Fe}(\text{H}_2\text{O})_6]^{3+}$ is an **octahedral complex** (coordination number 6). In an octahedral field, the d-orbitals split as:

$$d_{xy} = d_{xz} = d_{yz} < d_{x^2-y^2} = d_{z^2}$$

$[\text{Ni}(\text{Cl})_4]^{2-}$ is a **tetrahedral complex** (coordination number 4). In a tetrahedral field, the splitting is reversed compared to octahedral, i.e.:

$$d_{x^2-y^2} = d_{z^2} < d_{xy} = d_{xz} = d_{yz}$$

So, the given splittings match the correct orbital splitting patterns for these two geometries. Hence, **Statement II is also correct.**



Question24

Which of the following are true about the energy of the given d-orbitals of a tetrahedral complex?

A. $d_{xy} = d_{xz} > d_{x^2-y^2}$

B. $d_{xy} = d_{yz} > d_{z^2}$

C. $d_{x^2-y^2} > d_{z^2} > d_{xz}$

D. $d_{x^2-y^2} = d_{z^2} < d_{xz}$

Choose the correct answer from the given below :

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Options:

A.

A, B and D only

B.

A and B only

C.

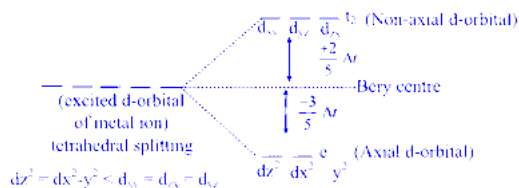
B and D only

D.

B, C and D only

Answer: A

Solution:



Question25

Match the LIST-I with LIST-II

List-I Electronic configuration of tetrahedral metal ion		List-II Crystal Field Stabilization Energy (Δ_t)	
A.	d^2	I.	-0.6
B.	d^4	II.	-0.8
C.	d^6	III.	-1.2
D.	d^8	IV.	-0.4

Choose the correct answer from the options given below:

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Options:

A.

A-III, B-IV, C-II, D-I

B.

A-III, B-I, C-IV, D-II

C.

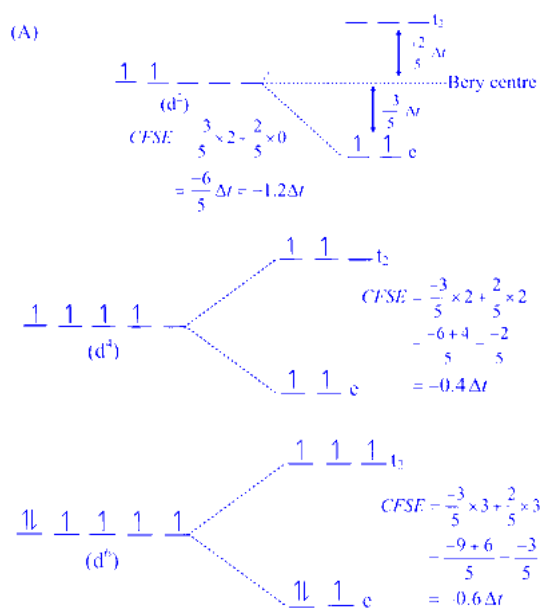
A-III, B-IV, C-I, D-II

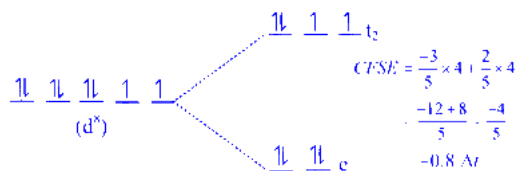
D.

A-II, B-I, C-IV, D-III

Answer: C

Solution:





Question 26

Which of the following sequences of hybridisation, geometry and magnetic nature are correct for the given coordination compounds?

- A. $[\text{NiCl}_4]^{2-}$ — sp^3 , tetrahedral, paramagnetic
- B. $[\text{Ni}(\text{NH}_3)_6]^{2+}$ — $\text{sp}^3 d^2$, octahedral, paramagnetic
- C. $[\text{Ni}(\text{CO})_4]$ — sp^3 , tetrahedral, paramagnetic
- D. $[\text{Ni}(\text{CN})_4]^{2-}$ — dsp^2 , square planar, diamagnetic

Choose the correct answer from the options given below :

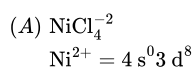
JEE Main 2026 (Online) 6th April Evening Shift

Options:

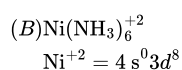
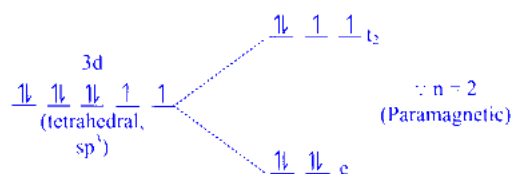
- A.
- A, B, C and D
- B.
- B, C and D only
- C.
- A, C and D only
- D.
- A, B and D only

Answer: D

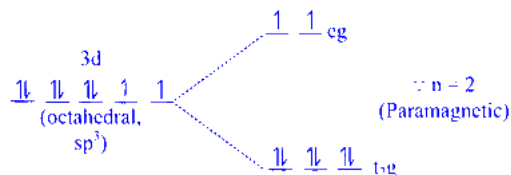
Solution:



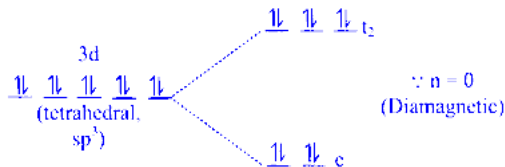
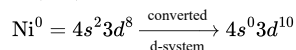
Cl: weak field ligand, high spin complex.



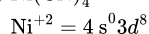
NH_3 : Weak field ligand, high spin complex.



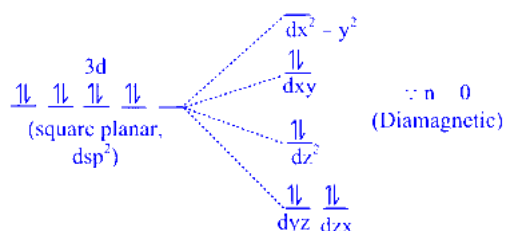
(C) $\text{Ni}(\text{CO})_4$



(D) $\text{Ni}(\text{CN})_4^{2-}$



CN: strong field ligand, low spin complex



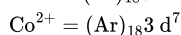
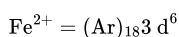
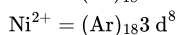
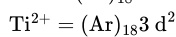
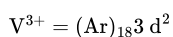
Question27

Identify the metal ions among Co^{2+} , Ni^{2+} , Fe^{2+} , V^{3+} and Ti^{2+} having a spin-only magnetic moment value more than 3.0 BM. The sum of unpaired electrons present in the high spin octahedral complexes formed by those metal ions is _____ .

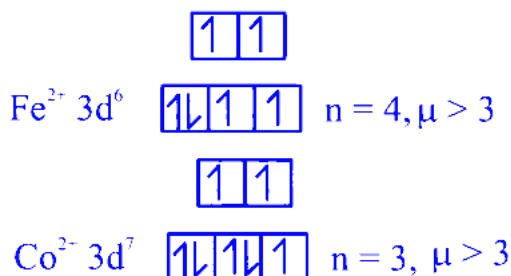
JEE Main 2026 (Online) 21st January Evening Shift

Answer: 7

Solution:



Only for Fe^{2+} and Co^{2+} μ is more than 3.0 B.M.



∴ Number of unpaired electrons = 4 + 3 = 7

Question28

The crystal field splitting energy of $[\text{Co}(\text{oxalate})_3]^{3-}$ complex is ' n ' times that of the $[\text{Cr}(\text{oxalate})_3]^{3-}$ complex. Here ' n ' is _____. (Assume $\Delta_0 \gg P$)

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Answer: 2

Solution:

Since $\Delta_0 \gg P$, we ignore pairing energy and decide the electron arrangement mainly using Δ_0 .

For $[\text{Co}(\text{ox})_3]^{3-}$:

Co^{+3} has configuration d^6 .

In an octahedral field (given $\Delta_0 \gg P$), all 6 electrons occupy t_{2g} first, so the distribution is $t_{2g}^{2,2,2} e_g^{0,0}$.

Each electron in t_{2g} gives CFSE of $(-0.4\Delta_0)$, so

$$\text{CFSE} = 6 \times (-0.4\Delta_0) = -2.4\Delta_0$$

For $[\text{Cr}(\text{ox}_3)]^{3-}$:

Cr^{+3} has configuration d^3 .

So the electrons occupy t_{2g} as $t_{2g}^{1,1,1} e_g^{0,0}$.

Hence,

$$\text{CFSE} = 3 \times (-0.4\Delta_0) = -1.2\Delta_0$$

Now,

$$\frac{(\text{CFSE})_{\text{Co}^{+3}}}{(\text{CFSE})_{\text{Cr}^{+3}}} = \frac{-2.4\Delta_0}{-1.2\Delta_0} = 2$$

So, $n = 2$.

Question29

Total number of unpaired electrons present in the central metal atoms/ions of

$[\text{Ni}(\text{CO})_4]$, $[\text{NiCl}_4]^{2-}$, $[\text{PtCl}_2(\text{NH}_3)_2]$, $[\text{Ni}(\text{CN})_4]^{2-}$ and $[\text{Pt}(\text{CN})_4]^{2-}$ is _____.

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Answer: 2

Solution:

In $[\text{Ni}(\text{CO})_4]$, Ni^0 : $3d^8 4s^2$

Hybridisation state: sp^3

Unpaired electron = 0

In $[\text{NiCl}_4]^{2-}$, Ni^{2+} : $3d^8$

Hybridisation state: sp^3

Unpaired electron = 2

In $[\text{PtCl}_4]^{2-}$, Pt^{2+} : 5 d^8

Hybridisation state : dsp^2

Unpaired electron = 0

In $[\text{Ni}(\text{CN})_4]^{2-}$, Ni^{2+} : 3 d^8

Hybridisation state: dsp^2

Unpaired electron = 0

In $[\text{Pt}(\text{NH}_3)_2\text{Cl}_2]$, Pt^{2+} : 5 d^8

Hybridisation state: dsp^2

Unpaired electron = 0

Question30

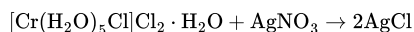
A chromium complex with a formula $\text{CrCl}_3 \cdot 6\text{H}_2\text{O}$ has a spin only magnetic moment value of 3.87 BM and its solution conductivity corresponds to 1 : 2 electrolyte. 2.75 g of the complex solution was initially passed through a cation exchanger. The solution obtained after the process was reacted with excess of AgNO_3 . The amount of AgCl formed in the above process is ____ g. (Nearest integer)

[Given: Molar mass in gmol^{-1} Cr : 52; Cl : 35.5, Ag : 108, O : 16, H : 1]

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Answer: 3

Solution:



$$\frac{2.75/266.5}{= 0.0103 \text{ moles}} \quad \quad \quad \frac{-}{0.02063 \text{ mole}}$$

$$\text{Mass of AgCl} = 0.02063 \times 143.5 = 2.96\text{gm}$$

Question31

X is the number of geometrical isomers exhibited by $[\text{Pt}(\text{NH}_3)(\text{H}_2\text{O})\text{BrCl}]$.

Y is the number of optically inactive isomer(s) exhibited by $[\text{CrCl}_2(\text{ox})_2]^{3-}$

Z is the number of geometrical isomers exhibited by $[\text{Co}(\text{NH}_3)_3(\text{NO}_2)_3]$.

The value of $X + Y + Z$ is ____ .

JEE Main 2026 (Online) 28th January Morning Shift

Answer: 6

Solution:

1) Find X for $[\text{Pt}(\text{NH}_3)(\text{H}_2\text{O})\text{BrCl}]$

- Pt in such complexes is usually Pt(II), which (as per NCERT) forms **square planar** complexes.
- The complex is of type $[\text{MABCD}]$ (four different ligands).

For square planar $[\text{MABCD}]$, the number of **geometrical isomers** is **3** (based on which ligand is trans to a chosen ligand; distinct trans-pairings give 3 arrangements).

So, $X = 3$.

2) Find Y for $[\text{CrCl}_2(\text{ox})_2]^{3-}$

- Here ox (oxalate) is a **bidentate** ligand.
- Two oxalate ligands occupy 4 coordination positions, and 2 positions are occupied by Cl^- .
- So the complex is **octahedral** of type $[\text{M}(\text{AA})_2\text{B}_2]$.

Such complexes show **cis–trans** isomerism:

- cis** form is **optically active** (exists as Δ and Λ enantiomers)
- trans** form is **optically inactive**

So optically inactive isomers = only **1 (trans)**.

Thus, $Y = 1$.

3) Find Z for $[\text{Co}(\text{NH}_3)_3(\text{NO}_2)_3]$

This is an octahedral complex of type $[\text{MA}_3\text{B}_3]$.

It shows **fac–mer** isomerism:

- fac**
- mer**

So, $Z = 2$.

Final Answer

$$X + Y + Z = 3 + 1 + 2 = 6.$$

6

Question32

The number of isoelectronic species among Sc^{3+} , Cr^{2+} , Mn^{3+} , Co^{3+} and Fe^{3+} is ‘n’. If ‘n’ moles of AgCl is formed during the reaction of complex with formula $\text{CoCl}_3(\text{en})_2\text{NH}_3$ with excess of AgNO_3 solution, then the number of electrons present in the t_{2g} orbital of the complex is _____.

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Answer: 6

Solution:

Sc^{+3}	18
Cr^{+2}	22
Mn^{+3}	22
Co^{+3}	24
Fe^{+3}	23

In the table, the number written for each ion is its total number of electrons.

Cr^{2+} and Mn^{3+} both have 22 electrons, so they are isoelectronic (same number of electrons).

So, the number of isoelectronic species is

$$n = 2$$

Now, for the complex $CoCl_3(en)_2NH_3$:

en and NH_3 are neutral ligands, so they do not change the charge.

In the given formula, total Cl are 3. If $n = 2$ moles of $AgCl$ are formed with excess $AgNO_3$, it means 2 chloride ions are outside the coordination sphere (they are ionisable and precipitate as $AgCl$).

So the complex must be:

Complex is : $[Co(en)_2NH_3Cl]Cl_2$

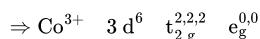
Here, 2 Cl^- are outside, so the complex ion has charge $+2$.

Let oxidation state of Co be x .

Inside the bracket: en and NH_3 contribute 0, and one coordinated Cl^- contributes -1 .

So, $x - 1 = +2 \Rightarrow x = +3$.

Thus cobalt is Co^{3+} , and its d -electron configuration is:



So, total electrons in t_{2g} orbitals = $2 + 2 + 2 = 6$.

Question 33

5.33 g of $CrCl_3 \cdot 6H_2O$, which is a 1 : 3 electrolyte, is dissolved in water and is passed through a cation exchanger. The chloride ions in the eluted solution, on treatment with $AgNO_3$ results in 8.61 g of $AgCl$. The ratio of moles of complex reacted and moles of $AgCl$ formed is _____ $\times 10^{-2}$. (Nearest integer)

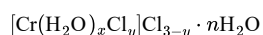
[Molar mass in $g\ mol^{-1}$ Cr : 52, Ag : 108, Cl : 35.5, H : 1, O : 16]

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Answer: 33

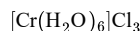
Solution:

Let the compound be of the type

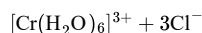


Since it is a 1 : 3 electrolyte, it gives a total of 4 ions in solution.

So, the complex must be:



because this dissociates as



Now calculate moles of the complex.

Molar mass of $CrCl_3 \cdot 6H_2O$:

$$= 52 + 3(35.5) + 6(18)$$

$$= 52 + 106.5 + 108 = 266.5\ g\ mol^{-1}$$

Moles of complex taken:

$$\frac{5.33}{266.5} = 0.02\ mol$$

Now, mass of $AgCl$ formed = 8.61 g

Molar mass of $AgCl$:

$$108 + 35.5 = 143.5\ g\ mol^{-1}$$

Moles of $AgCl$ formed:

$$\frac{8.61}{143.5} = 0.06\ mol$$

Required ratio:

$$\frac{\text{moles of complex reacted}}{\text{moles of AgCl formed}} = \frac{0.02}{0.06} = \frac{1}{3} = 0.3333$$

Now express this as

$$x \times 10^{-2}$$

So,

$$0.3333 = 33.33 \times 10^{-2}$$

Nearest integer = 33.

Final answer:

33

Question 34

The total number of unpaired electrons present in the d^3 , d^4 (low spin), d^5 (high spin), d^6 (high spin) and d^7 (low spin) octahedral complex systems is ____.

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Answer: 15

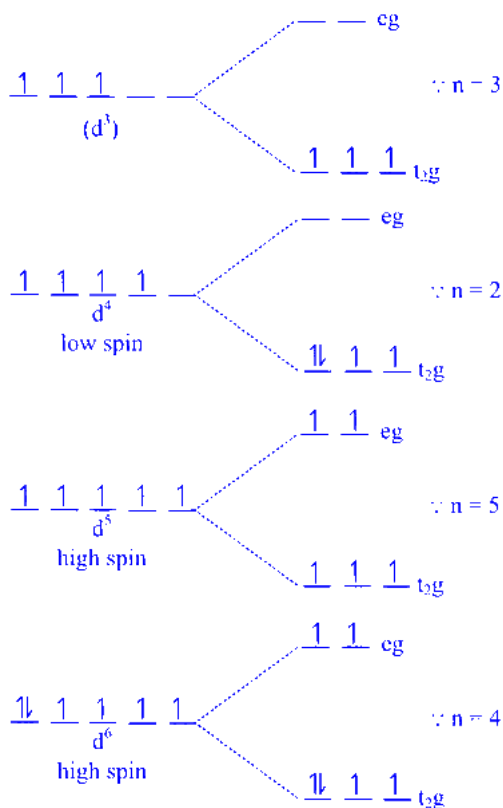
Solution:

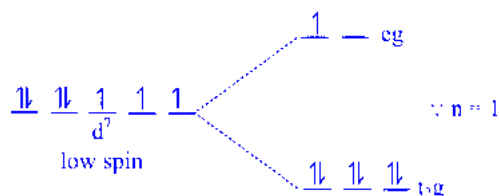
for oh complex,

In an octahedral (*oh*) complex, the five *d*-orbitals split into two sets: lower energy t_{2g} and higher energy e_g .

To find unpaired electrons, we fill electrons in these orbitals using Hund's rule. Then we decide whether pairing happens early (low spin) or not (high spin), as given in the question.

Below diagrams show the electron filling for each case (d^3 , d^4 low spin, d^5 high spin, d^6 high spin, and d^7 low spin). Count the unpaired electrons in each diagram.





Now add the number of unpaired electrons from each case shown in the diagrams:

$$\begin{aligned}\text{Total unpaired } e^- &= 3 + 2 + 5 + 4 + 1 \\ &= 15\end{aligned}$$

Question35

An excess of AgNO_3 is added to 100 mL of a 0.05 M solution of tetraaquadichloridochromium (III) chloride. The number of moles of AgCl precipitated will be $\text{---} \times 10^{-3}$.

(Nearest integer)

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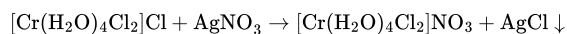
Answer: 5

Solution:

The given complex is $[\text{Cr}(\text{H}_2\text{O})_4\text{Cl}_2]\text{Cl}$.

When an excess of AgNO_3 is added, the Cl^- ion outside the coordination sphere reacts with Ag^+ to form a white precipitate of silver chloride (AgCl).

The reaction is:



From the reaction, 1 mole of the complex gives 1 mole of AgCl .

Now, the number of moles of the given complex is calculated as:

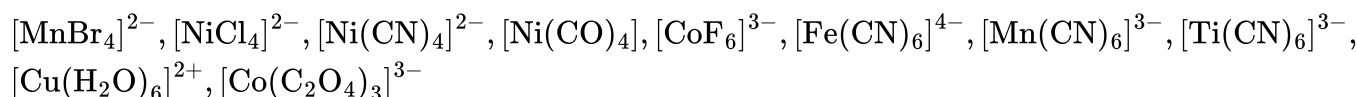
$$\text{Moles of complex} = M \times V = 0.05 \times 100 \times 10^{-3} = 5 \times 10^{-3} \text{ moles}$$

Hence, moles of AgCl formed = 5×10^{-3} moles.

Therefore, the required value of $x = 5$.

Question36

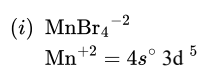
Number of paramagnetic complexes among the following is ____ .



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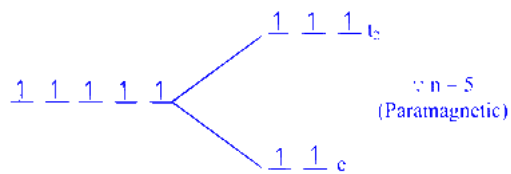
Answer: 6

Solution:



Br → weak field ligand

→ High spin complex

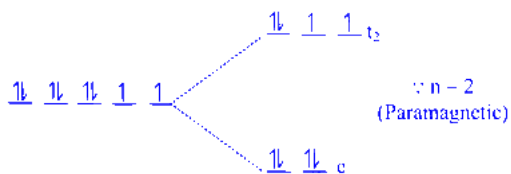


(ii) NiCl_4^{2-}

$\text{Ni}^{+2} = 4s^0 3d^8$

Cl → weak field ligand

→ High spin complex.

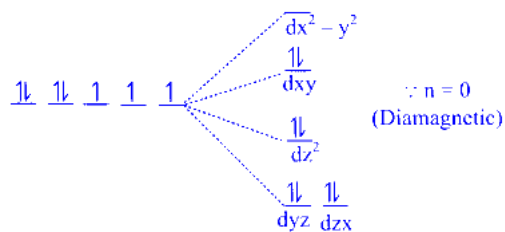


(iii) $\text{Ni}(\text{CN})_4^{2-}$

$\text{Ni}^{+2} = 4s^0 3d^8$

CN → strong field ligand

→ low spin complex.

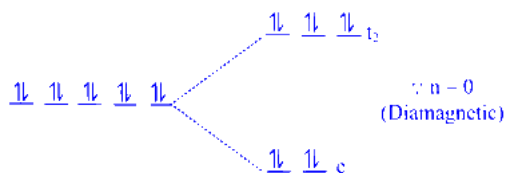


(iv) $\text{Ni}(\text{CO})_4$

$\text{Ni}^0 = 4s^2 3d^8 = 4s^0 3d^{10}$ (converted d-system)

Co → strong field ligand

→ low spin complex.

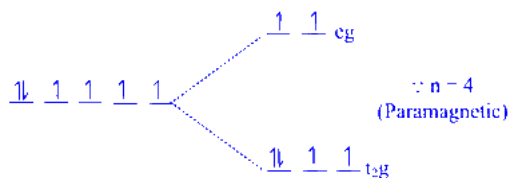


(v) CoF_6^{3-}

$\text{Co}^{+3} = 4s^0 3d^6$

F → weak field ligand

→ high spin complex.

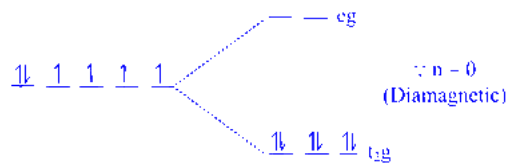


(vi) $\text{Fe}(\text{CN})_6^{4-}$

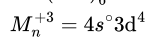
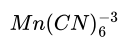
$\text{Fe}^{+2} = 4s^0 3d^6$

CN → strong field ligand

→ low spin complex.

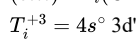
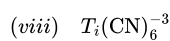
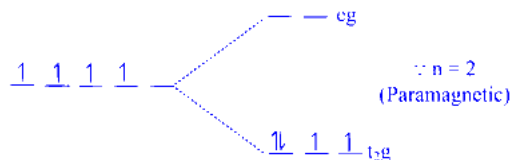


(vii)

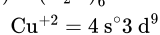
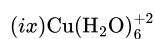
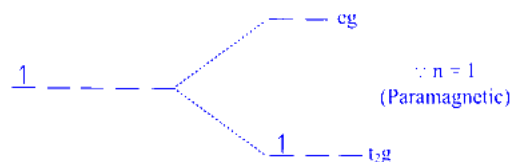


CN $\rightarrow$ strong field ligand.

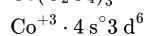
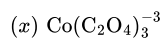
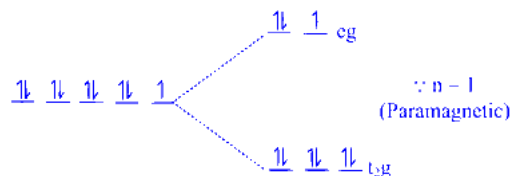
$\rightarrow$ low spin complex



CN $\rightarrow$ strong field ligand.

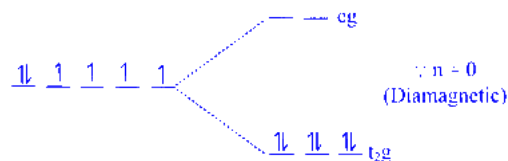


$H_2O \rightarrow$ weak field ligand.



$C_2O_4^{2-} \rightarrow$ strong field ligand.

$\rightarrow$ low spin complex.



Question37

The complex of Ni^{2+} ion and dimethyl glyoxime contains _____ number of Hydrogen (H) atoms.

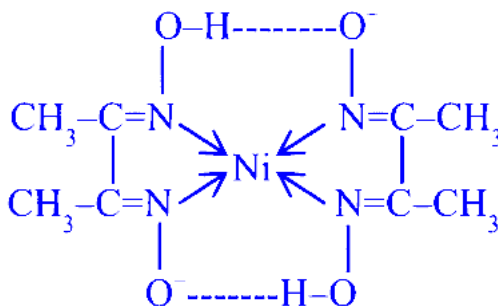
JEE Main 2025 (Online) 22nd January Evening Shift

Answer: 14

Solution:

Ni^{2+} with (dmg) forms $[\text{Ni}(\text{dmg})_2]^{2+}$ having 2 H-Bonds as shown :

The no. of H atoms = 14



Question38

Total number of molecules/species from following which will be paramagnetic is _____.

O_2 , O_2^+ , O_2^- , NO, NO_2 , CO, $\text{K}_2[\text{NiCl}_4]$, $[\text{Co}(\text{NH}_3)_6]\text{Cl}_3$, $\text{K}_2[\text{Ni}(\text{CN})_4]$

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Answer: 6

Solution:

To determine which molecules or species are paramagnetic, we need to identify those with unpaired electrons. Here's the analysis:

O_2 : This molecule has 2 unpaired electrons as predicted by the Molecular Orbital Theory (MOT), making it paramagnetic.

O_2^+ : This ion has 1 unpaired electron according to MOT, which also makes it paramagnetic.

O_2^- : This ion contains 1 unpaired electron as determined by MOT, indicating it is paramagnetic.

NO: As a species with an odd number of electrons, NO has unpaired electrons, making it paramagnetic.

NO_2 : This molecule is also an odd-electron species, which means it has unpaired electrons and is paramagnetic.

$\text{K}_2[\text{NiCl}_4]$: In this compound, Ni^{2+} (nickel with a +2 oxidation state) has an electron configuration of $3d^8$. Given the presence of weak field ligands and a coordination number of 4, this forms a tetrahedral complex. As a result, it is paramagnetic with 2 unpaired electrons.

Thus, the paramagnetic species in the provided list are O_2 , O_2^+ , O_2^- , NO, NO_2 , and $\text{K}_2[\text{NiCl}_4]$.

Question39

The spin only magnetic moment (μ) value (B.M.) of the compound with strongest oxidising power among Mn_2O_3 , TiO and VO is _____ B.M. (Nearest integer).

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Answer: 5

Solution:

The compound with the strongest oxidizing power among Mn_2O_3 , TiO, and VO is Mn_2O_3 . This conclusion is based on the standard electrode potential (E°) value:

$$E^\circ_{\text{Mn}^{+3}/\text{Mn}^{+2}} = +1.57 \text{ V}$$

For Mn^{3+} , the electronic configuration is d^4 . The spin-only magnetic moment (μ) is calculated as follows:

$$\mu = \sqrt{n(n+2)} \text{ BM}$$

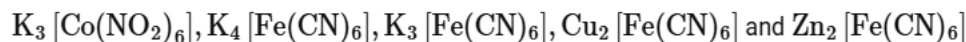
Where n is the number of unpaired electrons. Here, $n = 4$:

$$\mu = \sqrt{4(4+2)} = \sqrt{24} \text{ BM}$$

This calculates to approximately 4.89 BM, which rounds to an integer value of 5 BM.

Question 40

Consider the following low-spin complexes



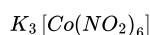
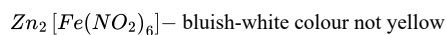
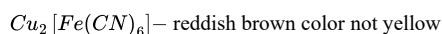
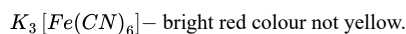
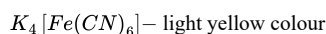
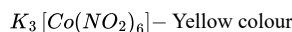
The sum of the spin-only magnetic moment values of complexes having yellow colour is _____ B.M.
(answer in nearest integer)

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Answer: 0

Solution:

Given low-spin complexes:



Oxidation state of Co : +3

Configuration:

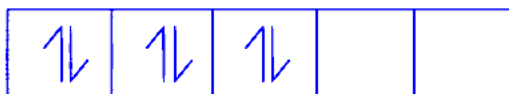


$$3(+1) + x + 6(-1) = 0$$

$$3 + x - 6 = 0$$

$$x = 6 - 3 = +3$$

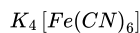
For low-spin d^6 configuration, no unpaired electrons are present :



spin-only magnetic moment, $\mu = \sqrt{n(n+2)}$

$$n = 0, \mu = 0$$

This complex shows yellow colour (bright yellow)



Oxidation state of Fe : +2

$$4(+1) + x + 6(-1) = 0$$

$$4 + x - 6 = 0$$

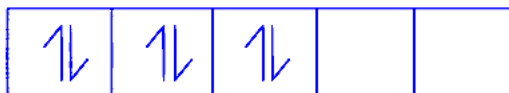
$$x = 6 - 4 = +2$$

Configuration:

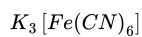


$$\mu = 0$$

For low-spin d^6 configuration, no unpaired electrons are present :



This complex shows yellow colour (light yellow)



Oxidation state of Fe : +3

$$3(+1) + x + 6(-1) = 0$$

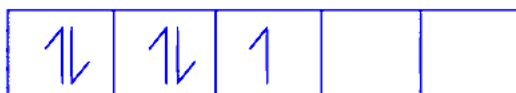
$$3 + x - 6 = 0$$

$$x = 6 - 3 = +3$$

Configuration:



low spin d^5



One unpaired electron is present.

$$\text{Spin-only magnetic moment, } \mu = \sqrt{n(n+2)}$$

$$n \text{ (number of unpaired electrons)} = 1$$

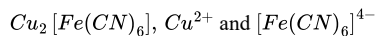
$$\text{So, } \mu = \sqrt{n(n+2)}$$

$$= \sqrt{1 \times 3}$$

$$= \sqrt{3}$$

$$= 1.732 \text{ BM}$$

Colour of the complex is not yellow.



$$x + 6(-1) = -4$$

$$x = -4 + 6$$

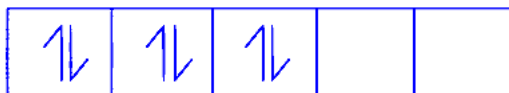
$$= +2$$

Oxidation state of Fe : +2

Configuration:

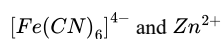
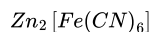


For low-spin d^6 configuration, no unpaired electrons are present



$$n = 0, \mu = \sqrt{n(n+2)} = 0$$

Colour of the complex is not yellow.



$$x + 6(-1) = -4$$

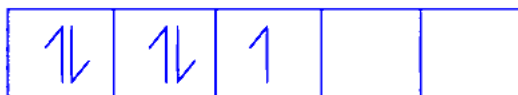
$$x = -4 + 6 = +2$$

Oxidation state of Fe : +2

Configuration:



For low-spin d^6 configuration, no unpaired electrons are present.



$$n = 0, \mu = \sqrt{n(n+2)} = 0$$

Colour of the complex is not yellow

The complexes having yellow colour are $K_3[Co(NO_2)_6]$ and $K_4[Fe(CN)_6]$

The sum of the spin-only magnetic moment values of complexes having yellow colour is 0 (zero).

Question41

A transition metal (M) among Mn, Cr, Co and Fe has the highest standard electrode potential (M^{3+}/M^{2+}). It forms a metal complex of the type $[M(CN)_6]^{4-}$. The number of electrons present in the e_g orbital of the complex is _____.

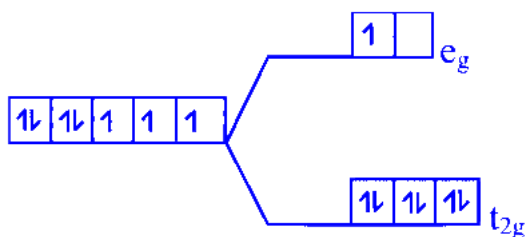
JEE Main 2025 (Online) 2nd April Morning Shift

Answer: 1

Solution:

Co has highest standard electrode potential (M^{3+}/M^{2+}) among Mn, Cr, Co, Fe

$\therefore$ Complex is $[Co(CN)_6]^{4-}$ and its splitting is as follows.



$\therefore$ electron in e_g orbital is one.

Question42

The spin-only magnetic moment value of M^{n+} ion formed among Ni, Zn, Mn and Cu that has the least enthalpy of atomisation is _____. (in nearest integer) Here n is equal to the number of diamagnetic complexes among $K_2[NiCl_4]$, $[Zn(H_2O)_6]Cl_2$, $K_3[Mn(CN)_6]$ and $[Cu(PPh_3)_3]I$

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Answer: 0

Solution:

$K_2[NiCl_4] \Rightarrow sp^3$, Paramagnetic

$[Zn(H_2O)_6]Cl_2 \Rightarrow sp^3 d^2$, Diamagnetic

$K_3[Mn(CN)_6] \Rightarrow d^2 sp^3$, Paramagnetic

$[\text{Cu}(\text{PPh}_3)_3\text{I}] \Rightarrow \text{sp}^3$, Diamagnetic

Hence the value of n is 2

Least value of enthalpy of atomisation among Ni,

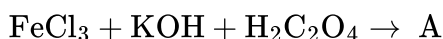
Zn, Mn and Cu is of Zn

$\text{Zn}^{+2} : -[\text{Ar}]3\text{d}^{10}$

$\mu = 0$

Question43

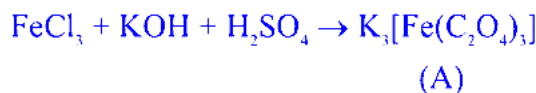
The number of optical isomers exhibited by the iron complex (A) obtained from the following reaction is _____.



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Answer: 2

Solution:



$\Rightarrow [\text{Fe}(\text{C}_2\text{O}_4)_3]^{3-}$ is $[\text{M}(\text{AA})_3]$ type complex.

So total optical isomers = 2

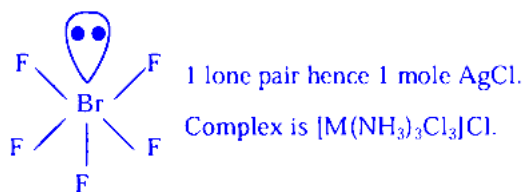
Question44

A metal complex with a formula $\text{MCl}_4 \cdot 3\text{NH}_3$ is involved in $\text{sp}^3 \text{d}^2$ hybridisation. It upon reaction with excess of AgNO_3 solution gives ' x ' moles of AgCl . Consider ' x ' is equal to the number of lone pairs of electron present in central atom of BrF_5 . Then the number of geometrical isomers exhibited by the complex is _____.

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Answer: 2

Solution:



It shows 2 geometrical isomers (Ma_3b_3 type)

facial (fac) & meridional (Mer)

Question45

The number of paramagnetic complexes among $[\text{FeF}_6]^{3-}$, $[\text{Fe}(\text{CN})_6]^{3-}$, $[\text{Mn}(\text{CN})_6]^{3-}$, $[\text{Co}(\text{C}_2\text{O}_4)_3]^{3-}$, $[\text{MnCl}_6]^{3-}$, and $[\text{CoF}_6]^{3-}$, which involved d^2sp^3 hybridization is _____.

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Answer: 2

Solution:

$[\text{FeF}_6]^{3-}$	Paramagnetic	$sp^3 d^2$
$[\text{Fe}(\text{CN})_6]^{3-}$	Paramagnetic	d^2sp^3
$[\text{Mn}(\text{CN})_6]^{3-}$	Paramagnetic	d^2sp^3
$[\text{Co}(\text{C}_2\text{O}_4)_3]^{3-}$	Diamagnetic	d^2sp^3
$[\text{MnCl}_6]^{3-}$	Paramagnetic	$sp^3 d^2$
$[\text{CoF}_6]^{3-}$	Paramagnetic	$sp^3 d^2$

Only $[\text{Fe}(\text{CN})_6]^{3-}$ and $[\text{Mn}(\text{CN})_6]^{3-}$ are paramagnetic and d^2sp^3 hybridisation of metal.

Question46

The number of paramagnetic metal complex species among $[\text{Co}(\text{NH}_3)_6]^{3+}$, $[\text{Co}(\text{C}_2\text{O}_4)_3]^{3-}$, $[\text{MnCl}_6]^{3-}$, $[\text{Mn}(\text{CN})_6]^{3-}$, $[\text{CoF}_6]^{3-}$, $[\text{Fe}(\text{CN})_6]^{3-}$ and $[\text{FeF}_6]^{3-}$ with same number of unpaired electrons is _____.

JEE Main 2025 (Online) 7th April Evening Shift

Answer: 2

Solution:

To determine which metal complexes are paramagnetic and have the same number of unpaired electrons, let's analyze each one:

$[\text{Co}(\text{NH}_3)_6]^{3+}$:

Cobalt ion: Co^{3+}

Electron configuration: $3d^6$

Orbital occupancy: $t_{2g}^{2,2,2} e_g^{0,0}$

Magnetic property: Diamagnetic (unpaired electrons = 0)

$[\text{Co}(\text{C}_2\text{O}_4)_3]^{3-}$:

Cobalt ion: Co^{3+}

Electron configuration: $3d^6$

Orbital occupancy: $t_{2g}^{2,2,2} e_g^{0,0}$

Magnetic property: Diamagnetic (unpaired electrons = 0)

$[\text{MnCl}_6]^{3-}$:

Manganese ion: Mn^{3+}

Electron configuration: $3d^4$

Orbital occupancy: $t_{2g}^{1,1,1} e_g^{1,0}$

Magnetic property: Paramagnetic (unpaired electrons = 4)

$[\text{Mn}(\text{CN})_6]^{3-}$:

Manganese ion: Mn^{3+}

Electron configuration: $3d^4$

Orbital occupancy: $t_{2g}^{2,1,1} e_g^{0,0}$

Magnetic property: Paramagnetic (unpaired electrons = 2)

$[\text{CoF}_6]^{3-}$:

Cobalt ion: Co^{3+}

Electron configuration: $3d^6$

Orbital occupancy: $t_{2g}^{2,1,1} e_g^{1,1}$

Magnetic property: Paramagnetic (unpaired electrons = 4)

$[\text{Fe}(\text{CN})_6]^{3-}$:

Iron ion: Fe^{3+}

Electron configuration: $3d^5$

Orbital occupancy: $t_{2g}^{2,2,1} e_g^{0,0}$

Magnetic property: Paramagnetic (unpaired electrons = 1)

$[\text{FeF}_6]^{3-}$:

Iron ion: Fe^{3+}

Electron configuration: $3d^5$

Orbital occupancy: $t_{2g}^{1,1,1} e_g^{1,1}$

Magnetic property: Paramagnetic (unpaired electrons = 5)

From this analysis, the paramagnetic complexes having the same number of unpaired electrons (4 unpaired electrons) are:

$[\text{MnCl}_6]^{3-}$

$[\text{CoF}_6]^{3-}$

These two complexes show the same paramagnetic property, each with 4 unpaired electrons.

Question 47

In which of the following complexes the CFSE, Δ_o will be equal to zero?

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Options:

A. $\text{K}_3 [\text{Fe}(\text{SCN})_6]$

B. $[\text{Fe}(\text{en})_3]\text{Cl}_3$

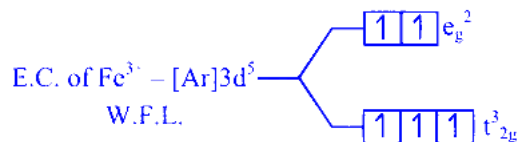
C. $[\text{Fe}(\text{NH}_3)_6]\text{Br}_2$

D. $\text{K}_4 [\text{Fe}(\text{CN})_6]$

Answer: A

Solution:

For complex $\text{K}_3 [\text{Fe}(\text{SCN})_6]$



Calculation of CFSE

$$= (-0.4 \times 3 + 0.6 \times 2)\Delta_0$$

$$= 0\Delta_0$$

Question48

From the magnetic behaviour of $[\text{NiCl}_4]^{2-}$ (paramagnetic) and $[\text{Ni}(\text{CO})_4]$ (diamagnetic), choose the correct geometry and oxidation state.

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Options:

- A. $[\text{NiCl}_4]^{2-}$: Ni(0), tetrahedral $[\text{Ni}(\text{CO})_4]$: Ni(0), square planar
- B. $[\text{NiCl}_4]^{2-}$: Ni^{II} , tetrahedral $[\text{Ni}(\text{CO})_4]$: Ni(0), tetrahedral
- C. $[\text{NiCl}_4]^{2-}$: Ni^{II} , tetrahedral $[\text{Ni}(\text{CO})_4]$: Ni^{II} , square planar
- D. $[\text{NiCl}_4]^{2-}$: Ni^{II} , square planar $[\text{Ni}(\text{CO})_4]$: Ni(0), square planar

Answer: B

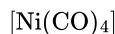
Solution:

The magnetic behavior of complexes can provide insights into their geometry and oxidation states. Given two complexes, $[\text{NiCl}_4]^{2-}$ and $[\text{Ni}(\text{CO})_4]$, we need to analyze the magnetic properties to deduce these characteristics.



Oxidation State: In $[\text{NiCl}_4]^{2-}$, nickel is in the +2 oxidation state (Ni^{2+}). The electron configuration for Ni^{2+} is $[\text{Ar}] 3d^8 4s^0$.

Geometry and Hybridization: The complex shows paramagnetic properties, indicating the presence of unpaired electrons. The hybridization in this case is sp^3 , which corresponds to a tetrahedral geometry. Thus, there are two unpaired electrons leading to its paramagnetism.



Oxidation State: In this complex, nickel is in the zero oxidation state ($\text{Ni}(0)$). The electron configuration is $[\text{Ar}] 3d^{10} 4s^0$ after rearrangement.

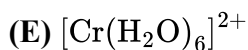
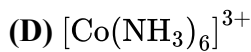
Geometry and Hybridization: The compound is diamagnetic, indicating all electrons are paired. Its hybridization is also sp^3 , meaning the geometry is tetrahedral, resulting in no unpaired electrons and confirming its diamagnetic nature.

In summary, $[\text{NiCl}_4]^{2-}$ is a tetrahedral complex with nickel in a +2 oxidation state and is paramagnetic. In contrast, $[\text{Ni}(\text{CO})_4]$ is a tetrahedral complex with nickel in a 0 oxidation state and is diamagnetic.

Question49

Identify the homoleptic complex(es) that is/are low spin.

- (A) $[\text{Fe}(\text{CN})_5\text{NO}]^{2-}$
- (B) $[\text{CoF}_6]^{3-}$
- (C) $[\text{Fe}(\text{CN})_6]^{4-}$



Choose the correct answer from the options given below :

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Options:

A. (C) and (D) only

B. (C) only

C. (A) and (C) only

D. (B) and (E) only

Answer: A

Solution:

(A) $[\text{Fe}(\text{CN})_5\text{NO}]^{2-}$: This is a heteroleptic complex with iron in the Fe^{2+} oxidation state, resulting in a $3d^6$ electronic configuration. It is a low spin complex with electron configuration $t_{2g}^6 e_g^0$ due to the strong field ligands (SFL).

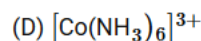
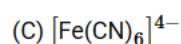
(B) $[\text{CoF}_6]^{3-}$: This homoleptic complex contains cobalt in the Co^{3+} oxidation state, leading to a $3d^6$ electron configuration. It is a high spin complex with an $sp^3 d^2$ hybridization due to weak field ligands (WFL).

(C) $[\text{Fe}(\text{CN})_6]^{4-}$: This homoleptic complex has iron in the Fe^{2+} state, giving a $3d^6$ configuration. It forms a low spin complex with hybridization $d^2 sp^3$ and electron configuration $t_{2g}^6 e_g^0$ due to the strong field nature of the cyanide ligands.

(D) $[\text{Co}(\text{NH}_3)_6]^{3+}$: This homoleptic complex contains cobalt in the Co^{3+} state, resulting in a $3d^6$ electron configuration. It forms a low spin complex with hybridization $d^2 sp^3$ and configuration $t_{2g}^6 e_g^0$ due to the strong field ligands.

(E) $[\text{Cr}(\text{H}_2\text{O})_6]^{2+}$: This homoleptic complex with chromium in the Cr^{2+} state results in a $3d^4$ electronic configuration. It is a high spin complex with hybridization $d^2 sp^3$ and configuration $t_{2g}^3 e_g^1$ due to weak field ligands.

In summary, the homoleptic complexes that are low spin are:

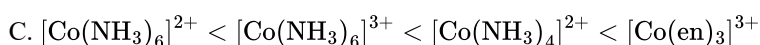
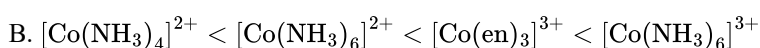
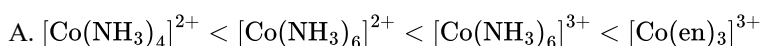


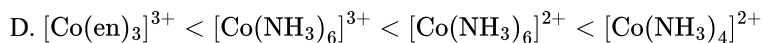
Question 50

The correct order of the following complexes in terms of their crystal field stabilization energies is :

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Options:





Answer: A

Solution:

The crystal field stabilization energy (CFSE) of a complex depends on two main factors:

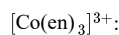
Charge/Oxidation State of the Central Metal Atom: Higher oxidation states generally lead to greater crystal field splitting energy (Δ).

Field Strength of the Ligand: Stronger field ligands also result in greater splitting. Additionally, chelating ligands typically increase the field strength.

For octahedral complexes, the CFSE is calculated as:

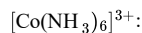
$$\text{CFSE} = [-0.4 t_{2g} + 0.6 e_g] \Delta_o$$

Considering the complexes:



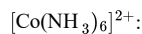
Co exists as Co^{3+} with electronic configuration $t_{2g}^6 e_g^0$.

$$\text{CFSE} = -2.4 (\Delta_o)_1$$



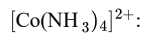
Co also as Co^{3+} with $t_{2g}^6 e_g^0$.

$$\text{CFSE} = -2.4 (\Delta_o)_2$$



Co with Co^{2+} configuration $t_{2g}^5 e_g^2$.

$$\text{CFSE} = -0.8 (\Delta_o)_3$$



Co as Co^{2+} with different configuration $e^4 t_2^3$.

$$\text{CFSE} = -1.2 \Delta_t, \text{ where } \Delta_t = \frac{4}{9} (\Delta_o)_3$$

The order of crystal field splitting energies, based on our criteria, is:

$$\Delta_t < (\Delta_o)_3 < (\Delta_o)_2 < (\Delta_o)_1$$

Therefore, the CFSEs of these complexes compare as follows:

$[\text{Co(en)}_3]^{3+}$ has the highest CFSE due to the Δ_o of the strongest field ligand and oxidation state.

$[\text{Co(NH}_3)_6]^{3+}$ follows with similarly high CFSE due to Co^{3+} .

$[\text{Co(NH}_3)_6]^{2+}$ has a moderate CFSE.

$[\text{Co(NH}_3)_4]^{2+}$ has the lowest CFSE with Δ_t .

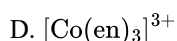
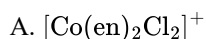
This hierarchy of CFSE is determined by both the oxidation state of the cobalt metal and the field strength of the ligands involved.

Question51

The complex that shows Facial - Meridional isomerism is :

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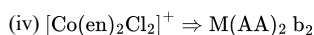
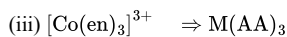
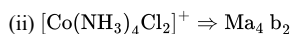
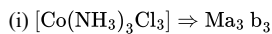
Options:



Answer: B

Solution:

Ma_3b_3 type complexes show Facial - Meridional isomerism



a, b, = NH_3 , Cl^-

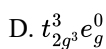
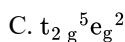
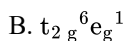
AA = en

Question52

The d-electronic configuration of an octahedral Co (II) complex having magnetic moment of 3.95 BM is:

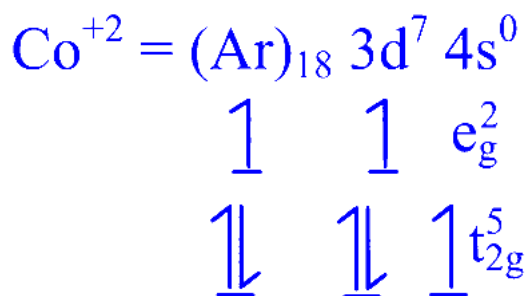
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Options:



Answer: C

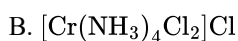
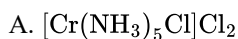
Solution:

**Question53**

$\text{CrCl}_3 \cdot x\text{NH}_3$ can exist as a complex. 0.1 molal aqueous solution of this complex shows a depression in freezing point of 0.558°C . Assuming 100% ionisation of this complex and coordination number of Cr is 6, the complex will be (Given $K_f = 1.86 \text{ K kg mol}^{-1}$)

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Options:





Answer: A

Solution:

Given the problem, we have the following information:

Depression in freezing point, $\Delta T_f = 0.558^\circ\text{C}$

Freezing point depression constant, $k_f = 1.86 \text{ K kg mol}^{-1}$

Molal concentration of the solution, 0.1 mol kg^{-1}

Coordination number of Cr is 6.

Using the formula for freezing point depression:

$$\Delta T_f = i \times k_f \times m$$

where i is the van 't Hoff factor, k_f is the freezing point depression constant, and m is the molality of the solution.

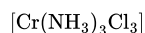
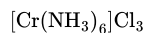
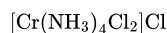
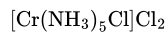
Substituting the known values:

$$0.558 = i \times 1.86 \times 0.1$$

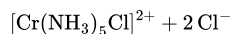
Solving for i :

$$i = \frac{0.558}{1.86 \times 0.1} = 3$$

The van 't Hoff factor $i = 3$ indicates that the complex ionizes into three particles in solution. Given the coordination number of Cr is 6, we analyze the possible complexes:



The complex that gives three ions upon dissociation could be $[\text{Cr}(\text{NH}_3)_5\text{Cl}]\text{Cl}_2$. This is because the complex will dissociate as:

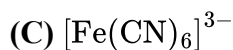
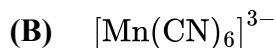


This results in three ions total, matching the van 't Hoff factor $i = 3$.

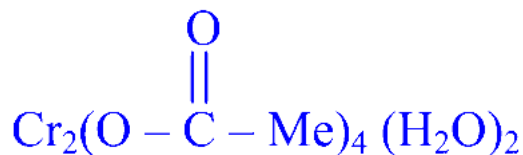
Therefore, the complex is $[\text{Cr}(\text{NH}_3)_5\text{Cl}]\text{Cl}_2$.

Question 54

Identify the coordination complexes in which the central metal ion has d^4 configuration.



(D)



Choose the correct answer from the options given below :

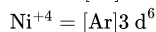
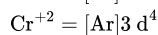
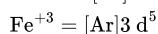
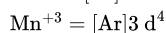
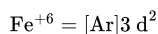
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Options:

- A. (B), (C) and (D) only
- B. (C) and (E) only
- C. (A), (B) and (E) only
- D. (B) and (D) only

Answer: D

Solution:



Question55

One mole of the octahedral complex compound $\text{Co}(\text{NH}_3)_5\text{Cl}_3$ gives 3 moles of ions on dissolution in water. One mole of the same complex reacts with excess of AgNO_3 solution to yield two moles of $\text{AgCl}_{(s)}$. The structure of the complex is:

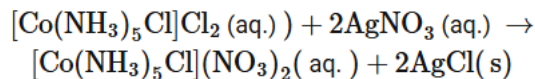
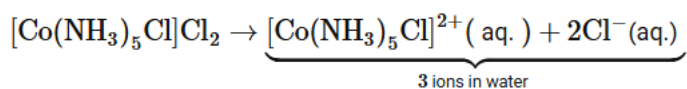
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Options:

- A. $[\text{Co}(\text{NH}_3)_5\text{Cl}]\text{Cl}_2$
- B. $[\text{Co}(\text{NH}_3)_4\text{Cl}] \cdot \text{Cl}_2 \cdot \text{NH}_3$
- C. $[\text{Co}(\text{NH}_3)_3\text{Cl}_3] \cdot 2\text{NH}_3$
- D. $[\text{Co}(\text{NH}_3)_4\text{Cl}_2] \cdot \text{Cl} \cdot \text{NH}_3$

Answer: A

Solution:



Question56

When Ethane-1,2-diamine is added progressively to an aqueous solution of Nickel (II) chloride, the sequence of colour change observed will be:

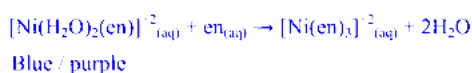
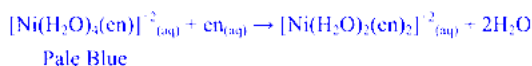
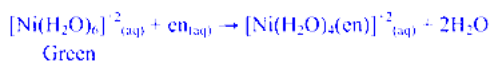
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Options:

- A. Pale Blue → Blue → Green → Violet
B. Pale Blue → Blue → Violet → Green
C. Green → Pale Blue → Blue → Violet
D. Violet → Blue → Pale Blue → Green

Answer: C

Solution:



Question 57

The conditions and consequence that favours the $t_{2g}^3 e_g^1$ configuration in a metal complex are :

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Options:

- A. strong field ligand, low spin complex
B. strong field ligand, high spin complex
C. weak field ligand, high spin complex
D. weak field ligand, low spin complex

Answer: C

Solution:

For $3d^4$

If ligand is SFL : $t_{2g}^4 e_g^0$ (Low spin)

If ligand is WFL : $t_{2g}^3 e_g^1$ (High spin)

Question 58

Match List - I with List - II.

List - I (Complex)	List - II (Hybridisation of central metal ion)
(A) $[\text{CoF}_6]^{3-}$	(I) d^2sp^3
(B) $[\text{NiCl}_4]^{2-}$	(II) sp^3
(C) $[\text{Co}(\text{NH}_3)_6]^{3+}$	(III) sp^3d^2
(D) $[\text{Ni}(\text{CN})_4]^{2-}$	(IV) dsp^2

Choose the correct answer from the options given below:

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Options:

A.

(A)-(I), (B)-(II), (C)-(III), (D)-(IV)

B.

(A)-(III), (B)-(IV), (C)-(I), (D)-(II)

C.

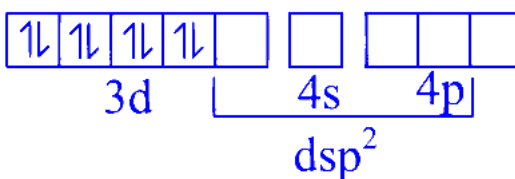
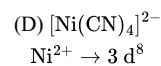
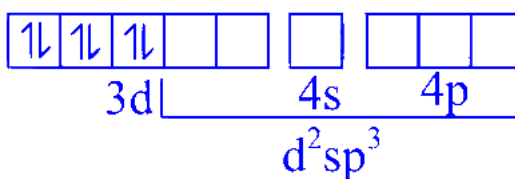
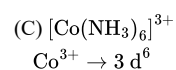
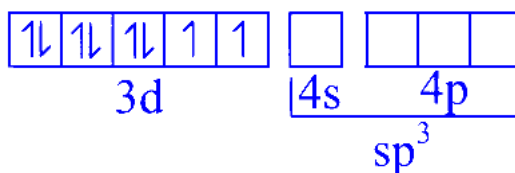
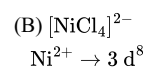
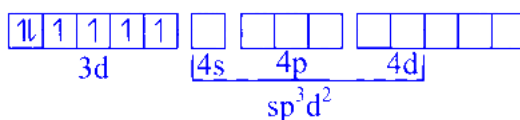
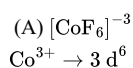
(A)-(I), (B)-(IV), (C)-(III), (D)-(II)

D.

(A)-(III), (B)-(II), (C)-(I), (D)-(IV)

Answer: D

Solution:



Question 59

Match List - I with List - II.

List - I (Complex)	List - II (Hybridisation & Magnetic characters)
(A) $[\text{MnBr}_4]^{2-}$	(I) d^2sp^3 & diamagnetic
(B) $[\text{FeF}_6]^{3-}$	(II) sp^3d^2 & paramagnetic
(C) $[\text{Co}(\text{C}_2\text{O}_4)_3]^{3-}$	(III) sp^3 & diamagnetic
(D) $[\text{Ni}(\text{CO})_4]$	(IV) sp^3 & paramagnetic

Choose the correct answer from the options given below :

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Options:

A.

(A)-(III), (B)-(I), (C)-(I), (D)-(IV)

B.

(A)-(III), (B)-(II), (C)-(III), (D)-(IV)

C.

(A)-(IV), (B)-(II), (C)-(I), (D)-(III)

D.

(A)-(IV), (B)-(I), (C)-(II), (D)-(III)

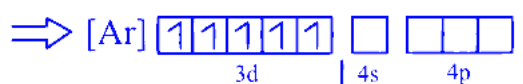
Answer: C

Solution:

(A) $[\text{MnBr}_4]^{2-}$

$\text{Mn}^{+2} \Rightarrow [\text{Ar}]3d^5$

In presence of ligand field

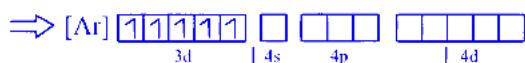


$\Rightarrow sp^3$ hybridization, paramagnetic in nature

(B) $[\text{FeF}_6]^{3-}$

$\text{Fe}^{+3} \Rightarrow [\text{Ar}]3d^5$

In presence of ligand field

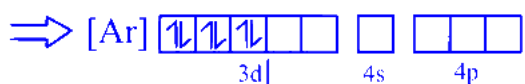


$\Rightarrow sp^3d^2$ hybridization, paramagnetic in nature

(C) $[\text{Co}(\text{C}_2\text{O}_4)_3]^{3-}$

$\text{Co}^{+3} \Rightarrow [\text{Ar}]3d^6$

In presence of ligand field

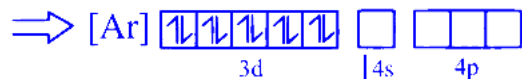


$\Rightarrow d^2sp^3$ hybridization, diamagnetic in nature

(D) $[\text{Ni}(\text{CO})_4]$



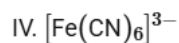
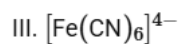
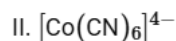
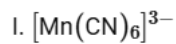
In presence of ligand field



$\Rightarrow sp^3$ hybridization, diamagnetic in nature

Question60

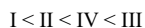
The correct increasing order of stability of the complexes based on Δ_0 value is :



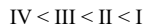
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Options:

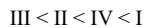
A.



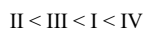
B.



C.



D.



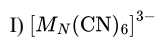
Answer: A

Solution:

Δ_0 value of the octahedral complexes can be calculated using the formula

$\text{CASE} = (-0.4\Delta_0) \times \text{number of electrons in } f_{2g} \text{ orbitals} + (+0.6\Delta_0) \times \text{number of electrons in eg orbitals}.$

The ligand CN^- is a strong field ligand and all the complexes form low-spin complex (pairing occurs).



Oxidation state of $M_n : +3$

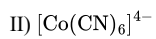
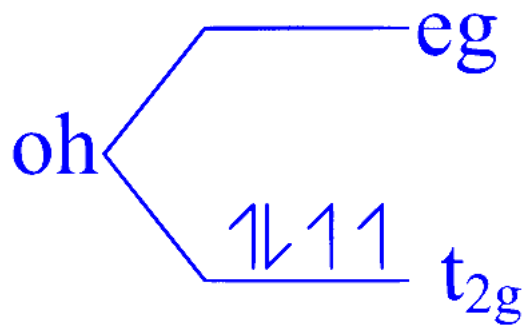
Mn configuration $\rightarrow d^5 s^2$

Mn^{+3} configuration $\rightarrow d^4$

$$x + 6(-1) = -3$$

$$x = -3 + 6 = +3$$

$$\begin{aligned} CFSE &= -0.4\Delta_0 \times 4 + (0.6\Delta_0) \times 0 \\ &= -1.6\Delta_0 \end{aligned}$$



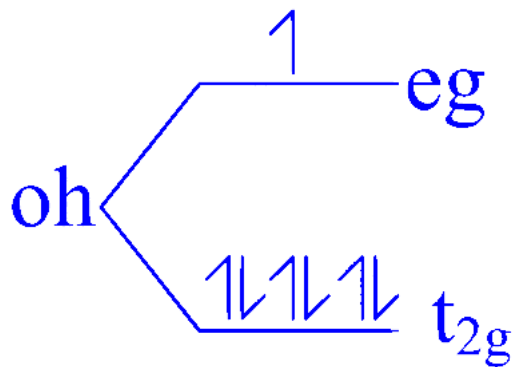
Oxidation state of Co : +2

Co configuration $\rightarrow d^7 s^2$

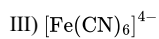
Co^{+2} configuration $\rightarrow d^7$

$$\begin{aligned}\text{CFSE} &= -0.4\Delta_0 \times 6 + (0.6\Delta_0) \times 1 \\ &= -2.4\Delta_0 + 0.6\Delta_0 = -1.8\Delta_0\end{aligned}$$

$$\begin{aligned}x + 6(-1) &= -4 \\ x &= -4 + 6 = +2\end{aligned}$$



$$\begin{aligned}\text{CFSE} &= -0.4\Delta_0 \times 6 + (0.6\Delta_0) \times 1 \\ &= -2.4\Delta_0 + 0.6\Delta_0 = -1.8\Delta_0\end{aligned}$$

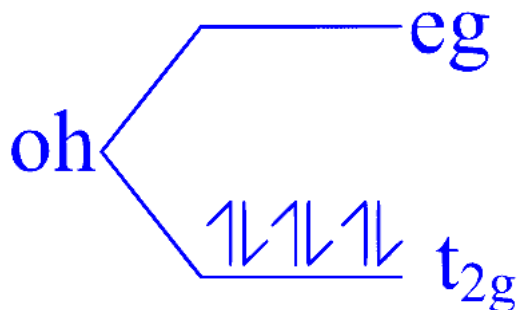


Oxidation state of Fe : +2

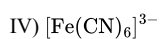
Fe configuration $\rightarrow d^6 s^2$

Fe^{+2} configuration $\rightarrow d^6$

$$\begin{aligned}x + (-1) \times 6 &= -4 \\ x &= -4 + 6 \\ &= +2\end{aligned}$$



$$\begin{aligned}\text{CFSE} &= -0.4\Delta_0 \times 6 + 0.6\Delta_0 \times 0 \\ &= -2.4\Delta_0\end{aligned}$$

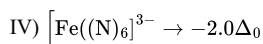
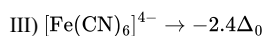
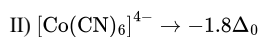
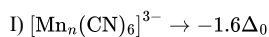
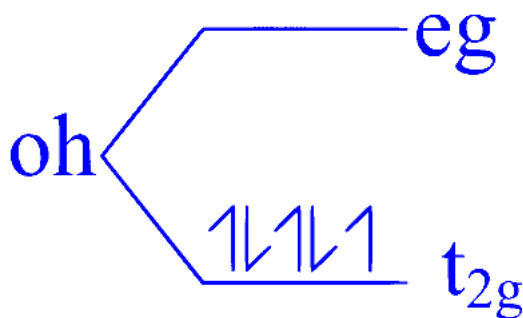


Oxidation state of Fe : +3

Fe configuration $\rightarrow d^6 s^2$

Fe^{+3} configuration $\rightarrow d^5$

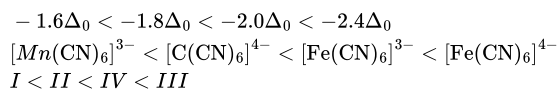
$$\begin{aligned}
 x + 6(-1) &= -3 \\
 x &= -3 + 6 \\
 &= +3
 \end{aligned}$$



Stability of complexes based on Δ_0 value, greater the CFSE value, more stable the complex.

The more negative the CFSE value, the more stable the complex.

So, complex with CFSE -2.4Δ is most stable and complex with CFSE $-1.6\Delta_0$, is least stable. Stability order:



Answer: Option (1)

Question61

The calculated spin-only magnetic moments of $K_3[\text{Fe}(\text{OH})_6]$ and $K_4[\text{Fe}(\text{OH})_6]$ respectively are :

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Options:

A.

4.90 and 4.90 B.M.

B.

4.90 and 5.92 B.M.

C.

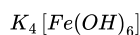
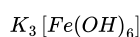
5.92 and 4.90 B.M.

D.

3.87 and 4.90 B.M.

Answer: C

Solution:



Spin-only magnetic moment formula is $\sqrt{n(n+2)}$, where n is the number of unpaired electrons in the central metal $K_3[\text{Fe}(\text{OH})_6]$

Also written as $[\text{Fe}(\text{OH})_6]^{3-}$

Oxidation state of $Fe \rightarrow +3$

Configuration of $Fe \rightarrow d^6 s^2$

Configuration of $Fe^{3+} \rightarrow d^5$

Since OH^- is a weak ligand, no pairing of electrons occurs.



Unpaired electrons, $n = 5$

So, spin-only magnetic moment

$$= \sqrt{n(n+2)}$$

$$= \sqrt{5(5+2)}$$

$$= \sqrt{5 \times 7}$$

$$= \sqrt{35}$$

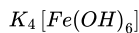
$$= 5.92 \text{ BM}$$

$$3(+1) + x + 6(-1) = 0$$

$$3 + x - 6 = 0$$

$$x - 3 = 0$$

$$x = +3$$



$$4(+1) + x + 6(-1) = 0$$

$$4 + x - 6 = 0$$

$$x = 6 - 4 = +2$$

Also written as $[Fe(OH)_6]^{4-}$

Oxidation state of $Fe \rightarrow +2$

Configuration of $Fe \rightarrow d^6 s^2$

Configuration of $Fe^{+2} \rightarrow d^6$

Since OH^- is a weak field ligand, no pairing of electrons occurs.



Unpaired electrons = 4

Spin-only magnetic moment

$$= \sqrt{n(n+2)}$$

$$= \sqrt{4(4+2)}$$

$$= \sqrt{4 \times 6}$$

$$= \sqrt{24}$$

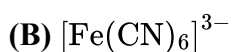
$$= 4.90 \text{ BM}$$

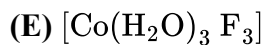
Spin-only magnetic moments of $K_3 [Fe(OH)_6]$ and $K_4 [Fe(OH)_6]$ respectively are 5.92 and 4.90 B.M.

Correct answer is option (3) 5.92 and 4.90 B.M.

Question62

Identify the homoleptic complexes with odd number of d electrons in the central metal :





Choose the correct answer from the options given below :

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Options:

A.

(C) and (E) only

B.

(A), (B) and (D) only

C.

(B) and (D) only

D.

(A), (C) and (E) only

Answer: C

Solution:

Homoleptic complexes are complexes with identical ligands attached to the central metal atom.

(A) $[\text{FeO}_4]^{2-}$ - Homoleptic

Central metal : Fe

IN this complex, Fe is in +6 oxidation state.

$$\{\text{oxidation state} : x + 4(-2) = -2$$

$$x - 8 = -2$$

$$x = -2 + 8 = +6$$

Configuration of $\text{Fe} - d^6s^2$

Configuration of $\text{Fe}(+6) - d^2$

Iron has even number of d electrons. So, it is not correct.

(B) $[\text{Fe}(\text{N})_6]^{3-}$ - Homoleptic

Central metal : Fe

In this complex, the oxidation state of Fe is +3

Oxidation state :

$$x + 6(-1) = -3$$

$$x - 6 = -3$$

$$x = -3 + 6 = +3$$

Configuration of $\text{Fe} - d^6s^2$

Configuration of $\text{Fe}(+3) - d^5$

Iron has odd number of d electrons.

So, it is correct.

(C) $[\text{Fe}(\text{CN})_5\text{NO}]^{2-}$ - Not homoleptic

This complex has different type of ligands.

$[CoCl_4]^{2-}$ - Homoleptic

Central metal - Co

In this complex, the oxidation state of Co is +2

Oxidation state:

$$x + 4(-1) = -2$$

$$x - 4 = -2$$

$$x = -2 + 4 = +2$$

Configuration of Co - $d^7 s^2$

Configuration of Co(+2) - d^7

Cobalt has odd number of d electrons.

So, it is correct.

(E) $[Co(H_2O)_3F_3]$ - Not homoleptic,

It has different type of ligands.

Complexes B and D are homoleptic complexes with odd number of d electrons in the central metal. So, correct answer is

(B) and (D) only.

Question63

Given below are two statements :

Statement (I) : In octahedral complexes, when $\Delta_0 < P$ high spin complexes are formed. When $\Delta_0 > P$ low spin complexes are formed.

Statement (II) : In tetrahedral complexes because of Δ_t

In the light of the above statements, choose the most appropriate answer from the options given below :

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Options:

- A. Both Statement I and Statement II are incorrect
- B. Statement I is correct but Statement II is incorrect
- C. Statement I is incorrect but Statement II is correct
- D. Both Statement I and Statement II are correct

Answer: D

Solution:

In octahedral complexes, which have a coordination number (CN) of 6, the formation of high spin or low spin complexes depends on the relative magnitude of the crystal field splitting energy, denoted as Δ_o , and the pairing energy (P.E.):

If Δ_o is less than the pairing energy ($\Delta_o < P.E.$), high spin complexes are formed.

If Δ_o is greater than the pairing energy ($\Delta_o > P.E.$), low spin complexes are formed.

In tetrahedral complexes, where the coordination number is 4:

The crystal field splitting energy (Δ_t) is typically less than the pairing energy ($\Delta_t < P.E.$), resulting in the formation of mainly high spin complexes. Low spin complexes are rarely observed in tetrahedral coordination due to this energy consideration.

Question64

The d-orbital electronic configuration of the complex among $[\text{Co}(\text{en})_3]^{3+}$, $[\text{CoF}_6]^{3-}$, $[\text{Mn}(\text{H}_2\text{O})_6]^{2+}$ and $[\text{Zn}(\text{H}_2\text{O})_6]^{2+}$ that has the highest CFSE is :

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Options:

A. $t_{2g}^6 e_g^0$

B. $t_{2g}^3 e_g^2$

C. $t_{2g}^4 e_g^2$

D. $t_{2g}^6 e_g^4$

Answer: A

Solution:

Identify the oxidation state and d-electron count for each complex:

For $[\text{Co}(\text{en})_3]^{3+}$:

Ethylenediamine (en) is neutral.

Oxidation state of cobalt is +3.

Cobalt (atomic number 27) in the +3 state gives a d^6 configuration.

With strong/moderate field ligands like en, the complex is low spin, leading to a configuration of $t_{2g}^6 e_g^0$.

For $[\text{CoF}_6]^{3-}$:

Each fluoride ion is F^- so, with six of them, Co must be in the +3 state again.

However, since fluoride is a weak field ligand, this complex adopts a high spin configuration for d^6 , resulting in $t_{2g}^4 e_g^2$.

For $[\text{Mn}(\text{H}_2\text{O})_6]^{2+}$:

Water is neutral.

Manganese in the +2 state gives a d^5 configuration.

With weak field water, the d^5 configuration is high spin: $t_{2g}^3 e_g^2$.

For $[\text{Zn}(\text{H}_2\text{O})_6]^{2+}$:

Zinc in the +2 state has a d^{10} configuration.

In an octahedral field, all 10 electrons fill the orbitals as $t_{2g}^6 e_g^4$.

Calculate the crystal field stabilization energy (CFSE) for each case:

The CFSE in an octahedral field can be estimated as:

$$\text{CFSE} = (n_{t_{2g}} \times -0.4\Delta_o) + (n_{e_g} \times 0.6\Delta_o)$$

For $t_{2g}^6 e_g^0$ (low-spin d^6 in $[\text{Co}(\text{en})_3]^{3+}$):

$$\text{CFSE} = 6(-0.4\Delta_o) + 0(0.6\Delta_o) = -2.4\Delta_o$$

For $t_{2g}^4 e_g^2$ (high-spin d^6 in $[\text{CoF}_6]^{3-}$):

$$\text{CFSE} = 4(-0.4\Delta_o) + 2(0.6\Delta_o) = -1.6\Delta_o + 1.2\Delta_o = -0.4\Delta_o$$

For $t_{2g}^3 e_g^2$ (high-spin d^5 in $[\text{Mn}(\text{H}_2\text{O})_6]^{2+}$):

$$\text{CFSE} = 3(-0.4\Delta_o) + 2(0.6\Delta_o) = -1.2\Delta_o + 1.2\Delta_o = 0$$

For $t_{2g}^6 e_g^4$ (for d^{10} in $[\text{Zn}(\text{H}_2\text{O})_6]^{2+}$):

$$\text{CFSE} = 6(-0.4\Delta_o) + 4(0.6\Delta_o) = -2.4\Delta_o + 2.4\Delta_o = 0$$

Compare the CFSE values:

$t_{2g}^6 e_g^0$ gives a CFSE of $-2.4\Delta_o$ (most negative, hence highest stabilization).

The others result in less stabilization (or net zero).

Conclusion:

The complex with the highest CFSE is the one with the configuration $t_{2g}^6 e_g^0$, which corresponds to $[\text{Co}(\text{en})_3]^{3+}$.

Thus, the answer is:

Option A: $t_{2g}^6 e_g^0$.

Question65

The type of hybridization and the magnetic property of $[\text{MnCl}_6]^{3-}$ are,

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Options:

A. sp^3d^2 , paramagnetic with four unpaired electrons.

B. d^2sp^3 , paramagnetic with four unpaired electrons.

C. sp^3d^2 , paramagnetic with two unpaired electrons.

D. d^2sp^3 , paramagnetic with two unpaired electrons.

Answer: A

Solution:

Determine the oxidation state of manganese in $[\text{MnCl}_6]^{3-}$:

Chloride (Cl^-) has a charge of -1.

With six chloride ions, the total charge contributed by the ligands is $6(-1) = -6$.

Let the oxidation state of Mn be x . Then:

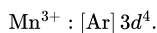
$$x - 6 = -3 \implies x = +3.$$

So, manganese is in the +3 oxidation state.

Find the d-electron count for Mn(III):

The neutral manganese atom has the electronic configuration $[\text{Ar}] 3d^5 4s^2$.

Removing three electrons (first from the 4s, then from the 3d) gives:



Therefore, Mn(III) is a d^4 system.

Predict the spin state in an octahedral complex:

Chloride (Cl^-) is a weak-field ligand, meaning it produces a relatively small crystal field splitting (Δ_o).

When Δ_o is small, the pairing energy is higher than Δ_o , so electrons prefer to remain unpaired.

For a d^4 configuration in a high-spin octahedral complex, the electrons will be distributed as:

Three electrons in the three t_{2g} orbitals (one in each)

One electron in one of the e_g orbitals

This results in a total of 4 unpaired electrons, making the complex paramagnetic.

Determine the type of hybridization:

In octahedral complexes, the two common hybridizations are:

d^2sp^3 (inner orbital complex), typically seen in low-spin complexes where inner 3d orbitals are available because the electrons are paired.

sp^3d^2 (outer orbital complex), seen in high-spin complexes where the inner 3d orbitals are occupied by unpaired electrons.

Since $[\text{MnCl}_6]^{3-}$ is high-spin (because of Cl^- being a weak-field ligand), the inner 3d orbitals are not available for hybridization.

Therefore, the complex uses the outer orbitals (the 4s, 4p, and 4d orbitals), leading to an sp^3d^2 hybridization.

Conclusion:

The hybridization of $[\text{MnCl}_6]^{3-}$ is sp^3d^2 .

It is paramagnetic with four unpaired electrons.

Thus, the correct answer is:

Option A: sp^3d^2 , paramagnetic with four unpaired electrons.

Question66

Match the LIST-I with LIST-II

LIST-I (Molecules/ion)		LIST-II (Hybridisation of central atom)	
A.	PF_5	I	dsp^2
B	SF_6	II	$sp^3 d$
C	$Ni(CO)_4$	III	$sp^3 d^2$
D	$[PtCl_4]^{2-}$	IV	sp^3

Choose the correct answer from the options given below:

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Options:

A. A-IV, B-I, C-II, D-III

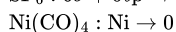
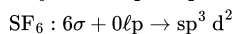
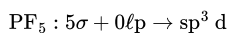
B. A-III, B-I, C-IV, D-II

C. A-II, B-III, C-IV, D-I

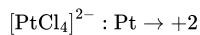
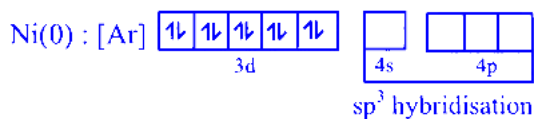
D. A-I, B-II, C-III, D-IV

Answer: C

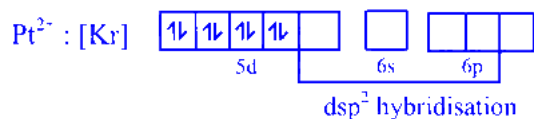
Solution:



In presence of ligand field :-



In presence of ligand field :-



Question67

The correct order of the complexes $[Co(NH_3)_5(H_2O)]^{3+}$ (A), $[Co(NH_3)_6]^{3+}$ (B), $[Co(CN)_6]^{3-}$ (C) and $[CoCl(NH_3)_5]^{2+}$ (D) in terms of wavelength of light absorbed is

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Options:

A. $D > C > B > A$

B. $C > B > A > D$

C. $D > A > B > C$

D. $C > B > D > A$

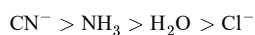
Answer: C

Solution:

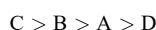
The energy of absorbed light by a complex can be expressed as $E = hv = \frac{hc}{\lambda}$, where E is the energy, h is Planck's constant, v is the frequency, C is the speed of light, and λ is the wavelength of the absorbed light. This formula demonstrates that energy (E) is inversely proportional to the wavelength (λ).

In this context, all cobalt complexes are in the +3 oxidation state. The Crystal Field Stabilization Energy (CFSE) is influenced by the strength of the ligands surrounding the metal ion. As the ligand field strength increases, the CFSE also increases.

The order of field strength for the ligands in this case is:



This implies the following order for CFSE in the given complexes:

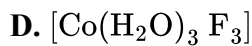
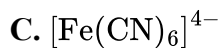
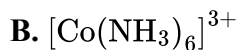
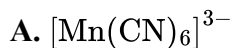


Because energy is inversely proportional to wavelength, complexes absorbing at higher energies will absorb at shorter wavelengths. Therefore, the order of the complexes in terms of the wavelength of light absorbed, from longest to shortest wavelength, is:



Question68

Identify the diamagnetic octahedral complex ions from below ;



Choose the correct answer from the options given below:

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Options:

A. B and C Only

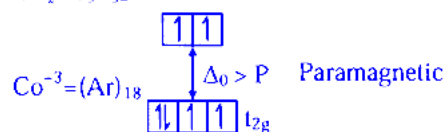
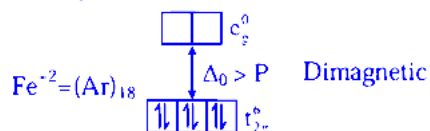
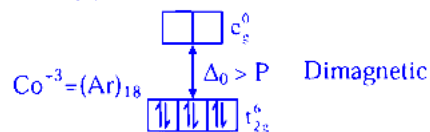
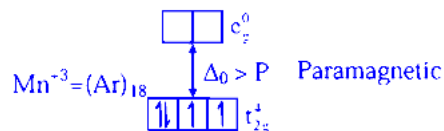
B. A and C Only

C. B and D Only

D. A and D Only

Answer: A

Solution:



Question69

Number of stereoisomers possible for the complexes, $[\text{CrCl}_3(\text{py})_3]$ and $[\text{CrCl}_2(\text{ox})_2]^{3-}$ are respectively (py = pyridine, ox = oxalate)

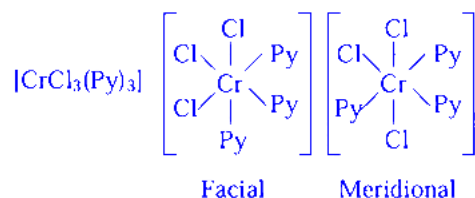
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Options:

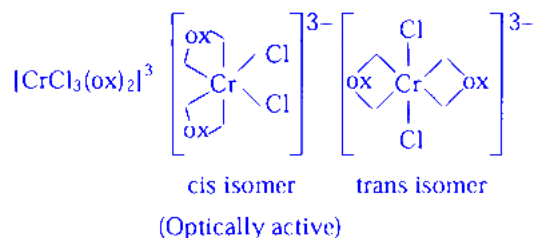
- A. 3 & 3
- B. 2 & 3
- C. 1 & 2
- D. 2 & 2

Answer: B

Solution:



So total stereo isomer = 2



Geometrical isomer = 2 (1 cis + 1 trans)

Optical isomer = 3 (2 optically active + 1 optically inactive)

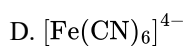
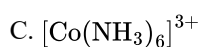
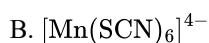
Stereoisomer = 3

Question 70

Which one of the following complexes will have $\Delta_o = 0$ and $\mu = 5.96$ B.M.?

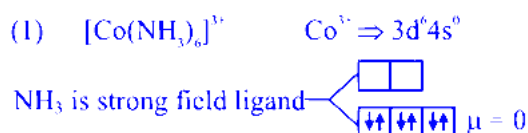
JEE Main 2025 (Online) 4th April Morning Shift

Options:

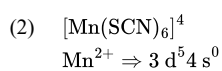


Answer: B

Solution:



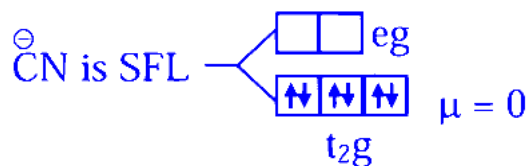
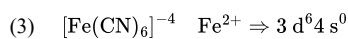
$$= [-0.4 \times 6 + 0.6 \times (0)]\Delta_o = -2.4\Delta_o$$



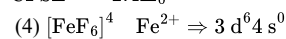
$$\mu = \sqrt{35} \text{ B.M.} = 5.92 \text{ B.M.}$$

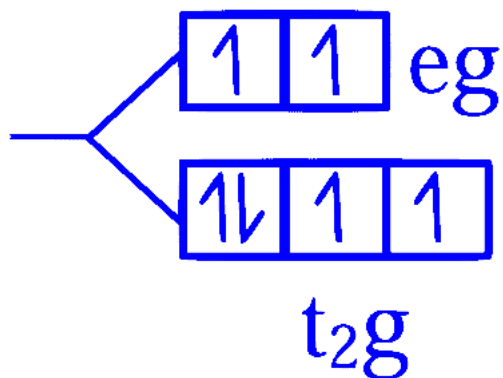
$$\text{CFSE} = (-0.4 \times 3 + 0.6 \times 2)\Delta_o$$

$$\text{So } \Delta_o = 0$$



$$\text{CFSE} = -2.4\Delta_o$$





$$\mu = \sqrt{24} \text{ B.M.} = 4.89 \text{ B.M.}$$

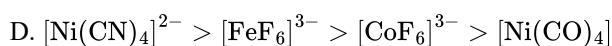
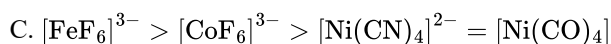
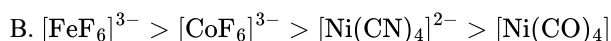
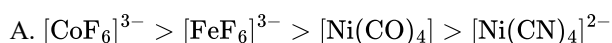
$$\text{CFSE} = (-0.4 \times 4 + 0.6 \times 2)\Delta_0 = -1.2\Delta_0$$

Question71

The correct order of $[\text{FeF}_6]^{3-}$, $[\text{CoF}_6]^{3-}$, $[\text{Ni}(\text{CO})_4]$ and $[\text{Ni}(\text{CN})_4]^{2-}$ complex species based on the number of unpaired electrons present is:

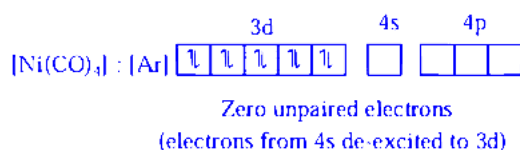
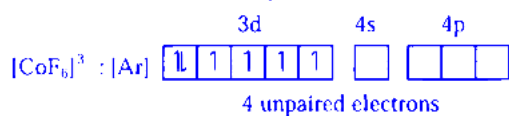
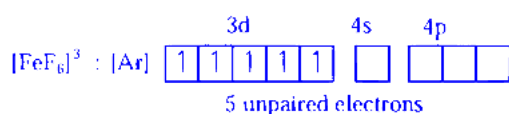
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Options:



Answer: C

Solution:



Question72

' X ' is the number of electrons in t_{2g} orbitals of the most stable complex ion among $[\text{Fe}(\text{NH}_3)_6]^{3+}$, $[\text{FeCl}_6]^{3-}$, $[\text{Fe}(\text{C}_2\text{O}_4)_3]^{3-}$ and $[\text{Fe}(\text{H}_2\text{O})_6]^{3+}$. The nature of oxide of vanadium of the type V_2O_x is :

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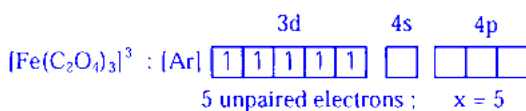
Options:

- A. Amphoteric
- B. Acidic
- C. Basic
- D. Neutral

Answer: A

Solution:

Most stable is $[\text{Fe}(\text{C}_2\text{O}_4)_3]^{3-}$ due to Chelation effect.



V_2O_5 is amphoteric.

Question73

An octahedral complex having molecular composition $\text{Co} \cdot 5\text{NH}_3 \cdot \text{Cl}^2 \cdot \text{SO}_4$ has two isomers A and B. The solution of A gives a white precipitate with AgNO_3 solution and the solution of B gives white precipitate with BaCl_2 solution. The type of isomerism exhibited by the complex is,

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Options:

- A. Co-ordinate isomerism
- B. Ionisation isomerism
- C. Geometrical isomerism
- D. Linkage isomerism

Answer: B

Solution:

In this octahedral complex with the molecular composition $\text{Co} \cdot 5\text{NH}_3 \cdot \text{Cl}^2 \cdot \text{SO}_4$, two isomers, A and B, are present.

Isomer A is represented as $[\text{Co}(\text{NH}_3)_5(\text{SO}_4)]\text{Cl}$. When this isomer is dissolved in solution, it reacts with AgNO_3 , forming a white precipitate of silver chloride (AgCl).

Isomer B is written as $[\text{Co}(\text{NH}_3)_5\text{Cl}]\text{SO}_4$. In solution, it reacts with BaCl_2 , forming a white precipitate of barium sulfate (BaSO_4).

These two isomers demonstrate ionisation isomerism. This type of isomerism occurs when the exchange of ions inside and outside the coordination sphere of a complex leads to the formation of isomers with different chemical behaviors in solution.

Question74

'X' is the number of acidic oxides among VO_2 , V_2O_3 , CrO_3 , V_2O_5 and Mn_2O_7 . The primary valency of cobalt in $[\text{Co}(\text{H}_2\text{NCH}_2\text{CH}_2\text{NH}_2)_3]_2(\text{SO}_4)_3$ is Y. The value of $X + Y$ is _____.

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Options:

A.

3

B.

4

C.

2

D.

5

Answer: D

Solution:

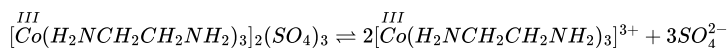
In the given compounds, we evaluate which are acidic oxides:

CrO_3 is acidic.

Mn_2O_7 is acidic.

Therefore, the number of acidic oxides, X , is 2.

Next, for the complex $[\text{Co}(\text{H}_2\text{NCH}_2\text{CH}_2\text{NH}_2)_3]_2(\text{SO}_4)_3$, we consider its dissociation:



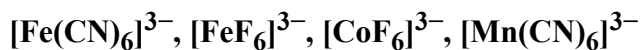
The primary valency of cobalt (Y) in this context is 3.

Thus, the value of $X + Y$ is:

$$X + Y = 2 + 3 = 5$$

Question 75

The number of unpaired electrons responsible for the paramagnetic nature of the following complex species are respectively :



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Options:

A.

1, 5, 4, 2

B.

1, 4, 4, 2

C.

1, 5, 5, 2

D.

1, 1, 4, 2

Answer: A

Solution:

The number of unpaired electrons responsible for the paramagnetic nature of each complex species is calculated as follows:

Complex: $[Fe(CN)_6]^{3-}$

Ion: Fe^{3+}

Electronic Configuration: $3d^5$

Orbital Population:

$$t_{2g}^{2,2,1}$$

$$e_g^{0,0}$$

Unpaired Electrons: 1

Complex: $[FeF_6]^{3-}$

Ion: Fe^{3+}

Electronic Configuration: $3d^5$

Orbital Population:

$$t_{2g}^{1,1,1}$$

$$e_g^{1,1}$$

Unpaired Electrons: 5

Complex: $[CoF_6]^{3-}$

Ion: Co^{3+}

Electronic Configuration: $3d^6$

Orbital Population:

$$t_{2g}^{2,1,1}$$

$$e_g^{1,0}$$

Unpaired Electrons: 4

Complex: $[Mn(CN)_6]^{3-}$

Ion: Mn^{3+}

Electronic Configuration: $3d^4$

Orbital Population:

$$t_{2g}^{2,1,1}$$

$$e_g^{0,0}$$

Unpaired Electrons: 2

Each complex's paramagnetic nature is due to the specific number of unpaired electrons as indicated above.

Question76

Match List - I with List - II.

List - I (Complex)	List - II (Primary valency and Secondary valency)
(A) $[Co(en)_2Cl_2]Cl$	(I) 3, 6

(B) $[\text{Pt}(\text{NH}_3)_2\text{Cl}(\text{NO}_2)]$	(II) 3, 4
(C) $\text{Hg} [\text{Co}(\text{SCN})_4]$	(III) 2, 6
(D) $[\text{Mg}(\text{EDTA})]^{2-}$	(IV) 2, 4

Choose the correct answer from the options given below :

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Options:

A.

(A)-(I), (B)-(III), (C)-(II), (D)-(IV)

B.

(A)-(II), (B)-(III), (C)-(IV), (D)-(I)

C.

(A)-(I), (B)-(IV), (C)-(II), (D)-(III)

D.

(A)-(III), (B)-(I), (C)-(II), (D)-(IV)

Answer: C

Solution:

Primary valency = Oxidation state

Secondary valency = Co-ordination number

	Complex	Primary valency	Secondary
(A)	$[\text{Co}(\text{en})_2\text{Cl}_2]\text{Cl}$	3	6
(B)	$[\text{Pt}(\text{NH}_3)_2\text{Cl}(\text{NO}_2)]$	2	4
(C)	$\text{Hg} [\text{Co}(\text{SCN})_4]$	3	4
(D)	$[\text{Mg}(\text{EDTA})]^{2-}$	2	6

Question 77

Given below are two statements:

Statement I: A homoleptic octahedral complex, formed using monodentate ligands, will not show stereoisomerism.

Statement II: cis- and trans- platin are heteroleptic complexes of Pd.

In the light of the above statements, choose the correct answer from the options given below:

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Options:

A.

Statement I is true but Statement II is false

B.

Both Statement I and Statement II are true

C.

Both Statement I and Statement II are false

D.

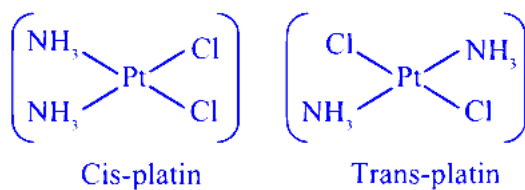
Statement I is false but Statement II is true

Answer: A

Solution:

Homoleptic complex of type $[Ma_6]$ (Where $a \Rightarrow$ monodentate ligand) cannot show geometrical as well as optical isomerism.

Cis-platin and trans-platin has formula $[Pt(NH_3)_2Cl_2]$ which is a heteroleptic complex of platinum.



Question78

The number of species from the following that are involved in sp^3d^2 hybridization is :

$[Co(NH_3)_6]^{3+}$, SF_6 , $[CrF_6]^{3-}$, $[CoF_6]^{3-}$, $[Mn(CN)_6]^{3-}$, and $[MnCl_6]^{3-}$

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Options:

A.

4

B.

3

C.

5

D.

6

Answer: B

Solution:

In $[Co(NH_3)_6]^{3+}$, $Co^{+3} : [Ar]3d^6$, NH_3 is S.F.L

Hybridisation state of Co^{3+} is d^2sp^3

In SF_6 , Hybridisation state of sulphur is sp^3d^2

In $[CrF_6]^{3-}$, $Cr^{+3} : [Ar]3d^3$

Hybridisation state of Cr^{3+} is d^2sp^3

$[\text{CoF}_6]^{3-}$, Co^{+3} : $[\text{Ar}]3d^6$ F^- is W.F.L

Hybridisation state of Co^{3+} is $sp^3 d^2$

$[\text{Mn}(\text{CN})_6]^{3-}$, Mn^{+3} : $[\text{Ar}]3d^4$ CN^- is S.F.L

Hybridisation state of Mn^{3+} is $d^2 sp^3$

$[\text{MnCl}_6]^{3-}$, Mn^{+3} : $[\text{Ar}]3d^4$ Cl^- is W.F.L

Hybridisation state of Cl^- is $sp^3 d^2$

Total number of $sp^3 d^2$ hybridized molecules is 3

Question79

Match the LIST-I with LIST-II

LIST-I (Complex/ Species)	LIST-II (Shape & magnetic moment)
A. $[\text{Ni}(\text{CO})_4]$	I. Tetrahedral, 2.8 BM
B. $[\text{Ni}(\text{CN})_4]^{2-}$	II. Square planar, 0 BM
C. $[\text{NiCl}_4]^{2-}$	III. Tetrahedral, 0 BM
D. $[\text{MnBr}_4]^{2-}$	IV. Tetrahedral, 5.9 BM

Choose the correct answer from the options given below:

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Options:

A.

A-I, B-II, C-III, D-IV

B.

A-IV, B-I, C-III, D-II

C.

A-III, B-II, C-I, D-IV

D.

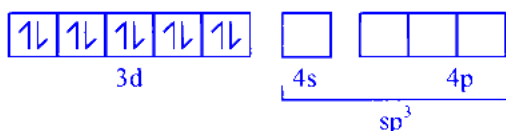
A-III, B-IV, C-II, D-I

Answer: C

Solution:

(A) $[\text{Ni}(\text{CO})_4]$, Ni^0 : $[\text{Ar}]3d^8 4s^2$

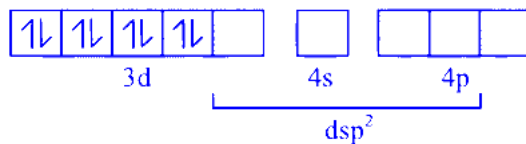
Valence orbitals of Ni^0 in pre-hybridisation state :



Tetrahedral, Diamagnetic, $\mu = 0$ B.M.

(B) $[\text{Ni}(\text{CN})_4]^{2-}$, Ni^{+2} : $[\text{Ar}]3d^8 4s^0$

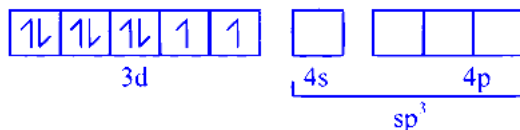
Valence orbitals of Ni^{+2} in pre-hybridisation state :



Square planar, Diamagnetic, $\mu = 0$ B.M.

(C) $[\text{NiCl}_4]^{2-}$, Ni^{+2} : $[\text{Ar}]3d^8 4s^0$

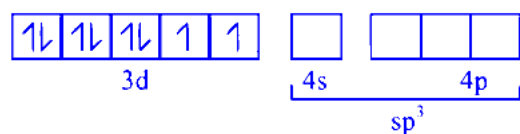
Valence orbitals of Ni^{+2} in ground state :



Tetrahedral, paramagnetic, $\mu = \sqrt{8} = 2.8$ B.M.

(D) $[\text{MnBr}_4]^{2-}$, Mn^{+2} : $[\text{Ar}]3d^5$

Valence orbitals of Mn^{+2} in ground state :



Tetrahedral, paramagnetic, $\mu = \sqrt{35} = 5.9$ B.M.

Question80

Yellow compound of lead chromate gets dissolved on treatment with hot NaOH solution. The product of lead formed is a :

[27-Jan-2024 Shift 1]

Options:

- A. Tetraanionic complex with coordination number six
- B. Neutral complex with coordination number four
- C. Dianionic complex with coordination number six
- D. Dianionic complex with coordination number four

Answer: D

Question81

Identity the incorrect pair from the following:

[27-Jan-2024 Shift 2]

Options:

- A. Photography - AgBr
- B. Polythene preparation - TiCl_4 , $\text{Al}(\text{CH}_3)_3$
- C. Haber process - Iron

D. Wacker process –Pt Cl₂

Answer: D

Question82

Identify from the following species in which d² sp³ hybridization is shown by central atom:
[27-Jan-2024 Shift 2]

Options:



Answer: A

Question83

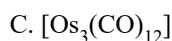
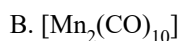
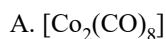
The Spin only magnetic moment value of square planar complex [Pt(NH₃)₂ Cl(NH₂CH₃)] Cl is _____ B.M. (Nearest integer)
(Given atomic number for Pt = 78)
[27-Jan-2024 Shift 2]

Answer: 0

Question84

In which one of the following metal carbonyls, CO forms a bridge between metal atoms?
[29-Jan-2024 Shift 1]

Options:



D. $[\text{Ru}_3(\text{CO})_{12}]$

Answer: A

Question85

The correct IUPAC name of K_2MnO_4 is

[29-Jan-2024 Shift 2]

Options:

- A. Potassium tetraoxopermanganate (VI)
- B. Potassium tetraoxidomanganate (VI)
- C. Dipotassium tetraoxidomanganate (VII)
- D. Potassium tetraoxidomanganese (VI)

Answer: B

Question86

Aluminium chloride in acidified aqueous solution forms an ion having geometry

[30-Jan-2024 Shift 1]

Options:

- A. Octahedral
- B. Square Planar
- C. Tetrahedral
- D. Trigonal bipyramidal

Answer: A

Question87

Choose the correct Statements from the following:

- (A) Ethane-1,2-diamine is a chelating ligand.
- (B) Metallic aluminium is produced by electrolysis of aluminium oxide in presence of cryolite.
- (C) Cyanide ion is used as ligand for leaching of silver.
- (D) Phosphine act as a ligand in Wilkinson catalyst.
- (E) The stability constants of Ca^{2+} and Mg^{2+} are similar with EDTA complexes.

Choose the correct answer from the options given below:

[30-Jan-2024 Shift 1]

Options:

- A. (B), (C), (E) only
- B. (C), (D), (E) only
- C. (A), (B), (C) only
- D. (A), (D), (E) only

Answer: C

Question88

**The molecule/ion with square pyramidal shape is:
[30-Jan-2024 Shift 2]**

Options:

- A. $[\text{Ni}(\text{CN})_4]^{2-}$
- B. PCl_5
- C. BrF_5
- D. PF_5

Answer: C

Question89

**The coordination geometry around the manganese in decacarbonyldimanganese(0)
[30-Jan-2024 Shift 2]**

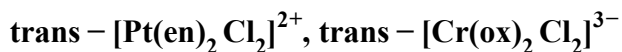
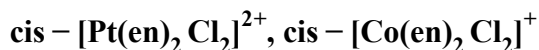
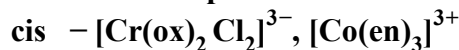
Options:

- A. Octahedral
- B. Trigonal bipyramidal
- C. Square pyramidal
- D. Square planar

Answer: A

Question90

Number of complexes which show optical isomerism among the following is



[30-Jan-2024 Shift 2]

Answer: 4

Question91

The correct statements from following are:

A. The strength of anionic ligands can be explained by crystal field theory.

B. Valence bond theory does not give a quantitative interpretation of kinetic stability of coordination compounds.

C. The hybridization involved in formation of $[\text{Ni}(\text{CN})_4]^{2-}$ complex is dsp^2 .

D. The number of possible isomer(s) of cis- $[\text{PtCl}_2(\text{en})_2]^{2+}$ is one

Choose the correct answer from the options given below:

[31-Jan-2024 Shift 1]

Options:

A. A, D only

B. A, C only

C. B, D only

D. B, C only

Answer: D

Question92

Select the option with correct property -

[31-Jan-2024 Shift 2]

Options:

A. $[\text{Ni}(\text{CO})_4]$ and $[\text{NiCl}_4]^{2-}$ both diamagnetic

B. $[\text{Ni}(\text{CO})_4]$ and $[\text{NiCl}_4]^{2-}$ both paramagnetic

C. $[\text{NiCl}_4]^{2-}$ diamagnetic, $[\text{Ni}(\text{CO})_4]$ paramagnetic

D. $[\text{Ni}(\text{CO})_4]$ diamagnetic, $[\text{NiCl}_4]^{2-}$ paramagnetic

Answer: D

Question93

Choose the correct answer from the options given below :

	LIST - I (Complex ion)		LIST - II Electronic Configuration
A.	$[\text{Cr}(\text{H}_2\text{O})_6]^{3+}$	I.	$t_{2g}^2 e_g^0$
B.	$[\text{Fe}(\text{H}_2\text{O})_6]^{3+}$	II.	$t_{2g}^3 e_g^0$
C.	$[\text{Ni}(\text{H}_2\text{O})_6]^{2+}$	III.	$t_{2g}^3 e_g^2$
D.	$[\text{V}(\text{H}_2\text{O})_6]^{3+}$	IV.	$t_{2g}^6 e_g^2$

[31-Jan-2024 Shift 2]

Options:

A. A-III, B-II, C-IV, D-I

B. A-IV, B-I, C-II, D-III

C. A-IV, B-III, C-I, D-II

D. A-II, B-III, C-IV, D-I

Answer: D

Question94

Which of the following complex is homoleptic?

[1-Feb-2024 Shift 1]

Options:

A. $[\text{Ni}(\text{CN})_4]^{2-}$

B. $[\text{Ni}(\text{NH}_3)_2 \text{Cl}_2]$

C. $[\text{Fe}(\text{NH}_3)_4 \text{Cl}_2]^+$

D. $[\text{Co}(\text{NH}_3)_4 \text{Cl}_2]^+$

Answer: A

Question95

Given below are two statements:

Statement (I): A solution of $[\text{Ni}(\text{H}_2\text{O})_6]^{2+}$ is green in colour.

Statement (II): A solution of $[\text{Ni}(\text{CN})_4]^{2-}$ is colourless.

In the light of the above statements, choose the most appropriate answer from the options given below:
[1-Feb-2024 Shift 1]

Options:

- A. Both Statement I and Statement II are incorrect
- B. Both Statement I and Statement II are correct
- C. Statement I is incorrect but Statement II is correct
- D. Statement I is correct but Statement II is incorrect

Answer: B

Question96

Given below are two statements :

Statement (I) : Dimethyl glyoxime forms a sixmembered covalent chelate when treated with NiCl_2 solution in presence of NH_4OH .

Statement (II) : Prussian blue precipitate contains iron both in (+2) and (+3) oxidation states. In the light of the above statements, choose the most appropriate answer from the options given below:
[1-Feb-2024 Shift 2]

Options:

- A. Statement I is false but Statement II is true
- B. Both Statement I and Statement II are true
- C. Both Statement I and Statement II are false
- D. Statement I is true but Statement II is false

Answer: A

Question97

$[\text{Co}(\text{NH}_3)_6]^{3+}$ and $[\text{CoF}_6]^{3-}$ are respectively known as:

[1-Feb-2024 Shift 2]

Options:

- A. Spin free Complex, Spin paired Complex
- B. Spin paired Complex, Spin free Complex

C. Outer orbital Complex, Inner orbital Complex

D. Inner orbital Complex, Spin paired Complex

Answer: B

Question98

1L, 0.02M solution of $[\text{Co}(\text{NH}_3)_5 \text{SO}_4] \text{Br}$ is mixed with 1L, 0.02M solution of $[\text{Co}(\text{NH}_3)_5 \text{Br}] \text{SO}_4$. The resulting solution is divided into two equal parts (X) and treated with excess AgNO_3 solution and BaCl_2 solution respectively as shown below:

1L Solution(X) + AgNO_3 solution (excess) $\rightarrow$ Y

1L Solution (X) + BaCl_2 solution (excess) $\rightarrow$ Z

The number of moles of Y and Z respectively are
[30-Jan-2023 Shift 2]

Options:

A. 0.02, 0.02

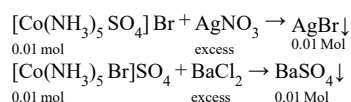
B. 0.01, 0.01

C. 0.02, 0.01

D. 0.01, 0.02

Answer: B

Solution:



Question99

The $\text{Cl} - \text{Co} - \text{Cl}$ bond angle values in a fac $[\text{Co}(\text{NH}_3)_3 \text{Cl}_3]$ complex is/are:
[30-Jan-2023 Shift 2]

Options:

A. 90° & 180°

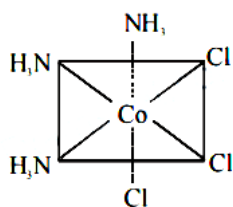
B. 90°

C. 180°

D. 90° & 120°

Answer: B

Solution:



The Cl – Co – Cl bond angle in above octahedral complex is 90°

Question100

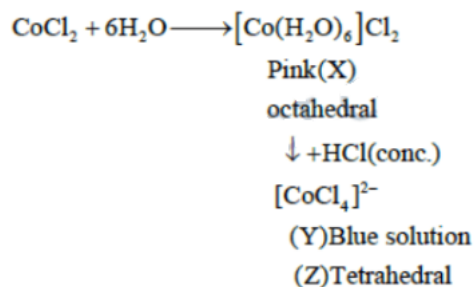
Cobalt chloride when dissolved in water forms pink colored complex X which has octahedral geometry. This solution on treating with cone HCl forms deep blue complex, Y which has a Z geometry. X, Y and Z, respectively, are
[31-Jan-2023 Shift 1]

Options:

- A. $X = [\text{Co}(\text{H}_2\text{O})_6]^{2+}$, $Y = [\text{CoCl}_4]^{2-}$, Z = Tetrahedral
- B. $X = [\text{Co}(\text{H}_2\text{O})_6]^{2+}$, $Y = [\text{CoCl}_6]^{3-}$, Z = Octahedral
- C. $X = [\text{Co}(\text{H}_2\text{O})_6]^{3+}$, $Y = [\text{CoCl}_6]^{3-}$, Z = Octahedral
- D. $X = [\text{Co}(\text{H}_2\text{O})_4\text{Cl}_2]^+$, $Y = [\text{CoCl}_4]^{2-}$, Z = Tetrahedral

Answer: A

Solution:



Question101

If the CFSE of $[\text{Ti}(\text{H}_2\text{O})_6]^{3+}$ is -96.0 kJ/mol , this complex will absorb maximum at wavelength ____ nm. (nearest integer)

Assume Planck's constant (h) = $6.4 \times 10^{-34} \text{ Js}$ Speed of light (c) = $3.0 \times 10^8 \text{ m/s}$ and Avogadro's constant (N_A) = $6 \times 10^{23} / \text{mol}$

[31-Jan-2023 Shift 2]

Answer: 480

Solution:

$$\begin{aligned}
 &(\text{Ti}^{+3}(\text{H}_2\text{O})_6)^{3+} \\
 &\text{Ti}^{+3} : 3d^1 \\
 &\text{C.F.S.E.} = -0.4 \times \Delta_0 \\
 &= -\frac{96 \times 10^3}{N_0} \text{J} \\
 &\Delta_0 = \frac{96 \times 10^3}{0.4 \times 6 \times 10^{23}} \\
 &\Rightarrow \frac{hc}{\lambda} = \frac{96 \times 10^3}{0.4 \times 6 \times 10^{23}} \\
 &\lambda = \frac{0.4 \times 6 \times 10^{23} \times 6.4 \times 10^{-34} \times 3 \times 10^8}{96 \times 10^3} \\
 &= 0.48 \times 10^{-6} \text{m} \\
 &= 480 \times 10^{-9} \text{m} \\
 &= 480 \text{ nm}
 \end{aligned}$$

Question102

Match List I with List II

List I Coordination entity	List II Wavelength of light absorbed in nm
A. $[\text{CoCl}(\text{NH}_3)_5]^{2+}$	I. 310
B. $[\text{Co}(\text{NH}_3)_6]^{3+}$	II. 475
C. $[\text{Co}(\text{CN})_6]^{3-}$	III. 535
D. $[\text{Cu}(\text{H}_2\text{O})_4]^{2+}$	IV. 600

Choose the correct answer from the options given below :-
[25-Jan-2023 Shift 2]

Options:

- A. A-IV, B-I, C-III, D-II
- B. A-III, B-II, C-I, D-IV
- C. A-III, B-I, C-II, D-IV
- D. A- II, B-III, C-IV, D-I

Answer: B

Solution:

List I Coordination entity	List II Wavelength of light absorbed in nm
A. $[\text{CoCl}(\text{NH}_3)_5]^{2+}$	I. 535
B. $[\text{Co}(\text{NH}_3)_6]^{3+}$	II. 475
C. $[\text{Co}(\text{CN})_6]^{3-}$	III. 310
D. $[\text{Cu}(\text{H}_2\text{O})_4]^{2+}$	IV. 600

$$\begin{aligned}
 E &= \frac{hc}{\lambda} \Rightarrow E \propto \frac{1}{\lambda} \\
 \Rightarrow \Delta (\text{CFSE}) &\propto \frac{1}{\lambda_{\text{absorb}}} \propto \text{strength of ligand.}
 \end{aligned}$$

Question103

Total number of moles of AgCl precipitated on addition of excess of AgNO_3 to one mole each of the following complexes $[\text{Co}(\text{NH}_3)_4 \text{Cl}_2] \text{Cl}$, $[\text{Ni}(\text{H}_2\text{O})_6]\text{Cl}_2$, $[\text{Pt}(\text{NH}_3)_2 \text{Cl}_2]$ and $[\text{Pd}(\text{NH}_3)_4]\text{Cl}_2$ is
[25-Jan-2023 Shift 2]

Answer: 5

Solution:

$[\text{Co}(\text{NH}_3)_4 \text{Cl}_2] \text{Cl} \Rightarrow$ Gives 1 mole AgCl

$[\text{Ni}(\text{H}_2\text{O})_6]\text{Cl}_2 \Rightarrow$ Gives 2 moles AgCl

$[\text{Pt}(\text{NH}_3)_2 \text{Cl}_2] \Rightarrow$ Gives No AgCl

$[\text{Pd}(\text{NH}_3)_4]\text{Cl}_2 \Rightarrow$ Gives 2 moles AgCl

Total number of moles of AgCl = 5 mole.

Question104

Chiral complex from the following is:

Here en = ethylene diamine

[29-Jan-2023 Shift 1]

Options:

A. $\text{cis} - [\text{PtCl}_2(\text{en})_2]^{2+}$

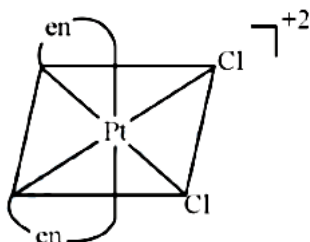
B. $\text{trans} - [\text{PtCl}_2(\text{en})_2]^{2+}$

C. $\text{cis} - [\text{PtCl}_2(\text{NH}_3)_2]$

D. $\text{trans} - [\text{Co}(\text{NH}_3)_4 \text{Cl}_2]^+$

Answer: D

Solution:



this is chiral complex form

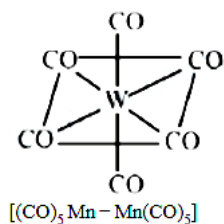
Question105

The sum of bridging carbonyls in $\text{W}(\text{CO})_6$ and $\text{Mn}_2(\text{CO})_{10}$ is _____.

[29-Jan-2023 Shift 1]

Answer: 0

Solution:



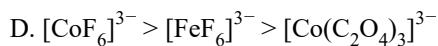
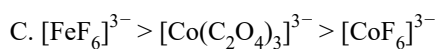
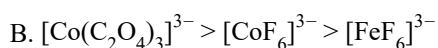
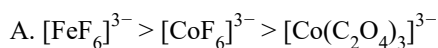
Question106

Correct order of spin only magnetic moment of the following complex ions is:

(Given At. No. Fe: 26, Co:27)

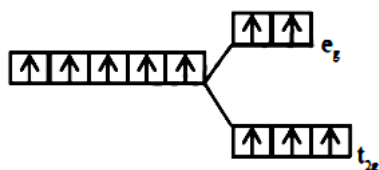
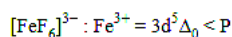
[29-Jan-2023 Shift 2]

Options:

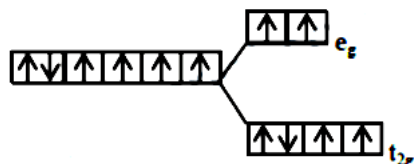
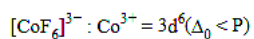


Answer: C

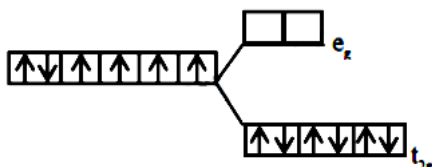
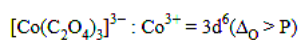
Solution:



Number of unpaired $e^- = 5 \therefore \mu = \sqrt{35} \text{ BM}$



Number of unpaired $e^- = 4 \therefore \mu = \sqrt{24} \text{ BM}$



Number of unpaired $e^- = 0 \therefore \mu = 0 \text{ BM}$

Question107

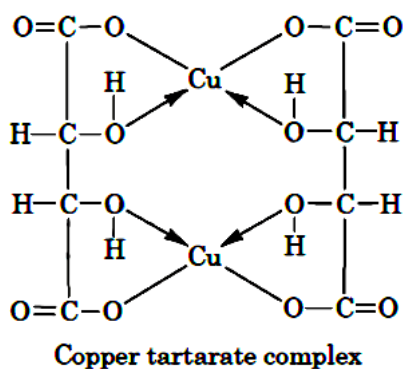
The denticity of the ligand present in the Fehling's reagent is _____.

Denticity = 2

[29-Jan-2023 Shift 2]

Answer: 4

Solution:



Question108

Which of the following is correct order of ligand field strength?

[30-Jan-2023 Shift 1]

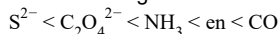
Options:

- A. $\text{CO} < \text{en} < \text{NH}_3 < \text{C}_2\text{O}_4^{2-} < \text{S}^{2-}$
- B. $\text{S}^{2-} < \text{C}_2\text{O}_4^{2-} < \text{NH}_3 < \text{en} < \text{CO}$
- C. $\text{NH}_3 < \text{en} < \text{CO} < \text{S}^{2-} < \text{C}_2\text{O}_4^{2-}$
- D. $\text{S}^{2-} < \text{NH}_3 < \text{en} < \text{CO} < \text{C}_2\text{O}_4^{2-}$

Answer: A

Solution:

The increasing order of field strength of ligands (according to spectrochemical series)



Question109

To inhibit the growth of tumours, identify the compounds used from the following:

- (A) EDTA
- (B) Coordination Compounds of Pt
- (C) D-Penicillamine
- (D) Cis - Platin

Choose the correct answer from the option given below:
[30-Jan-2023 Shift 1]

Options:

- A. B and D Only
- B. C and D Only
- C. A and B Only
- D. A and C Only

Answer: B

Solution:

Cis - Platin is used in chemotherapy to inhibits the growth of tumors. (cis $[\text{Pt}(\text{NH}_3)_2 \text{Cl}_2]$)

Question110

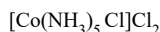
The primary and secondary valencies of cobalt respectively in $[\text{Co}(\text{NH}_3)_5 \text{Cl}]\text{Cl}_2$ are :
[24-Jan-2023 Shift 1]

Options:

- A. 3 and 5
- B. 2 and 6
- C. 2 and 8
- D. 3 and 6

Answer: D

Solution:



Oxidation number of Co is +3.

So primary valency is 3 .

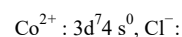
It is an octahedral complex so secondary valency 6 or Co-ordination number 6 .

Question111

The d-electronic configuration of $[\text{CoCl}_4]^{2-}$ in tetrahedral crystal field is $e^m t_2^n$. Sum of ' m ' and 'number of unpaired electrons is _____
[24-Jan-2023 Shift 1]

Answer: 7

Solution:



Configuration $e^4 t_2^3 : m = 4$

Number of unpaired electrons = 3

So, answer = 7

Question112

Which of the following cannot be explained by crystal field theory?
[24-Jan-2023 Shift 2]

Options:

- A. The order of spectrochemical series
- B. Magnetic properties of transition metal complexes
- C. Colour of metal complexes
- D. Stability of metal complexes

Answer: A

Solution:

Crystal field theory introduce spectrochemical series based upon the experimental values of Δ but can't explain it's order. While other three points are explained by CFT. Specially when the CFSE increases thermodynamic stability of the complex increases.

Question113

Identify the correct statements about alkali metals.

- A. The order of standard reduction potential ($M^+ | M$) for alkali metal ions is $Na > Rb > Li$.
- B. CsI is highly soluble in water.
- C. Lithium carbonate is highly stable to heat.
- D. Potassium dissolved in concentrated liquid ammonia is blue in colour and paramagnetic.
- E. All the alkali metal hydrides are ionic solids.

Choose the correct answer from the options given below
[24-Jan-2023 Shift 2]

Options:

- A. A, B, D only
- B. C and E only
- C. A and E only
- D. A, B and E only

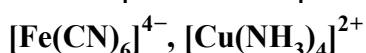
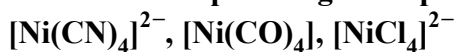
Answer: C

Solution:

(1) $Na > Cs > Li$ - true { If considered with sign } The low solubility of CsI is due to smaller hydration enthalpy of it's two ions
 Li_2CO_3 is highly stable to heat - false
In Conc. NH_3 , K formed blue solution - true
All the alkali metal hydrides are ionic solid (True).

Question114

The number of paramagnetic species from the following is ____.

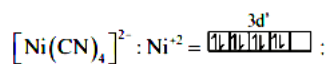




[25-Jan-2023 Shift 1]

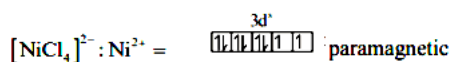
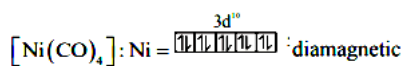
Answer: 4

Solution:

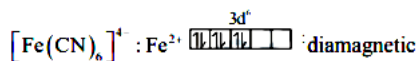


diamagnetic

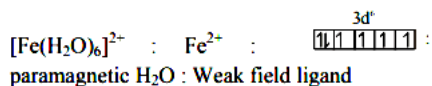
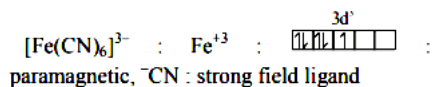
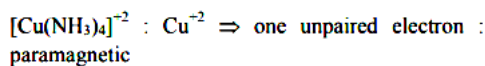
CN^- : strong field ligand



Cl^- : weak field ligand



CN^- : strong field ligand

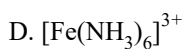
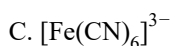
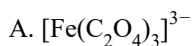


Question115

Which of the following complex will show largest splitting of d-orbitals?

[1-Feb-2023 Shift 1]

Options:



Answer: C

Solution:

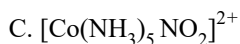
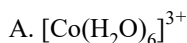
CN^- is a strong field ligand so maximum splitting in d orbitals take place.

Question116

The complex cation which has two isomers is :

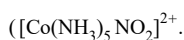
[1-Feb-2023 Shift 2]

Options:



Answer: C

Solution:



Two linkage isomers possible

$\text{NO}_2 \rightarrow$ Ambidentate ligand

Question117

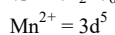
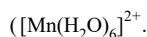
The spin only magnetic moment of $[\text{Mn}(\text{H}_2\text{O})_6]^{2+}$ complexes is _____ B.M. (Nearest integer)

(Given : Atomic no. of Mn is 25)

[1-Feb-2023 Shift 2]

Answer: 6

Solution:



$\mu = \sqrt{5(5+2)} = 5.91 \text{ BM}$

Question118

Given below are two statements, one is labelled as Assertion A and the other is labelled as Reason R.

Assertion A: The spin only magnetic moment value for $[\text{Fe}(\text{CN})_6]^{3-}$ is 1.74 BM, whereas for

$[\text{Fe}(\text{H}_2\text{O})_6]^{3+}$ is 5.92 BM.

Reason R : In both complexes, Fe is present in +3 oxidation state.

In the light of the above statements, choose the correct answer from the options given below:

[6-Apr-2023 shift 1]

Options:

A. Both A and R are true but R is NOT the correct explanation of A

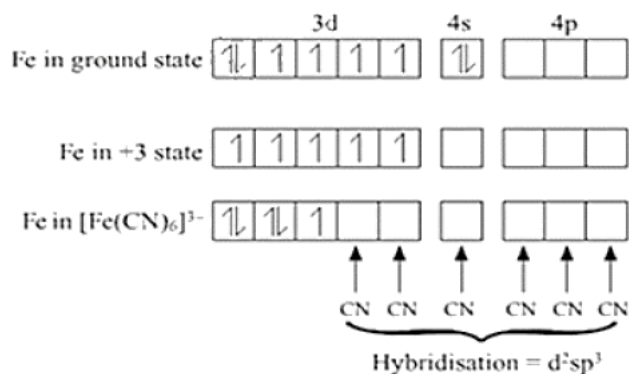
B. A is false but R is true

C. A is true but R is false

D. Both A and R are true and R is the correct explanation of A

Answer: A

Solution:



Unpaired electron = 1

$$\mu = \sqrt{n(n+2)} = \sqrt{1 \times 3} = 1.74 \text{ B.M.}$$

$[\text{Fe}(\text{H}_2\text{O})_6]^{3+}$ No pairing because H_2O is WFL

Number of unpaired electrons = 5, $\mu = 5.92 \text{ BM}$

Assertion is true, Reason is true but not correct explanation.

Question119

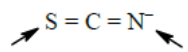
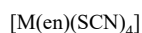
Number of ambidentate ligands in a representative metal complex $[\text{M}(\text{en})(\text{SCN})_4]$ is

[en = ethylenediamine]

[6-Apr-2023 shift 1]

Answer: 4

Solution:



Ambidentate ligand means two ligand site, so ambidentate ligand is SCN^- .

Ans: 4

Question120

The IUPAC name of $\text{K}_3[\text{Co}(\text{C}_2\text{O}_4)_3]$ is :-

[6-Apr-2023 shift 2]

Options:

A. Potassium trioxalatocobaltate(III)

B. Potassium tris(oxalato)cobalt(III)

C. Potassium tris(oxalato)cobaltate(III)

D. Potassium trioxalatocobalt(III)

Answer: A

Solution:

IUPAC name of $\text{K}_3[\text{Co}(\text{C}_2\text{O}_4)_3]$ is Potassium trioxalatocobaltate(III)

Question121

Given below are two statements : one is labelled as "Assertion A" and the other is labelled as "Reason R".

Assertion A : In the complex $\text{Ni}(\text{CO})_4$ and $\text{Fe}(\text{CO})_5$, the metals have zero oxidation state.

Reason R : Low oxidation states are found when a complex has ligands capable of π -donor character in addition to the σ -bonding.

In the light of the above statements, choose the most appropriate answer from the option given below.
[6-Apr-2023 shift 2]

Options:

- A. A is correct but R is not correct
- B. A is not correct but R is correct
- C. Both A and R are correct but R is NOT the correct explanation of A
- D. Both A and R are correct and R is the correct explanation of A.

Answer: A

Question122

Element not present in Nessler's reagent is:-
[6-Apr-2023 shift 2]

Options:

- A. Hg
- B. I
- C. K
- D. N

Answer: D

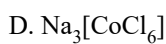
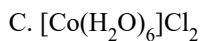
Solution:

Nessler reagent is $-\text{K}_2[\text{HgI}_4]$

Question123

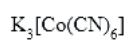
Which of the following complex is octahedral, diamagnetic and the most stable?
[8-Apr-2023 shift 1]

Options:



Answer: A

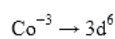
Solution:



$$+3 + x - 6 = 0$$

$$x = +3$$

$\Downarrow$



1↓	1	1	1	1
----	---	---	---	---

$\therefore \text{CN}^-$ is SFL so pairing occurs so

1↓	1↓	1↓		
----	----	----	--	--

$$u-e = 0$$

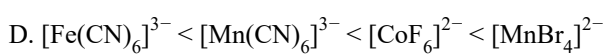
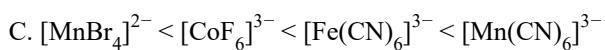
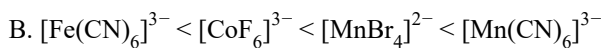
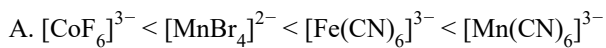
$\downarrow$

So diamagnetic

Question 124

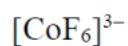
The correct order of spin only magnetic moments for the following complex ions is [8-Apr-2023 shift 1]

Options:



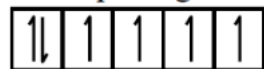
Answer: D

Solution:

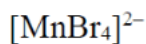


F^- WFL

So no pairing

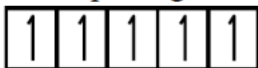


u.e = 4



Br^- WFL

So no pairing

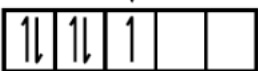
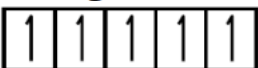


u.e = 5

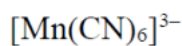


CN^- SFL

Pairing occur

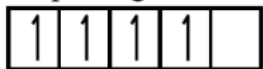


u.e = 1

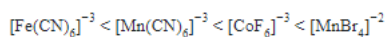


CN^- is SFL

So pairing occur



u.e = 2



Question125

Match list I with list II

LIST - I	LIST - II
Coordination complex	Number of unpaired electrons
A. $[\text{Cr}(\text{CN})_6]^{3-}$	I. 0
B. $[\text{Fe}(\text{H}_2\text{O})_6]^{2+}$	II. 3
C. $[\text{Co}(\text{NH}_3)_6]^{3+}$	III. 2
D. $[\text{Ni}(\text{NH}_3)_6]^{2+}$	IV. 4

Choose the correct answer from the options given below:
[8-Apr-2023 shift 2]

Options:

A. A-II, B-IV, C-I, D-III

B. A-IV, B-III, C-II, D-I

C. A-II, B-I, C-IV, D-III

D. A-III, B-IV, C-I, D-II

Answer: A

Solution:

For option (A)

$\text{Cr}^{+3} : 3d^3$

$\text{CN}^- \rightarrow \text{SFL}$

$\Rightarrow$ No. of unpaired electrons = 3

For option (B)

$\text{Fe}^{+2} : 3d^6$

$\text{H}_2\text{O} : \text{WFL}$

No. of unpaired electrons = 4

For option (C)

$\text{Co}^{+3} : 3d^6$

$\text{NH}_3 : \text{SFL}$

No. of unpaired electrons = 0

For option (D)

$\text{Ni}^{+2} : 3d^8$

$\text{NH}_3 : \text{SFL}$

No. of unpaired electrons = 2

Question126

The observed magnetic moment of the complex $[\text{Mn}(\text{x NCS})_6]^{x-}$ is 6.06 BM. The numerical value of x is

[8-Apr-2023 shift 2]

Answer: 4

Solution:

$[\text{Mn}(\text{NCS})_6]^{x-}$

Number of unpaired electron = 5

So, Mn must be in +2 oxidation state (Mn^{+2})

$\Rightarrow 2 + (-6) = -x \Rightarrow -4 = -x \Rightarrow x = 4$

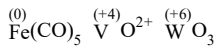
Question127

The sum of oxidation state of the metals in $\text{Fe}(\text{CO})_5$, VO^{2+} and WO_3 is _____

[8-Apr-2023 shift 2]

Answer: 10

Solution:

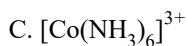


So, Sum of oxidation state = $0 + 4 + 6 = 10$

Question128

The octahedral diamagnetic low spin complex among the following is
[10-Apr-2023 shift 1]

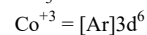
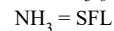
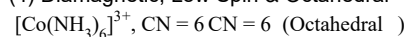
Options:



Answer: C

Solution:

- (1) Paramagnetic, High Spin & Tetrahedral
- (2) Paramagnetic, High Spin & Octahedral
- (3) Paramagnetic, High Spin & Octahedral
- (4) Diamagnetic, Low Spin & Octahedral



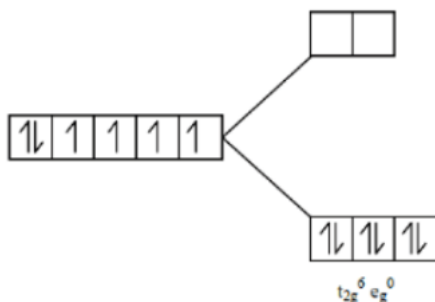
Diamagnetic & Low spin complex

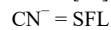
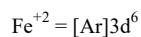
Question129

In potassium ferrocyanide, there are _____ pairs of electrons in the t_{2g} set of orbitals.
[10-Apr-2023 shift 1]

Answer: 3

Solution:





t_{2g} contain 6 electron so it become 3 pairs

Question130

Match List I with List II

LIST I Complex	LIST II Crystal Field splitting energy (Δ_o)
A $[\text{Ti}(\text{H}_2\text{O})_6]^{2+}$	I. -1.2
B $[\text{V}(\text{H}_2\text{O})_6]^{2+}$	II. -0.8
C $[\text{Mn}(\text{H}_2\text{O})_6]^{3+}$	III. 0
D $[\text{Fe}(\text{H}_2\text{O})_6]^{3+}$	IV. -0.8

Choose the correct answer from the options given below:
[10-Apr-2023 shift 2]

Options:

A. A-IV, B-I, C-II, D-III

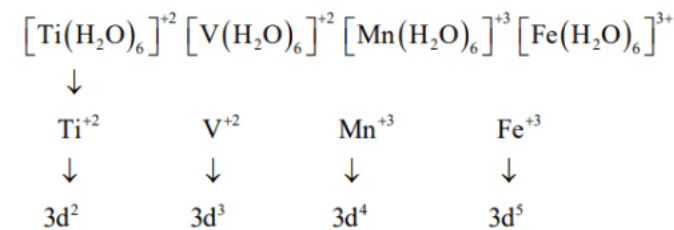
B. A-IV, B-I, C-III, D-II

C. A-II, B-IV, C-III, D-I

D. A-II, B-IV, C-I, D-III

Answer: A

Solution:



$$\text{CFSE} = -0.4 \times t_{2g} + 0.6 \times e_g + x p$$

$$= -0.4 \times 2 + 0.6 \times 0 + x p$$

$$(A) = -0.8 \rightarrow \text{Ti}^{+2}$$

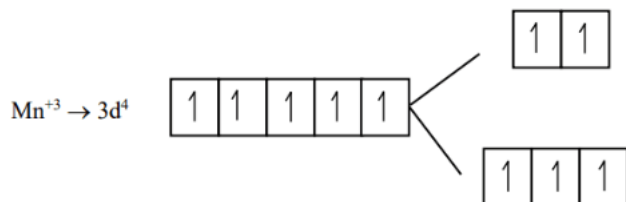
$$(B) \text{V}^{+2} \rightarrow 3d^3$$

$$\text{CFSE} = -0.4 \times t_{2g} + 0.6 \times e_g + x p$$

$$= -0.4 \times 3 + 0.6 \times 0 + x p$$

$$= -1.2$$

(C)

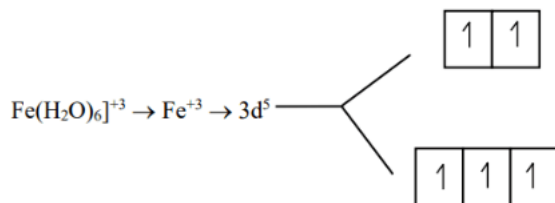


$$\text{CFSE} = -0.4 \times t_{2g} + 0.6 \times e_g + x p$$

$$= -0.4 \times 3 + 0.6 \times 1 + x p$$

$$= -1.2 + 0.6 = -0.6$$

(D)



$$\begin{aligned}\text{CFSE} &= -0.4 \times t_{2g} + 0.6 \times e_g + x p \\ &= -0.4 \times 3 + 0.6 \times 2 \\ &= -1.2 + 1.2 \\ &= 0\end{aligned}$$

Question131

The correct order of the number of unpaired electrons in the given complexes is

- A. $[\text{Fe}(\text{CN})_6]^{3-}$
- B. $[\text{FeF}_6]^{3-}$
- C. $[\text{CoF}_6]^{3-}$
- D. $[\text{Cr}(\text{oxalate})_3]^{3-}$
- E. $[\text{Ni}(\text{CO})_4]$

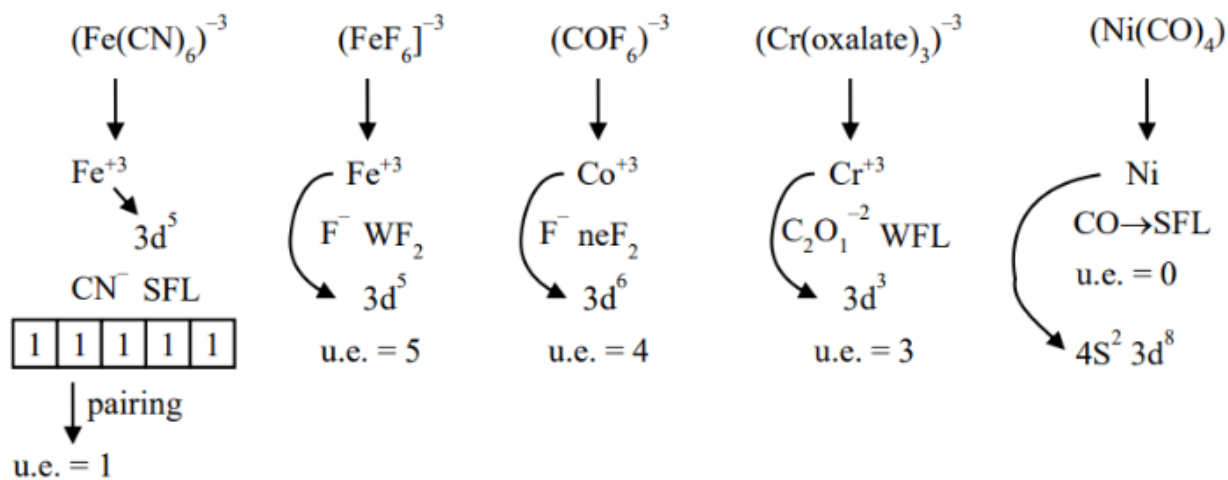
Choose the correct answer from the options given below:
[10-Apr-2023 shift 2]

Options:

- A. $E < A < D < C < B$
- B. $A < E < C < B < D$
- C. $A < E < D < C < B$
- D. $E < A < B < D < C$

Answer: A

Solution:



Question132

**The complex that dissolves in water is
[11-Apr-2023 shift 1]**

Options:

- A. $[\text{Fe}_3(\text{OH})_2(\text{OAc})_6] \text{Cl}$
- B. $\text{Fe}_4[\text{Fe}(\text{CN})_6]_3$
- C. $\text{K}_3[\text{Co}(\text{NO}_2)_6]$
- D. $(\text{NH}_4)_3[\text{As}(\text{Mo}_3\text{O}_{10})_4]$

Answer: A

Solution:

$\text{Fe}_4[\text{Fe}(\text{CN})_6]_3$ Prussian Blue-water insoluble
 $\text{K}_3[\text{Co}(\text{NO}_2)_6]$ very poorly water soluble
 $(\text{NH}_4)_3[\text{As}(\text{Mo}_3\text{O}_{10})_4]$ water insoluble
 ammonium arseno molybdate
 $[\text{Fe}_3(\text{OH})_2(\text{OAc})_6] \text{Cl}$ is water soluble.

Question133

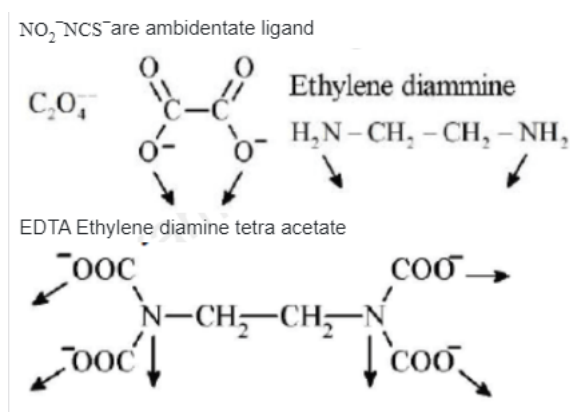
**The set which does not have ambidentate ligand (s) is
[11-Apr-2023 shift 1]**

Options:

- A. $\text{C}_2\text{O}_4^{2-}$, NO_2^- , NCS^-
- B. EDTA^{4-} , NCS^- , $\text{C}_2\text{O}_4^{2-}$
- C. NO_2^- , $\text{C}_2\text{O}_4^{2-}$, EDTA^{4-}
- D. $\text{C}_2\text{O}_4^{2-}$, ethylene diamine, H_2O

Answer: D

Solution:



Question134

Which of the following complex has a possibility to exist as meridional isomer?
[11-Apr-2023 shift 1]

Options:

- A. $[\text{Co}(\text{en})_2 \text{Cl}_2]$
- B. $[\text{Pt}(\text{NH}_3)_2 \text{Cl}_2]$
- C. $[\text{Co}(\text{en})_3]$
- D. $[\text{Co}(\text{NH}_3)_3 (\text{NO}_2)_3]$

Answer: D

Solution:

$[\text{MA}_3\text{B}_3]$ type of compound exists as facial and meridional isomer.



Question135

The ratio of spin-only magnetic moment values $\mu_{\text{eff}} [\text{Cr}(\text{CN})_6]^{3-} / \mu_{\text{eff}} [\text{Cr}(\text{H}_2\text{O})_6]^{3+}$ is _____.
[11-Apr-2023 shift 1]

Answer: 1

Solution:

Spin magnetic moment of $[\text{Cr}(\text{CN})_6]^{3-} (t_{2g}^3 e_g^0)$

$$\mu_1 = \sqrt{3(3+2)} = \sqrt{15} \text{ BM}$$

Spin magnetic moment of $[\text{Cr}(\text{H}_2\text{O})_6]^{3+} (t_{2g}^3 e_g^0)$

$$\mu_2 = \sqrt{3(3+2)} = \sqrt{15} \text{ BM}$$

$$\frac{\mu_1}{\mu_2} = \frac{\sqrt{15}}{\sqrt{15}} = 1$$

Question136

If Ni^{2+} is replaced by Pt^{2+} in the complex $[\text{NiCl}_2\text{Br}_2]^{2-}$, which of the following properties are expected to get changed?

- A. Geometry
- B. Geometrical isomerism
- C. Optical isomerism
- D. Magnetic properties

[11-Apr-2023 shift 2]

Options:

- A. A, B and C

- B. A and D
- C. B and C
- D. A, B and D

Answer: D

Solution:

$[\text{NiBr}_2\text{Cl}_2]^{2-} \rightarrow$ This complex species is tetrahedral as $\text{Br}^\ominus$ & $\text{Cl}^\ominus$ are weak field ligands.

$[\text{PtBr}_2\text{Cl}_2]^{2-} \rightarrow$ As Pt belongs to 5d series. This complex species is square planar.

Both the complex species are optically inactive.

$[\text{NiBr}_2\text{Cl}_2]^{2-}$, being tetrahedral does not show Geometrical Isomerism.

$[\text{PtBr}_2\text{Cl}_2]^{2-}$ shows two Geometrical Isomers.

Question137

Match List I with List II

List I Complex	List I Colour
A. $\text{Mg}(\text{NH}_4)\text{PO}_4$	I. Brown
B. $\text{K}_3[\text{Co}(\text{NO}_2)_6]$	II. White
C. $\text{MnO}(\text{OH})_2$	III. Yellow
D. $\text{Fe}_4[\text{Fe}(\text{CN})_6]_3$	IV. blue

Choose the correct answer from the options given below:
[11-Apr-2023 shift 2]

Options:

- A. A-II, B-III, C-IV, D-I
- B. A-II, B-IV, C-I, D-III
- C. A-III, B-IV, C-II, D-I
- D. A-II, B-III, C-I, D-IV

Answer: D

Solution:

$\text{Mg}(\text{NH}_4)\text{PO}_4 \Rightarrow$ White

$\text{K}_3[\text{Co}(\text{NO}_2)_6] \Rightarrow$ Yellow

$\text{MnO}(\text{OH})_2 \Rightarrow$ Brown

$\text{Fe}_4[\text{Fe}(\text{CN})_6]_3 \Rightarrow$ Blue

Question138

Given below are two statements, one is labelled as Assertion A and the other is labelled as Reason R.

Assertion A : $[\text{CoCl}(\text{NH}_3)_5]^{2+}$ absorbs at lower wavelength of light with respect to $[\text{CoCl}(\text{NH}_3)_5(\text{H}_2\text{O})]^{3+}$

Reason R : It is because the wavelength of the light absorbed depends on the oxidation state of the metal ion.

In the light of the above statements, choose the correct answer from the options given below:
[11-Apr-2023 shift 2]

Options:

- A. Both A and R are true but R is NOT the correct explanation of A
- B. A is true but R is false
- C. Both A and R are true and R is the correct explanation of A
- D. A is false but R is true

Answer: D

Solution:

Since H_2O is strong field ligand compared to chloride and Co^{3+} ion is present.
 $\therefore$ CFSE is higher for $[\text{Co}(\text{NH}_3)_5\text{H}_2\text{O}]^{+3}$, hence it will absorb at lower wavelength.

Question139

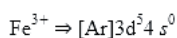
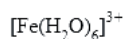
The magnetic moment is measured in Bohr Magneton (BM). Spin only magnetic moment of Fe in $[\text{Fe}(\text{H}_2\text{O})_6]^{3+}$ and $[\text{Fe}(\text{CN})_6]^{3-}$ complexes respectively is:
[11-Apr-2023 shift 2]

Options:

- A. 3.87 B. M. and 1.732 B.M.
- B. 6.92 B.M. in both
- C. 5.92 B.M. and 1.732 B.M.
- D. 4.89 B.M. and 6.92 B.M.

Answer: C

Solution:



No pairing

1	1	1	1	1
---	---	---	---	---

$$\therefore \text{Unpaired } e^- = 5$$

$$\mu = \sqrt{n(n+2)}$$

$$= \sqrt{5(5+2)}$$

$$\mu = \sqrt{35} = 5.92 \text{ B.M.}$$



1↓	1↓	1	1	1
----	----	---	---	---

$$\therefore \text{Unpaired } e^- = 1$$

$$\mu = \sqrt{n(n+2)}$$

$$= \sqrt{1(1+2)} = \sqrt{3} = 1.732 \text{ B.M.}$$

Question140

Match List I with List II

LIST I		LIST II	
Complex		CFSE (Δ_o)	
A.	$[\text{Cu}(\text{NH}_3)_6]^{2+}$	I.	-0.6
B.	$[\text{Ti}(\text{H}_2\text{O})_6]^{3+}$	II.	-2.0
C.	$[\text{Fe}(\text{CN})_6]^{3-}$	III.	-1.2
D.	$[\text{NiF}_6]^{4-}$	IV.	-0.4

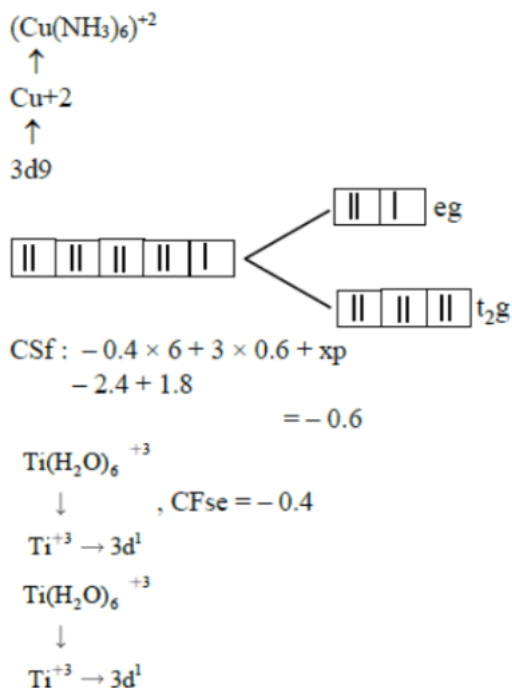
Choose the correct answer from the options given below:
[12-Apr-2023 shift 1]

Options:

- A. A-III, B-IV, C-I, D-II
- B. A-II, B-III, C-I, D-IV
- C. A-I, B-IV, C-II, D-III
- D. A-I, B-II, C-IV, D-III

Answer: C

Solution:



Question141

The mismatched combinations are

- A. Chlorophyll - Co
- B. Water hardness - EDTA
- C. Photography - $[\text{Ag}(\text{CN})_2]^-$

D. Wilkinson catalyst - $[(\text{Ph}_3\text{P})_3\text{RhCl}]$

E. Chelating ligand - D-Penicillamine

Choose the correct answer from the options given below :

[13-Apr-2023 shift 1]

Options:

A. A and C Only

B. D and E Only

C. A and E Only

D. A, C, and E Only

Answer: A

Solution:

Mg is present in chlorophyll and in black and white photography the developed film is fixed by washing with hypo solution which dissolves the undecomposed AgBr to form a complex ion $[\text{Ag}(\text{S}_2\text{O}_3)_2]^{3-}$

Question142

The covalency and oxidation state respectively of boron in $[\text{BF}_4]^-$, are :

[13-Apr-2023 shift 2]

Options:

A. 3 and 5

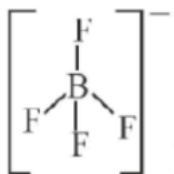
B. 4 and 3

C. 4 and 4

D. 3 and 4

Answer: B

Solution:



Number of covalent bond formed by Boron is 4

Oxidation number of fluorine is -1 ,

Oxidation number of B + $4 \times (-1) = -1$,

Thus, Oxidation number of B = +3

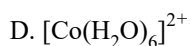
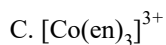
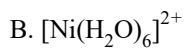
Question143

Which of the following complexes will exhibit maximum attraction to an applied magnetic field?

[13-Apr-2023 shift 2]

Options:

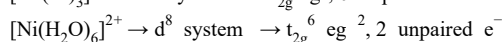
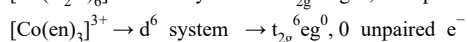
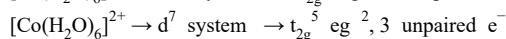
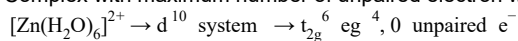
A. $[\text{Zn}(\text{H}_2\text{O})_6]^{2+}$



Answer: D

Solution:

Complex with maximum number of unpaired electron will exhibit maximum attraction to an applied magnetic field



Question144

The total number of stereoisomers for the complex $[\text{Cr}(\text{ox})_2 \text{ClBr}]^{3-}$ (where ox = oxalate) is :
[13-Apr-2023 shift 2]

Options:

A. 3

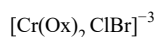
B. 1

C. 4

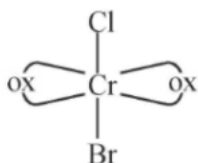
D. 2

Answer: A

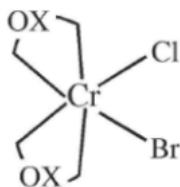
Solution:



- No. of isomers-



- This structure has plane of symmetry, So no optical isomerism will be shown.

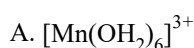


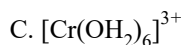
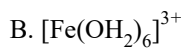
- This structure does not contain plane of symmetry, So two forms d as well as 1 will be shown.

Question145

The complex with highest magnitude of crystal field splitting energy (Δ_0) is
[15-Apr-2023 shift 1]

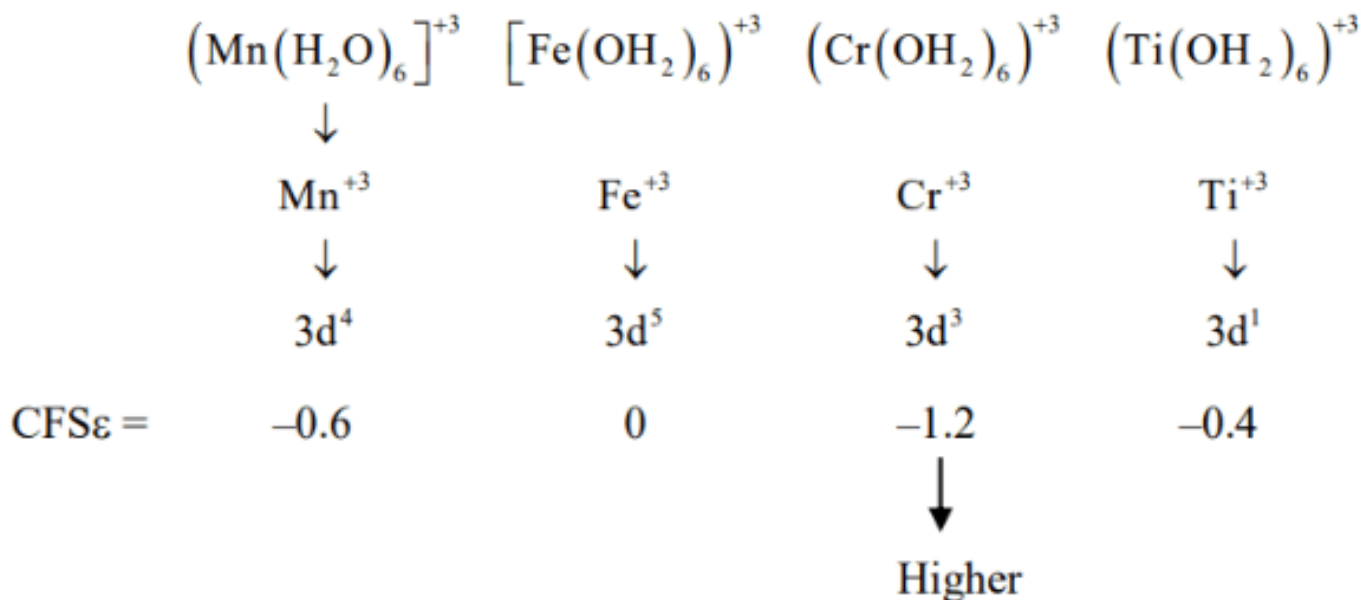
Options:





Answer: C

Solution:



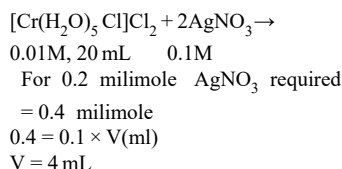
Question146

The volume (in mL) of 0.1M AgNO_3 required for complete precipitation of chloride ions present in 20 mL of 0.01M solution of $[\text{Cr}(\text{H}_2\text{O})_5\text{Cl}]\text{Cl}_2$ as silver chloride is _____.

[15-Apr-2023 shift 1]

Answer: 4

Solution:



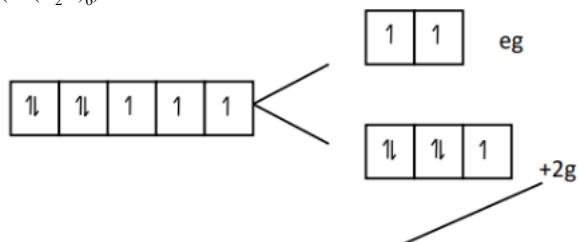
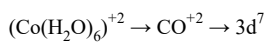
Question147

The homoleptic and octahedral complex of Co^{2+} and H_2O has _____ unpaired electrons(s) in the t_{2g} set of orbitals.

[15-Apr-2023 shift 1]

Answer: 1

Solution:



u.e. in t_{2g} is = (1)

Question148

List-1		List -II	
(A)	$[\text{PtCl}_4]^{2-}$	I	sp^3d
(B)	BrF_5	II	d^2sp^3
(C)	PCl_5	III	dsp^2
(D)	$[\text{CoNH}_3]^{3+}$	IV	sp^3d^2

Choose the most appropriate answer from the options given below :
[24-Jun-2022-Shift-1]

Options:

- A. (A)-(II), (B)-(IV), (C)-(I), (D)-(III)
B. (A)-(III), (B)-(IV), (C)-(I), (D)-(II)
C. (A)-(III), (B)-(I), (C)-(IV), (D)-(II)
D. (A)-(II), (B)-(I), (C)-(IV), (D)-(III)

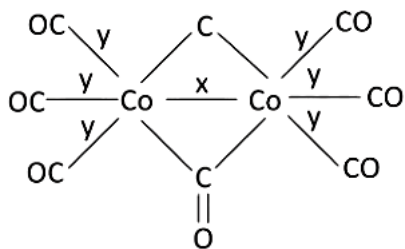
Answer: B

Question149

In the cobalt-carbonyl complex : $[\text{Co}_2(\text{CO})_8]$, number of Co – Co bonds is "X" and terminal CO ligands is "Y". $X + Y =$
[24-Jun-2022-Shift-1]

Answer: 7

Solution:



$$\begin{aligned}
 x &= 1 \\
 y &= 6 \\
 \therefore x + y &= 7
 \end{aligned}$$

Question150

Transition metal complex with highest value of crystal field splitting (Δ_0) will be
[24-Jun-2022-Shift-2]

Options:

- A. $[\text{Cr}(\text{H}_2\text{O})_6]^{3+}$
- B. $[\text{Mo}(\text{H}_2\text{O})_6]^{3+}$
- C. $[\text{Fe}(\text{H}_2\text{O})_6]^{3+}$
- D. $[\text{Os}(\text{H}_2\text{O})_6]^{3+}$

Answer: D

Solution:

Crystal field splitting (Δ_0) for octahedral complexes depends on oxidation state of the metal as well as to which transition series the metal belongs. For the same oxidation state, the crystal field splitting (Δ_0) increases as we move from 3d $\rightarrow$ 4d $\rightarrow$ 5d

Cr^{3+} and Fe^{3+} belong to 3d series, Mo^{3+} belongs to 4d series and Os^{3+} belongs to 5d series.

Therefore crystal field splitting (Δ_0) is highest for $[\text{Os}(\text{H}_2\text{O})_6]^{3+}$.

Question151

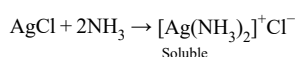
White precipitate of AgCl dissolves in aqueous ammonia solution due to formation of :
[25-Jun-2022-Shift-1]

Options:

- A. $[\text{Ag}(\text{NH}_3)_4]\text{Cl}_2$
- B. $[\text{Ag}(\text{Cl})_2(\text{NH}_3)_2]$
- C. $[\text{Ag}(\text{NH}_3)_2] \text{Cl}$
- D. $[\text{Ag}(\text{NH}_3) \text{Cl}] \text{Cl}$

Answer: C

Solution:



Question152

If $[\text{Cu}(\text{H}_2\text{O})_4]^{2+}$ absorbs a light of wavelength 600 nm for d – d transition, then the value of octahedral crystal field splitting energy for $[\text{Cu}(\text{H}_2\text{O})_6]^{2+}$ will be $\text{____} \times 10^{-21} \text{ J}$. [Nearest Integer]

(Given : $h = 6.63 \times 10^{-34} \text{ Js}$ and $c = 3.08 \times 10^8 \text{ ms}^{-1}$)
[25-Jun-2022-Shift-1]

Answer: 745

Solution:

$[\text{Cu}(\text{H}_2\text{O})_4]^{2+}$ is tetrahedral

$[\text{Cu}(\text{H}_2\text{O})_6]^{2+}$ is octahedral

$$\therefore \Delta_t = \frac{4}{9} \times \Delta_0$$

$$\Delta_t = \frac{6.63 \times 10^{-34} \times 3.08 \times 10^8}{600 \times 10^{-9}}$$

$$\Delta_0 = \frac{9}{4} \times \frac{6.63 \times 10^{-34} \times 3.08 \times 10^8}{600 \times 10^{-9}} \approx 765 \times 10^{-21} \text{ J}$$

Question153

Amongst $\text{FeCl}_3 \cdot 3\text{H}_2\text{O}$, $\text{K}_3[\text{Fe}(\text{CN})_6]$ and $[\text{Co}(\text{NH}_3)_6]\text{Cl}_3$, the spin-only magnetic moment value of the innerorbital complex that absorbs light at shortest wavelength is _____ B.M. [nearest integer]
[25-Jun-2022-Shift-2]

Answer: 2

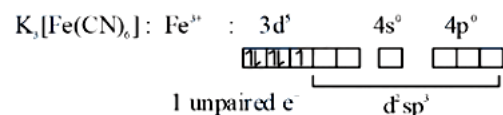
Solution:

$[\text{Fe}(\text{H}_2\text{O})_3\text{Cl}_3]$, $\text{K}_3[\text{Fe}(\text{CN})_6]$, $[\text{Co}(\text{NH}_3)_6]\text{Cl}_3$

inner orbital complexes

$\text{K}_3[\text{Fe}(\text{CN})_6]$ has more value of Δ_0 than that of $[\text{Co}(\text{NH}_3)_6]\text{Cl}_3$; as CN^- is stronger ligand.

More $\Delta_0 \Rightarrow$ smaller value of absorbed λ



Spin only magnetic moment (μ) = $\sqrt{3} \text{ BM} = 1.732 \text{ BM}$

Rounding off $\Rightarrow 2$

Question154

Which statement is not true with respect to nitrate ion test?
[26-Jun-2022-Shift-1]

Options:

A. A dark brown ring is formed at the junction of two solutions.

B. Ring is formed due to nitroferrous sulphate complex.

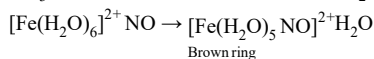
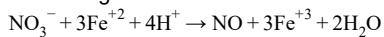
C. The brown complex is $[\text{Fe}(\text{H}_2\text{O})_5(\text{NO})]\text{SO}_4$.

D. Heating the nitrate salt with conc. H_2SO_4 , light brown fumes are evolved.

Answer: B

Solution:

Brown ring test

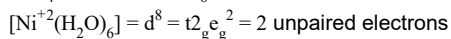
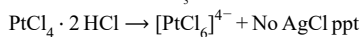
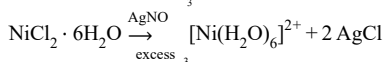
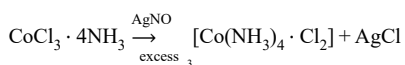


Question 155

The spin-only magnetic moment value of an octahedral complex among $\text{CoCl}_3 \cdot 4\text{NH}_3$, $\text{NiCl}_2 \cdot 6\text{H}_2\text{O}$ and $\text{PtCl}_4 \cdot 2\text{HCl}$, which upon reaction with excess of AgNO_3 gives 2 moles of AgCl is ____ B.M. (Nearest integer)
[26-Jun-2022-Shift-1]

Answer: 3

Solution:



$$\text{Magnetic moment} = \sqrt{2(2+2)}$$

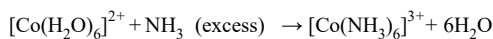
$$2\sqrt{8} \approx 3$$

Question 156

Reaction of $[\text{Co}(\text{H}_2\text{O})_6]^{2+}$ with excess ammonia and in the presence of oxygen results into a diamagnetic product. Number of electrons present in t_{2g} -orbitals of the product is ____
[26-Jun-2022-Shift-2]

Answer: 6

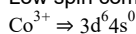
Solution:



Diamagnetic

↓

Low spin complex



Total number electrons = 6

Question157

Which of the following will have maximum stabilization due to crystal field?
[27-Jun-2022-Shift-1]

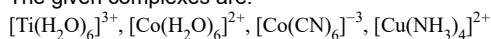
Options:

- A. $[\text{Ti}(\text{H}_2\text{O})_6]^{3+}$
- B. $[\text{Co}(\text{H}_2\text{O})_6]^{2+}$
- C. $[\text{Co}(\text{CN})_6]^{3-}$
- D. $[\text{Cu}(\text{NH}_3)_4]^{2+}$

Answer: C

Solution:

The given complexes are:



CN^- is the strongest ligand among the given complexes CFSE value for the $[\text{Co}(\text{CN})_6]^{3-}$ complex will be highest as it has d^6 configuration with a CFSE value of $-2.40 \Delta_0 + 2P$, where P represents pairing energy and Δ_0 represents splitting energy in octahedral field.

The value of Δ_0 is high for cyanide complexes.

Question158

Arrange the following coordination compounds in the increasing order of magnetic moments. (Atomic numbers : Mn = 25; Fe = 26)

- A. $[\text{FeF}_6]^{3-}$
- B. $[\text{Fe}(\text{CN})_6]^{3-}$
- C. $[\text{MnCl}_6]^{3-}$ (high spin)
- D. $[\text{Mn}(\text{CN})_6]^{3-}$

Choose the correct answer from the options given below :
[27-Jun-2022-Shift-2]

Options:

- A. $A < B < D < C$
- B. $B < D < C < A$
- C. $A < C < D < B$
- D. $B < D < A < C$

Answer: B

Solution:

Coordination Compound	Number of unpaired e^- (n)	Magnetic moment (μ)(B . M)
A. $[\text{FeF}_6]^{3-} - d^5$	5	5.91
B. $[\text{Fe}(\text{CN})_6]^{3-} - d^5$	1	1.73
C. $[\text{MnCl}_6]^{3-} - d^4$	4	4.89
D. $[\text{Mn}(\text{CN})_6]^{3-} - d^4$	2	2.82

Hence, correct order of magnetic moment is $2 < 4 < 3 < 1$

Question159

Given below are two statements :

Statement I : $[\text{Ni}(\text{CN})_4]^{2-}$ is square planar and diamagnetic complex, with dsp^2 hybridization for Ni but $[\text{Ni}(\text{CO})_4]$ is tetrahedral, paramagnetic and with sp^3 -hybridization for Ni.

Statement II : $[\text{NiCl}_4]^{2-}$ and $[\text{Ni}(\text{CO})_4]$ both have same d-electron configuration have same geometry and are paramagnetic.

In light the above statements, choose the correct answer from the options given below :
[28-Jun-2022-Shift-1]

Options:

- A. Both Statement I and Statement II are true.
- B. Both Statement I and Statement II are false.
- C. Statement I is correct but Statement II is false.
- D. Statement I is false but Statement II is correct.

Answer: B

Solution:

$[\text{Ni}(\text{CN})_4]^{2-}$: d^8 configuration, SFL, sq. planar splitting (dsp^2), diamagnetic.

$[\text{Ni}(\text{CO})_4]$: d^{10} config (after excitation), SFL, tetrahedral splitting (sp^3), diamagnetic.

$[\text{NiCl}_4]^{2-}$: d^8 config, WFL, tetrahedral splitting (sp^3), paramagnetic (2. unpaired e^-).

Question160

Number of complexes which will exhibit synergic bonding amongst, $[\text{Cr}(\text{CO})_6]$, $[\text{Mn}(\text{CO})_5]$ and $[\text{Mn}_2(\text{CO})_{10}]$ is ____

[28-Jun-2022-Shift-1]

Answer: 3

Solution:

Carbonyl complex compounds have tendency to show synergic bonding.

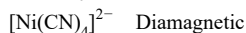
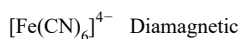
Question161



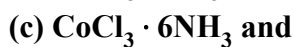
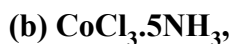
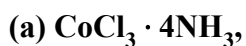
Among the given complexes, number of paramagnetic complexes is ____
[28-Jun-2022-Shift-2]

Answer: 2

Solution:



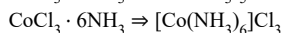
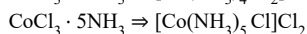
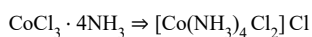
Question162



Number of complex(es) which will exist in cis-trans form is/are _____
[28-Jun-2022-Shift-2]

Answer: 1

Solution:



Only $[\text{Co}(\text{NH}_3)_4\text{Cl}_2]$ can show geometrical isomerism. Hence can exist in cis-trans form.

Question163

Given below are two statements.

- Statement I : In $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$, Cu – O bonds are present.

- Statement II : In $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$, ligands coordinating with Cu(II) ion are O-and S-based ligands.

In the light of the above statements, choose the correct answer from the options given below.

[29-Jun-2022-Shift-2]

Options:

A. Both Statement I and Statement II are correct.

B. Both Statement I and Statement II are incorrect.

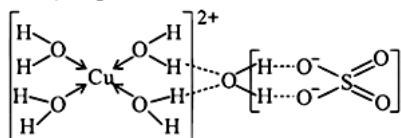
C. Statement I is correct but Statement II is incorrect.

D. Statement I is incorrect but Statement II is correct.

Answer: C

Solution:

Statement I is true but statement II is false. Only oxygen atom forms a Co-ordinate bond with Cu^{+2} in $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$



Question164

Spin only magnetic moment of $[\text{MnBr}_6]^{4-}$ is _____ B.M. (round off to the closest integer)

[29-Jun-2022-Shift-2]

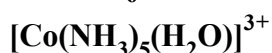
Answer: 6

Solution:

$$\begin{aligned}\text{Mn}^{2+} &\rightarrow t_{2g}^5 e_g^5 \\ \mu_s &= \sqrt{35} \\ &= 5.91 \\ &= 6\end{aligned}$$

Question165

Consider the following metal complexes:



The spin-only magnetic moment value of the complex that absorbs light with shortest wavelength is

B. M. (Nearest integer)
[25-Jul-2022-Shift-1]

Answer: 0

Solution:

$$\Delta_0 \propto \frac{1}{\lambda}$$

Here, CN^- being SFL will have maximum CFSE

So, $[\text{Co}(\text{CN})_6]^{3-}$ will be $d^2 sp^3$, $\mu = 0$

Question 166

The correct order of energy of absorption for the following metal complexes is

A : $[\text{Ni}(\text{en})_3]^{2+}$, B : $[\text{Ni}(\text{NH}_3)_6]^{2+}$, C : $[\text{Ni}(\text{H}_2\text{O})_6]^{2+}$

[25-Jul-2022-Shift-2]

Options:

A. $C < B < A$

B. $B < C < A$

C. $C < A < B$

D. $A < C < B$

Answer: A

Solution:

Stronger is ligand attached to metal ion, greater will be the splitting between t_{2g} and e_g (hence greater will be ΔU)

$\therefore$ greater will be absorption of energy.

Hence correct order

$[\text{Ni}(\text{en})_3]^{2+} > [\text{Ni}(\text{NH}_3)_6]^{2+} > [\text{Ni}(\text{H}_2\text{O})_6]^{2+}$

Question 167

The difference between spin only magnetic moment values of $[\text{Co}(\text{H}_2\text{O})_6]\text{Cl}_2$ and $[\text{Cr}(\text{H}_2\text{O})_6]\text{Cl}_3$ is

[26-Jul-2022-Shift-1]

Answer: 0

Solution:

$\text{Co} \rightarrow 4s^2 3d^7$

H_2O is weak field ligand.

$\text{Co}^{+2} \rightarrow 3d^7$

$n = 3$ $\mu_{\text{spin only}} = \sqrt{n(n+2)}$ B.M.

$\text{Cr} \rightarrow 4s^1 3d^5$

$$\text{Co}^{+3} \rightarrow 3d^3$$

$$n = 3 \quad \mu_2 = \sqrt{15} \text{ B.M.}$$

$$\mu_1 - \mu_2 = 0$$

Question168

The metal complex that is diamagnetic is (Atomic number: Fe, 26; Cu, 29)
[26-Jul-2022-Shift-2]

Options:

- A. $\text{K}_3[\text{Cu}(\text{CN})_4]$
- B. $\text{K}_2[\text{Cu}(\text{CN})_4]$
- C. $\text{K}_3[\text{Fe}(\text{CN})_4]$
- D. $\text{K}_4[\text{FeCl}_6]$

Answer: A

Solution:

$\Rightarrow \text{K}_3[\text{Cu}(\text{CN})_4]$ is diamagnetic

$\text{Cu(I)} \Rightarrow d^{10}$ configuration $\Rightarrow$ No unpaired electrons.

$\Rightarrow \text{K}_2[\text{Cu}(\text{CN})_4]$, $\text{K}_3[\text{Fe}(\text{CN})_4]$ and $\text{K}_4[\text{FeCl}_6]$ are paramagnetic in nature

Question169

The conductivity of a solution of complex with formula $\text{CoCl}_3(\text{NH}_3)_4$ corresponds to 1 : 1 electrolyte, then the primary valency of central metal ion is _____.
[27-Jul-2022-Shift-1]

Answer: 3

Solution:

In 1: 1 type of electrolyte the ions have +1 and -1 charge on them

$\therefore$ Possible compound is $\rightarrow [\text{Co}(\text{NH}_3)_4 \text{Cl}_2]^+ \text{Cl}^-$

Oxidation state of central atom represents the total number of primary valency

$\therefore$ Primary valency will be 3 .

Question170

Low oxidation state of metals in their complexes are common when ligands:
[27-Jul-2022-Shift-2]

Options:

- A. have good π -accepting character
- B. have good σ -donor character
- C. are having good π -donating ability

D. are having poor σ -donating ability

Answer: A

Solution:

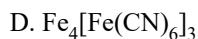
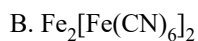
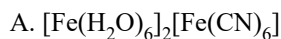
Ligands like :CO , are sigma donor and π -acceptor and they make stronger bond with lower oxidation state metal ion, in this case back bonding is more effective

Question171

Fe^{3+} cation gives a prussian blue precipitate on addition of potassium ferrocyanide solution due to the formation of :

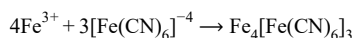
[27-Jul-2022-Shift-2]

Options:



Answer: D

Solution:



Question172

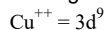
The spin only magnetic moment of the complex present in Fehling's reagent is _____ B.M. (Nearest integer).

[27-Jul-2022-Shift-2]

Answer: 2

Solution:

Fehling solution is a complex of Cu^{++}



No. of unpaired $e^- = 1$

$$\text{M.M} = \sqrt{1(1+2)} = \sqrt{3} = 1.73 \text{ BM}$$

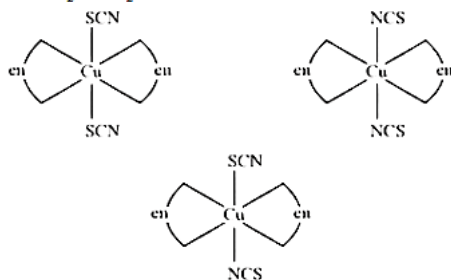
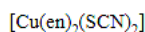
Question173

Total number of relatively more stable isomer(s) possible for octahedral complex $[\text{Cu}(\text{en})_2(\text{SCN})_2]$ will be _____.

[28-Jul-2022-Shift-1]

Answer: 3

Solution:



Question174

Match List I with List II

List - I (Complex)	List - II (Hybridization)
(A) $\text{Ni}(\text{CO})_4$	(I) sp^3
(B) $[\text{Ni}(\text{CN})_4]^{2-}$	(II) sp^3d^2
(C) $[\text{Co}(\text{CN})_6]^{3-}$	(III) $d^2 sp^3$
(D) $[\text{CoF}_6]^{3-}$	(IV) dsp^2

Choose the correct answer from the options given below:
[28-Jul-2022-Shift-2]

Options:

A. A-IV, B-I, C-III, D-II

B. A-I, B-IV, C-III, D-II

C. A-I, B-IV, C-II, D-III

D. A-IV, B-I, C-II, D-III

Answer: B

Solution:

$\text{Ni}(\text{CO})_4$ Hybridisation sp^3

$[\text{Ni}(\text{CN})_4]^{2-}$ Hybridisation dsp^2

$[\text{Co}(\text{CN})_6]^{3-}$ Hybridisation $d^2 sp^3$

$[\text{Co}(\text{F})_6]^{3-}$ Hybridisation sp^3d^2

Question175

$[\text{Fe}(\text{CN})_6]^{3-}$ should be an inner orbital complex. Ignoring the pairing energy, the value of crystal field stabilization energy for this complex is $(-)$ _____ Δ_0 . (Nearest integer)

[29-Jul-2022-Shift-1]

Answer: 2

Solution:

In $[\text{Fe}(\text{CN})_6]^{3-}$, Fe is present in (+3) oxidation state $\text{Fe(III)} \Rightarrow$ inner orbital complex $\Rightarrow d^5$ (with pairing)

Configuration $\Rightarrow t_{2g}^5$

$$\text{CFSE} = 5 \times \frac{-2}{5} \Delta_0 = -2\Delta_0$$

Question176

Octahedral complexes of copper(II) undergo structural distortion (Jahn-Teller). Which one of the given copper (II) complexes will show the maximum structural distortion? (en - ethylenediamine;

. $\text{H}_2\text{N}-\text{CH}_2-\text{CH}_2-\text{NH}_2$)

[29-Jul-2022-Shift-2]

Options:

A. $[\text{Cu}(\text{H}_2\text{O})_6]\text{SO}_4$

B. $[\text{Cu}(\text{en})(\text{H}_2\text{O})_4]\text{SO}_4$

C. cis- $[\text{Cu}(\text{en})_2\text{Cl}_2]$

D. trans- $[\text{Cu}(\text{en})_2\text{Cl}_2]$

Answer: A

Solution:

According to Jahn Teller any nonlinear molecular system in a degenerate electronic state will be unstable and will undergo some kind of distortion which will lower its symmetry and energy and split the degenerate state.

In case of octahedral d^9 configuration, the last electron may occupy either d^2 or $d_{x^2-y^2}$ orbitals of e_g set.

If it occupies dz^2 orbital most of the electron density will be concentrated between the metal and the two ligands on the z axis. Thus there will be greater electrostatic repulsion associated with these ligands than with the other four on xy plane.

The Jahn Teller effect is mostly observed in octahedral environments. The considerable distortions are usually observed in high spin d^4 , low spin d^7 and d^9 configuration.

Question177

Sum of oxidation state (magnitude) and coordination number of cobalt in $\text{Na} \cdot [\text{Co}(\text{bpy})\text{Cl}_4]$ is

[29-Jul-2022-Shift-2]

Answer: 9

Solution:

$\text{Na}[\text{Co}(\text{bpy})\text{Cl}_4]$

Oxidation state of cobalt = +3

Coordination number of cobalt = 6

[As bpy is bidentate]

So, sum = 9

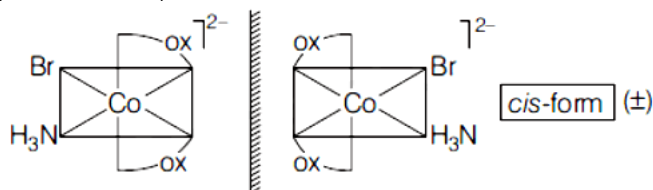
Question178

The number of stereoisomers possible for $[\text{Co}(\text{OX})_2(\text{Br})(\text{NH}_3)]^{2-}$ is [ox = Oxal ate].
[26 Feb 2021 Shift 2]

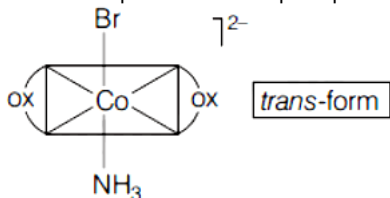
Answer: 3

Solution:

The coordination compound $[\text{Co}(\text{ox})_2(\text{Br})(\text{NH}_3)]^{2-}$ of general formula $[\text{M}(\text{A}-\text{A})_2\text{BC}]$ can show both geometrical and optical isomerism (stereoisomerism).



The cis-form produces non-superimposable mirror images, i.e. enantiomeric pairs (optically active)



The trans-form is optically inactive.

So, total number of stereoisomers possible
= cis(±) + trans = 3

Question179

Given below are two statements.

Statement I The identification of Ni^{2+} is carried out by dimethyl glyoxime in the presence of NH_4OH .

Statement II The dimethyl glyoxime is a bidentate neutral ligand.

In the light of the above statements, choose the correct answer from the options given below.
[25 Feb 2021 Shift 2]

Options:

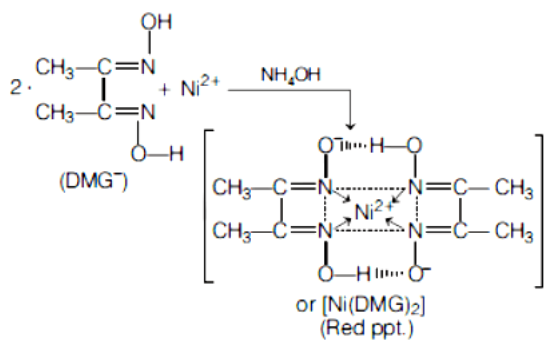
- A. Both statements I and II are true.
- B. Both statements I and II are false.
- C. Statement I is true but statement II is false.
- D. Statement I is false but statement II is true.

Answer: A

Solution:

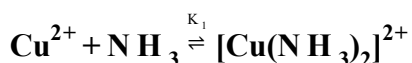
Both statements are true. Dimethyl glyoxime (DMG) is a neutral bidentate ligand (DMG^0).

Ni^{2+} ion is identified with DMG in presence of NH_4OH to give a red ppt. Its reaction is as follows.



Question 180

The stepwise formation of $[\text{Cu}(\text{NH}_3)_4]^{2+}$ is given below



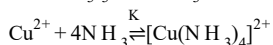
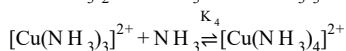
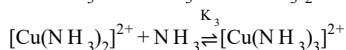
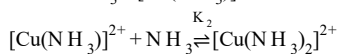
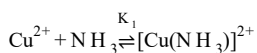
The value of stability constants K_1 , K_2 , K_3 and K_4 are 10^4 , 1.58×10^3 , 5×10^2 and 10^2 respectively. The overall equilibrium constants for dissociation of $[\text{Cu}(\text{NH}_3)_4]^{2+}$ is $x \times 10^{-12}$. The value of x is ____

(Rounded off to the nearest integer)

[24 Feb 2021 Shift 1]

Answer: 1

Solution:



$$\text{So, } K = K_1 \times K_2 \times K_3 \times K_4$$

$$= 10^4 \times 1.58 \times 10^3 \times 5 \times 10^2 \times 10^2$$

$$\therefore K = 7.9 \times 10^{11}$$

Where K = Equilibrium constant for formation of $[\text{Cu}(\text{NH}_3)_4]^{2+}$

So, equilibrium constant (K') for dissociation

of $[\text{Cu}(\text{NH}_3)_4]^{2+}$ is $\frac{1}{K}$

$$K' = \frac{1}{K}$$

$$K' = \frac{1}{7.9 \times 10^{11}} = 1.26 \times 10^{-12} = (x \times 10^{-12})$$

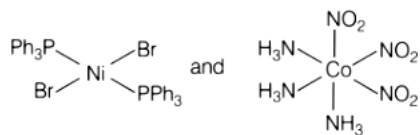
$$\text{So, } x = 1.26 \approx 1.0.$$

Question181

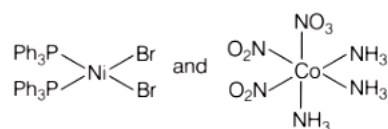
The correct structures of $\text{trans-[NiBr (PPh}_3)_2]$ and meridional- $[\text{Co(NH}_3)_3(\text{NO}_2)_3]$, respectively are [18 Mar 2021 Shift 1]

Options:

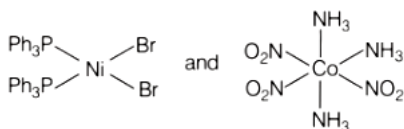
A.



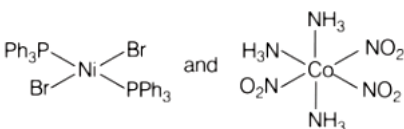
B.



C.



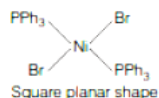
D.



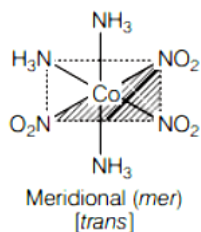
Answer: D

Solution:

Trans $[\text{NiBr}_2(\text{PPh}_3)_2]$



and meridional $[\text{Co(NH}_3)_3(\text{NO}_2)_3]$



Question182

Match List-I and List-II.

List I	List II
A. $[Co(NH_3)_6][Cr(CN)_6]$	1. Linkage isomerism
B. $[Co(NH_3)_3(NO_2)_3]$	2. Solvate isomerism
C. $[Cr(H_2O)_6]Cl_3$	3. Co-ordination isomerism
D. $cis - [CrCl_2(ox)_2]^{3-}$	4. Optical isomerism

[17 Mar 2021 Shift 2]**Options:**

A. A-3 ,B-1 ,C-2 ,D-4

B. A-4 ,B-2 ,C-3 ,D-1

C. A-2 ,B-1 ,C-3 ,D-4

D. A-1 ,B-2 ,C-3 ,D-4

Answer: A**Solution:**

A. Coordination isomerism occurs in coordination salts in which there is an interchange of ligands between the metal in the cationic coordination spheres and the metal in the anionic coordination sphere.

$[Co(NH_3)_6][Cr(CN)_6] \rightleftharpoons [Co(CN)_6][Cr(NH_3)_6]$ (Coordinate isomerism)

B. Linkage isomerism occurs in coordination compounds that have the same composition but differ in the connectivity of the ligand to the metal.

$[Co(NH_3)_3(NO_2)_3]$ shows linkage isomerism. NO_2 can connect through N and O donor sites.

C. Solvate isomerism is also known as 'hydrate isomerism' where water behaves as a ligand and water of crystallisation. e.g., The complex $[Cr(H_2O)_6]Cl_3$ (violet) contains water as a ligand and its solvate isomer $[Cr(H_2O)_5Cl]Cl_2 \cdot H_2O$ (grey-green) contains water as a water of crystallisation.

D. Optical isomerism occurs mainly in compounds that are mirror images of each other. $cis[CrCl_2(ox)_2]^{3-}$ is a chiral compound, hence shows optical isomerism.

Question 183**The correct pair(s) of the ambident nucleophiles is(are)****A. AgCN / K CN****B. RCOOAg / RCOOK****C. AgNO₂ / KNO₂****D. AgI / KI****[17 Mar 2021 Shift 2]****Options:**

A. B and C

B. Only A

C. A and C

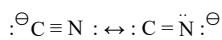
D. Only B

Answer: C**Solution:**

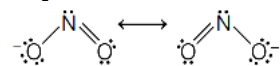
The correct pair(s) of ambident nucleophile are (A) KCN and AgCN and (C) AgNO₂ and KNO₂.

Structure are as follows:

$^-\text{CN}^-$ can bind through C and N atoms.



NO_2^- can bind through N and O.



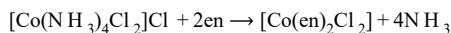
Question184

The equivalents of ethylene diamine required to replace the neutral ligands from the coordination sphere of the trans-complex of $\text{CoCl}_3 \cdot 4\text{NH}_3$ is (Round off to the nearest integer).

[16 Mar 2021 Shift 1]

Answer: 2

Solution:



NH_3 is the neutral monodentate ligand. Ethylene diamine is a neutral didentate ligand.



So, two ethylene diamine are equivalent to four NH_3 ligand.

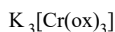
Question185

The total number of unpaired electrons present in the complex $\text{K}_3[\text{Cr}(\text{oxalate})_3]$ is _____.

[18 Mar 2021 Shift 1]

Answer: 3

Solution:



Let, x be the oxidation state of Cr in $\text{K}_3[\text{Cr}(\text{C}_2\text{O}_4)_3]$. Since, the overall charge on the complex is 0, the sum of oxidation states of all elements in it should be equal to 0. Therefore,

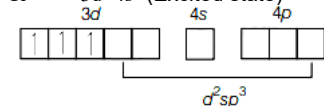
$$+3 + x + 3(-2) = 0 \text{ (oxalate is bidentate ligand and having } -2 \text{ charge)}$$

or, $x = +3$.

Hence, chromium is in +3 oxidation state.

$\text{Cr} \rightarrow 3d^5 4s^1$ (ground state electronic configuration)

$\text{Cr}^{3+} \rightarrow 3d^3 4s^0$ (Excited state)



$$\text{CN} = 6$$

$\therefore$ oxalate is weak field ligand

$\therefore$ 3 unpaired electron in d-orbital. The hybridisation of chromium in the complex is d^2sp^3 .

Question186

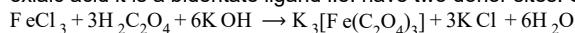
On complete reaction of FeCl_3 with oxalic acid in aqueous solution containing KOH , resulted in the formation of product A. The secondary valency of Fe in the product A is (Round off to the nearest

integer).
[17 Mar 2021 Shift 2]

Answer: 6

Solution:

Secondary valency is equal to coordination number of central metal ion. In this reaction, central metal atom iron is attached with 3-oxalato ligand, and oxalic acid is a bidentate ligand i.e. have two donor sites. So, coordination number or secondary valency = $3 \times 2 = 6$



Hence, product A have iron with secondary valency = 6

Question 187

What is the spin-only magnetic moment value (BM) of a divalent metal ion with atomic number 25, in its aqueous solution?
[17 Mar 2021 Shift 1]

Options:

- A. 5.92
- B. 5.0
- C. zero
- D. 5.26

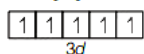
Answer: A

Solution:

Atomic number 25 is for manganese (Mn).

Configuration of Mn = $[\text{Ar}]4s^23d^5$

In aqueous medium, $\text{Mn}^{2+} = [\text{Ar}]3d^5$



Total number of unpaired electrons = 5

μ (magnetic moment) = $\sqrt{n(n+2)}$ BM

where, n = number of unpaired electron

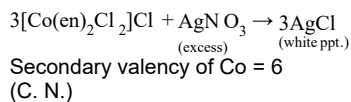
$$\therefore \mu = \sqrt{5(5+2)} = \sqrt{35} \text{ BM} = 5.92 \text{ BM}$$

Question 188

3 moles of metal complex with formula $\text{Co}(\text{en})_2\text{Cl}_3$ gives 3 moles of silver chloride on treatment with excess of silver nitrate. The secondary valency of Co in the complex is _____.
(Round off to the nearest integer)
[27 Jul 2021 Shift 2]

Answer: 6

Solution:



Question189

Given below are two statements:

Statement I : $[\text{Mn}(\text{CN})_6]^{3-}$, $[\text{Fe}(\text{CN})_6]^{3-}$ and $[\text{Co}(\text{C}_2\text{O}_4)_3]^{3-}$ are d^2sp^3 hybridised.

Statement II : $[\text{MnCl}_6]^{3-}$ and $[\text{FeF}_6]^{3-}$ are paramagnetic and have 4 and 5 unpaired electrons, respectively.

In the light of the above statements, choose the correct answer from the options given below :
[27 Jul 2021 Shift 2]

Options:

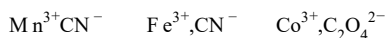
- A. Statement I is correct but statement II is false
- B. Both statement I and statement II are false
- C. Statement I is incorrect but statement II is true
- D. Both statement I and statement II are true

Answer: D

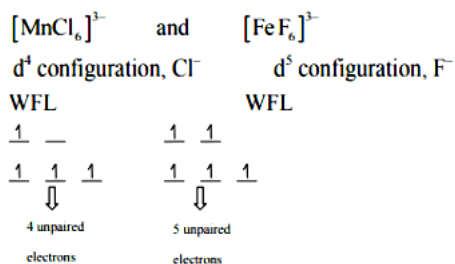
Solution:



↓



d^4 configuration, SFL d^5 configuration, SFL d^6 configuration, Chelating ligand
⇒ All will have larger splitting hence d^2sp^3 hybridisation

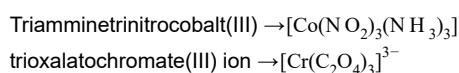


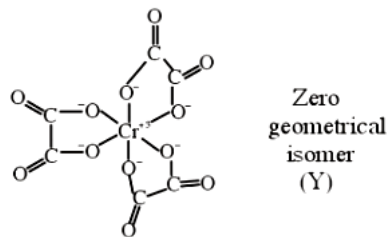
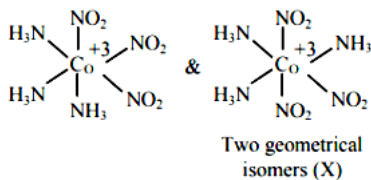
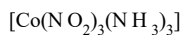
Question190

The number of geometrical isomers possible in triamminetrinitrocobalt (III) is X and in trioxalatochromate (III) is Y. Then the value of X + Y is _____.
[27 Jul 2021 Shift 1]

Answer: 2

Solution:





$$X + Y = 2 + 0 = 2.0$$

Question191

An organic compound is subjected to chlorination to get compound A using 5.0 g of chlorine. When 0.5 g of compound A is reacted with AgNO_3 [Carius Method], the percentage of chlorine in compound A is _____ when it forms 0.3849 g of AgCl . (Round off to the Nearest Integer)
(Atomic masses of Ag and Cl are 107.87 and 35.5 respectively)
[27 Jul 2021 Shift 1]

Answer: 19

Solution:

$$\begin{aligned} n_{\text{Cl}} \text{ in compound} &= n_{\text{AgCl}} = \frac{0.3849\text{g}}{(107.87 + 35.5)} \text{g/mol} \\ \Rightarrow \text{mass of chlorine} &= n_{\text{Cl}} \times 35.5 = 0.0953\text{gm} \\ \Rightarrow \% \text{ wt of chlorine} &= \frac{0.0953}{0.5} \times 100 \\ &= 19.06\% \end{aligned}$$

Question192

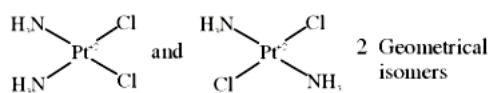
The number of geometrical isomers found in the metal complexes $[\text{PtCl}_2(\text{NH}_3)_2]$, $[\text{Ni}(\text{CO})_4]$, $[\text{Ru}(\text{H}_2\text{O})_3\text{Cl}_3]$ and $[\text{CoCl}_2(\text{NH}_3)_4]^+$ respectively, are :
[27 Jul 2021 Shift 1]

Options:

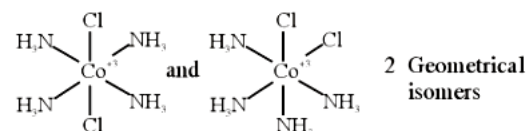
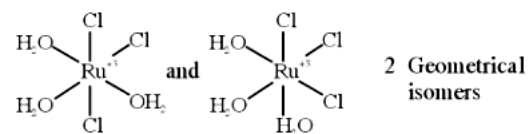
- A. 1, 1, 1, 1
- B. 2, 1, 2, 2
- C. 2, 0, 2, 2
- D. 2, 1, 2, 1

Answer: C

Solution:



$[\text{Ni}(\text{CO})_4] \rightarrow$ All ligands are same Zero Geometrical isomers



Question193

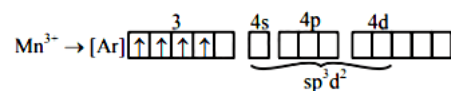
The type of hybridisation and magnetic property of the complex $[\text{MnCl}_6]^{3-}$, respectively, are:
[27 Jul 2021 Shift 1]

Options:

- A. sp^3d^2 and diamagnetic
- B. d^2sp^3 and diamagnetic
- C. d^2sp^3 and paramagnetic
- D. sp^3d^2 and paramagnetic

Answer: D

Solution:



Paramagnetic and having 4 unpaired electrons.

Question194

Which one of the following metal complexes is most stable?
[25 Jul 2021 Shift 2]

Options:

- A. $[\text{Co}(\text{en})(\text{NH}_3)_4]\text{Cl}_2$
- B. $[\text{Co}(\text{en})_3]\text{Cl}_2$
- C. $[\text{Co}(\text{en})_2(\text{NH}_3)_2]\text{Cl}_2$

D. $[\text{Co}(\text{NH}_3)_6]\text{Cl}_2$

Answer: B

Solution:

Complex $[\text{Co}(\text{en})_3]\text{Cl}_2$ is most stable complex among the given complex compounds because more number of chelate rings are present in this complex as compare to others.

(1) $[\text{Co}(\text{en})(\text{NH}_3)_4]\text{Cl}_2$	1 chelate ring
(2) $[\text{Co}(\text{en})_3]\text{Cl}_2$	3 chelate ring
(3) $[\text{Co}(\text{en})_2(\text{NH}_3)_2]\text{Cl}_2$	2 chelate ring
(4) $[\text{Co}(\text{NH}_3)_6]\text{Cl}_2$	0 chelate ring

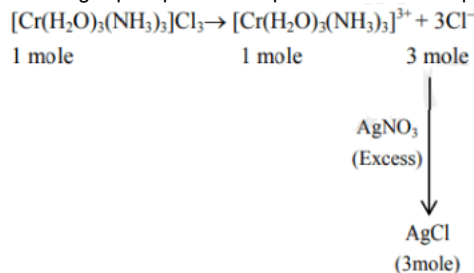
Question195

Three moles of AgCl get precipitated when one mole of an octahedral co-ordination compound with empirical formula $\text{CrCl}_3 \cdot 3\text{NH}_3 \cdot 3\text{H}_2\text{O}$ reacts with excess of silver nitrate. The number of chloride ions satisfying the secondary valency of the metal ion is _____.
[25 Jul 2021 Shift 1]

Answer: 0

Solution:

Mole of AgCl precipitated is equal the mole of Cl^- present in ionization sphere.



Since none of Cl^- is present in the co-ordination sphere. Therefore answer is zero.

Question196

Which one of the following species responds to an external magnetic field?
[25 Jul 2021 Shift 1]

Options:

A. $[\text{Fe}(\text{H}_2\text{O})_6]^{3+}$

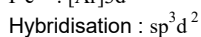
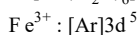
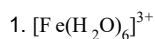
B. $[\text{Ni}(\text{CN})_4]^{2-}$

C. $[\text{Co}(\text{CN})_6]^{3-}$

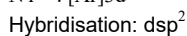
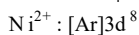
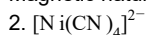
D. $[\text{Ni}(\text{CO})_4]$

Answer: A

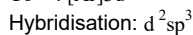
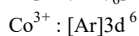
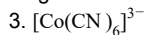
Solution:



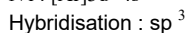
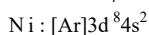
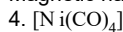
Magnetic nature : Paramagnetic (so this complex response to external magnetic field)



Magnetic nature : diamagnetic



Magnetic nature : diamagnetic



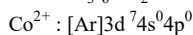
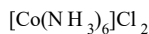
Magnetic nature : diamagnetic

Question197

The total number of unpaired electrons present in $[\text{Co}(\text{NH}_3)_6]\text{Cl}_2$ and $[\text{Co}(\text{NH}_3)_6]\text{Cl}_3$ is
[22 Jul 2021 Shift 2]

Answer: 1

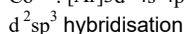
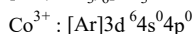
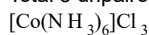
Solution:



For this complex $\Delta_0 < P.E.$, so pairing of electron does not take place.



Total 3 unpaired electrons are present.



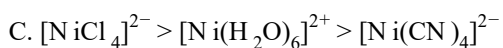
NH_3 acts as SFL because $\Delta_0 > P.E.$

So here all electrons becomes paired.

Question198

The correct order of intensity of colors of the compounds is:
[20 Jul 2021 Shift 1]

Options:



Answer: C

Solution:

	$[\text{NiCl}_4]^{2-} > [\text{Ni}(\text{H}_2\text{O})_6]^{2+} > [\text{Ni}(\text{CN})_4]^{2-}$
Splitting energy order	$\Delta_t < \Delta_0 < \Delta_{\text{sq}}$
absorbed energy order	$[\text{NiCl}_4]^{2-} < [\text{Ni}(\text{H}_2\text{O})_6]^{2+} < [\text{Ni}(\text{CN})_4]^{2-}$
intensity of colour of compound	$[\text{NiCl}_4]^{2-} > [\text{Ni}(\text{H}_2\text{O})_6]^{2+} > [\text{Ni}(\text{CN})_4]^{2-}$

Question199

According to the valence bond theory the hybridization of central metal atom is $d sp^2$ for which one of the following compounds?

[20 Jul 2021 Shift 1]

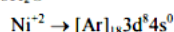
Options:

- A. $\text{NiCl}_2 \cdot 6\text{H}_2\text{O}$
- B. $\text{K}_2[\text{Ni}(\text{CN})_4]$
- C. $[\text{Ni}(\text{CO})_4]$
- D. $\text{Na}_2[\text{NiCl}_4]$

Answer: B

Solution:

1) $\text{NiCl}_2 \cdot 6\text{H}_2\text{O}$



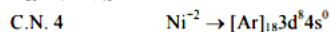
C.N. = 6 octahedral

splitting



Hybridisation
 $sp^3 d^2$

2) $\text{K}_2[\text{Ni}(\text{CN})_4]$

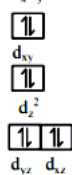


$\text{CN}^- \rightarrow$ Strong field
ligand



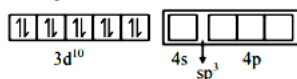
Hybridisation
 dsp^2

Square planar splitting



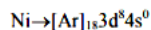
3) $\text{Ni}(\text{CO})_4$

CO - Strong field ligand



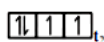
Hybridisation

4) $\text{Na}_2[\text{NiCl}_4]$



$\text{Cl}^- \rightarrow$ weak

field ligand



C.N. 4

tetrahedral
splitting



Hybridisation
 sp^3

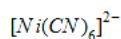
Question200

An aqueous solution of NiCl_2 was heated with excess sodium cyanide in presence of strong oxidizing agent to form $[\text{Ni}(\text{CN})_6]^{2-}$. The total change in number of unpaired electrons on metal centre is

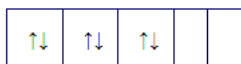
[20 Jul 2021 Shift 2]

Answer: 2

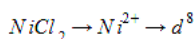
Solution:



$Ni^{+4} \rightarrow d^6$ strong field ligand



Pairing will be there zero unpaired electron



$\rightarrow$ two unpaired e^-

Change = 2

Question201

Spin only magnetic moment in BM of $[Fe(CO)_4(C_2O_4)]^+$ is
[31 Aug 2021 Shift 2]

Options:

- A. 5.92
- B. 0
- C. 1
- D. 1.73

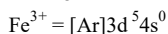
Answer: D

Solution:

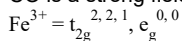
In $[Fe(CO)_4(C_2O_4)]^+$, oxidation number of Fe = +3

$$\{x + 4 \times 0 + (-2) = 1\}$$

$$\therefore x = 3$$



CO is a strong field ligand, so pairing is favoured



Number of unpaired electron (n) = 1

$$\mu = \sqrt{n(n+2)} \text{ BM}$$

$$= \sqrt{1(1+2)} = \sqrt{3} = 1.73 \text{ BM}$$

Question202

The denticity of an organic ligand, biuret is
[31 Aug 2021 Shift 1]

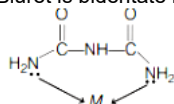
Options:

- A. 2
- B. 4
- C. 3
- D. 6

Answer: A

Solution:

Biuret is bidentate ligand and denticity of this organic ligand is 2.



Hence, correct option is (a).

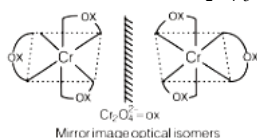
Question203

The number of optical isomers possible for $[\text{Cr}(\text{C}_2\text{O}_4)_3]^{3-}$ is ...
[27 Aug 2021 Shift 2]

Answer: 2

Solution:

The structure of $[\text{Cr}(\text{C}_2\text{O}_4)_3]^{3-}$ is



Oxalate is bidentate symmetrical ligand. The complex does not possess any element of symmetry and are optically active. The two optical isomers are shown above.

∴ Answer is two (2).

Question204

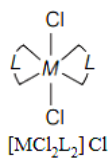
1 mol of an octahedral metal complex with formula $\text{MCl}_3 \cdot 2\text{L}$ on reaction with excess of AgNO_3 gives 1 mol of AgCl . The denticity of ligand L is (Integer answer)
[27 Aug 2021 Shift 1]

Answer: 2

Solution:

Since, the compound is octahedral, the central metal M will form 6 bonds. On reaction with excess of AgNO_3 , only 1 mol of AgCl is precipitated that means only 1Cl^- is present as counter ion and 2Cl^- ions are present in coordinate sphere.

So 'L' should be bidentate to form an octahedral complex along with 2Cl^- ions.



Question205

Indicate the complex/complex ion which did not show any geometrical isomerism.
[26 Aug 2021 Shift 2]

Options:

- A. $[CoCl_2(en)_2]$
- B. $[Co(CN)_5(NC)]^{3-}$
- C. $[Co(NH_3)_3(NO_2)_3]$
- D. $[Co(NH_3)_4Cl_2]^+$

Answer: B

Solution:

- (a) $[CoCl_2(en)_2]$ shows cis-trans isomerism.
- (b) $[Co(CN)_5(NC)]^{3-}$ is of MA_5B type complex, which will not show any isomerism.
- (c) $[Co(NH_3)_3(NO_2)_3]$ shows facial and meridional isomerism.
- (d) $[Co(NH_3)_4Cl_2]^+$ shows geometrical (cis-trans) isomerism. In cis isomer, two Cl ligands are adjacent to each other. In trans isomer two Cl ligands are opposite to each other.
∴ Option (b) is correct.

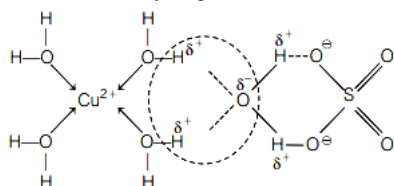
Question206

The number of hydrogen bonded water molecule(s) associated with stoichiometry $CuSO_4 \cdot 5H_2O$ is/are.
[31 Aug 2021 Shift 1]

Answer: 1

Solution:

The number of hydrogen bonded water molecule associated with stoichiometry $CuSO_4 \cdot 5H_2O$ is 1.



Question207

**In the structure of the dichromate ion, there is a
[31 Aug 2021 Shift 1]**

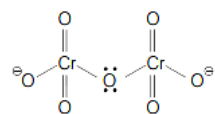
Options:

- A. linear symmetrical Cr – O – Cr bond.
- B. non-linear symmetrical Cr – O – Cr bond.
- C. linear unsymmetrical Cr – O – Cr bond.
- D. non-linear unsymmetrical Cr – O – Cr bond.

Answer: B

Solution:

The structure of dichromate ion ($\text{Cr}_2\text{O}_7^{2-}$) contains a non-linear symmetrical Cr – O – Cr bond.



Hence, correct option is (b).

Question208

**Acidic ferric chloride solution on treatment with excess of potassium ferrocyanide gives a prussian blue coloured colloidal species. It is
[27 Aug 2021 Shift 1]**

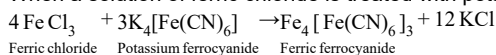
Options:

- A. $\text{Fe}_4[\text{Fe}(\text{CN})_6]_3$
- B. $\text{K}_5\text{Fe}[\text{Fe}(\text{CN})_6]_2$
- C. $\text{HFe}[\text{Fe}(\text{CN})_6]$
- D. $\text{KFe}[\text{Fe}(\text{CN})_6]$

Answer: A

Solution:

When a solution of ferric chloride is treated with potassium ferrocyanide solution, a prussian blue precipitate of ferric ferrocyanide is obtained.



Question209

**The overall stability constant of the complex ion $[\text{Cu}(\text{NH}_3)_4]^{2+}$ is 2.1×10^{13} . The overall dissociations constant is $y \times 10^{-14}$. Then, y is (Nearest integer)
[26 Aug 2021 Shift 2]**

Answer: 5

Solution:

Given, $K_f = 2.1 \times 10^{13}$

Dissociation constant (K_d) = $\frac{1}{K_f} = 4.7 \times 10^{-14}$

$\therefore y = 4.7 \approx 5$.

Question210

Arrange the following cobalt complexes in the order of increasing crystal field stabilisation energy (CFSE) value.

Complexes $[\text{CoF}_6]^{3-}$, $[\text{Co}(\text{H}_2\text{O})_6]^{3+}$, $[\text{Co}(\text{NH}_3)_6]^{3+}$ and $[\text{Co}(\text{en})_3]^{3+}$
(A) (B) (C) (D)

Choose the correct option.

[26 Aug 2021 Shift 2]

Options:

A. $A < B < C < D$

B. $B < A < C < D$

C. $B < C < D < A$

D. $C < D < B < A$

Answer: A

Solution:

For the same central metal ion, the CFSE increases with increase in strength of the ligand. The order of strength of ligand is given by the spectrochemical series:

$\text{I}^- < \text{Br}^- < \text{SCN}^- < \text{Cl}^- < \text{S}^{2-} < \text{F}^- < \text{OH}^-$

$< \text{C}_2\text{O}_4^{2-} < \text{H}_2\text{O} < \text{NCS}^- < \text{EDTA}^{4-} < \text{NH}_3 < \text{en} < \text{CN}^- < \text{CO}$

Thus, the order of CFSE of given complexes will be :

$[\text{CoF}_6]^{3-} < [\text{Co}(\text{H}_2\text{O})_6]^{3+} < [\text{Co}(\text{NH}_3)_6]^{3+} < [\text{Co}(\text{en})_3]^{3+}$.
(A) (B) (C) (D)

Question211

Which one of the following complexes is violet in colour?

[26 Aug 2021 Shift 1]

Options:

A. $[\text{Fe}(\text{CN})_6]^{4-}$

B. $[\text{Fe}(\text{SCN})_6]^{4-}$

C. $\text{Fe}_4[\text{Fe}(\text{CN})_6]_3 \cdot \text{H}_2\text{O}$

D. $[\text{Fe}(\text{CN})_5\text{NOS}]^{4-}$

Answer: D

Solution:

The complex $[\text{Fe}(\text{CN})_5\text{NOS}]^{4-}$ is violet in colour. It is formed when sodium nitroprusside reacts with sulphide ion to give violet colour as follows

$\text{S}^{2-} + [\text{Fe}(\text{CN})_5\text{NO}]^{2-} \rightarrow [\text{Fe}(\text{CN})_5\text{NOS}]^{4-}$
(Violet colour)

Question212

The potassium ferrocyanide solution gives a prussian blue colour, when added to
[1 Sep 2021 Shift 2]

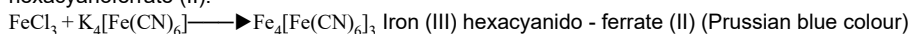
Options:

- A. CoCl_3
- B. FeCl_2
- C. CoCl_2
- D. FeCl_3

Answer: D

Solution:

Potassium ferrocyanide solution $\{\text{K}_4[\text{Fe}(\text{CN})_6]\}$ will give prussian blue colour when added to ferric chloride (FeCl_3) due to formation of iron (III) hexacyanoferrate (II).



Question213

The crystal field stabilisation energy (CFSE) and magnetic moment (spin - only) of an octahedral aqua complex of a metal ion (M^{2+}) are $-0.8\Delta_0$ and 3.87 BM, respectively.

Identify (M^{2+}).

[1 Sep 2021 Shift 2]

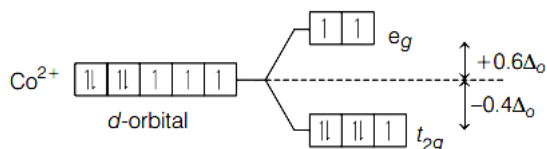
Options:

- A. V^{3+}
- B. Cr^{3+}
- C. Mn^{4+}
- D. Co^{2+}

Answer: D

Solution:

The crystal field stabilisation energy (CFSE) and magnetic moment (spin - only) of Co^{2+} octahedral aqua complex are $-0.8\Delta_0$ and 3.87 BM respectively.



$$\begin{aligned}\text{CFSE} &= [(e^- \text{ in } t_{2g}) \times (-0.4) + (e^- \text{ in } e_g) \times 0.6] \Delta_0 \\ &= [5 \times -0.4 + 2 \times 0.6] \Delta_0 \\ &= -0.8\Delta_0\end{aligned}$$

Magnetic moment (spin only) can be calculated as:

$$\mu = \sqrt{n(n+2)}$$

(where, μ = spin only magnetic moment and

n = number of unpaired electrons = 3

$$= \sqrt{3(3+2)} = \sqrt{15} = 3.87 \text{ BM}$$

Question214

The volume (in mL) of 0.125M AgN O₃ required to quantitatively precipitate chloride ions in 0.3g of [Co(N H₃)₆]Cl₃ is _____.

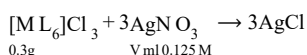
$$M_{[\text{Co}(\text{N H}_3)_6], \text{Cl}_3} = 267.46\text{g/mol}$$

$$M_{\text{AgN O}_3} = 169.87\text{g/mol}$$

[NV, Jan. 08 ,2020(I)]

Answer: 26.92

Solution:



$$\text{Number of moles of the complex} = \frac{0.3}{267.46}$$

$$\text{Number of moles of AgN O}_3 = 0.125 \times V \times 10^{-3}$$

$$\frac{0.3}{267.46} \times 3 = 0.125 \times V \times 10^{-3}$$

$$\text{Or, } V = \frac{0.3 \times 3 \times 1000}{267.46 \times 0.125} = 26.92\text{mL}$$

Question215

The correct order of the calculated spin-only magnetic moments of complexes (A) to (D) is:

(A) N i(CO)₄

(B) [N i(H₂O)₆]Cl₂

(C) N a₂[N i(CN)₄]

(D) P d Cl₂(PPh₃)₂

[Jan. 08, 2020(II)]

Options:

A. (A) $\approx$ (C) < (B) $\approx$ (D)

B. (C) < (D) < (B) < (A)

C. (C) $\approx$ (D) < (B) < (A)

D. (A) $\approx$ (C) $\approx$ (D) < (B)

Answer: D

Solution:

(A) Ni(CO)₄

$$\text{Ni} = 3d^8 4s^2$$

CO is strong field ligand. So pairing of electrons happens.

$$\therefore \text{Number of unpaired electrons} = 0$$

$$\therefore \mu_{\text{spin}} = 0$$

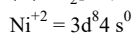
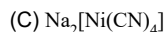
(B) [Ni(H₂O)₆]Cl₂

$$\text{Ni}^{+2} = 3d^8 4s^0$$

H₂O is weak field ligand. So no pairing of electrons happens.

$$\text{Number of unpaired electron} = 2$$

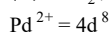
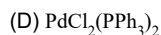
$$\therefore \mu_{\text{spin}} = \sqrt{n(n+2)} = \sqrt{2(2+2)} = \sqrt{8} \text{ B.M}$$



CN^- is strong field ligand. So pairing of electrons happens.

Number of unpaired electron = 0

$\therefore \mu_{\text{spin}} = 0$



This is dsp^2 complex. And shape is square planar.

$\therefore \mu_{\text{spin}} = 0$

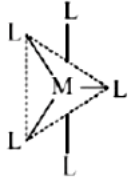
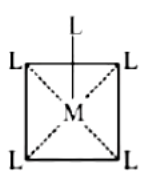
Question216

Complexes (ML_5) of metals Ni and Fe have ideal square pyramidal and trigonal bipyramidal geometries, respectively. The sum of the 90° , 120° and $180^\circ \text{L} - \text{M} - \text{L}$ angles in the two complexes is

[NV, Jan. 08, 2020(II)]

Answer: 20

Solution:

 (Trigonal bipyramidal) $\angle 120^\circ = 3; \angle 90^\circ = 6;$ $\angle 180^\circ = 1$ $\Rightarrow \text{Total} = 10$	 (Square pyramidal) $\angle 90^\circ = 8; \angle 180^\circ = 2$ $\Rightarrow \text{Total} = 10$
--	--

Total number of 180° , 90° and $120^\circ \text{L} - \text{M} - \text{L}$ bond angles = $10 + 10 = 20$

Question217

The theory that can completely\properly explain the nature of bonding in $[\text{Ni}(\text{CO})_4]$ is:

[Jan. 07, 2020(I)]

Options:

- A. Werner's theory
- B. Molecular orbital theory
- C. Crystal field theory
- D. Valence bond theory

Answer: B

Solution:

The covalent character of the bonding ($\text{M} - \text{C}\sigma$ and $\text{M} - \text{C}\pi$ bonding) which exists between the metal and the carbon atom of the CO can only be explained by the molecular orbital theory.

Question218

Complex X of composition $\text{Cr}(\text{H}_2\text{O})_6\text{Cl}_n$ has a spin only magnetic moment of 3.83 BM. It reacts with AgNO_3 and shows geometrical isomerism. The IUPAC nomenclature of X is:

[Jan. 09,2020 (I)]

Options:

- A. Hexaaqua chromium (III) chloride
- B. Tetraaquadichlorido chromium (IV) chloride dihydrate
- C. Dichloridotetraqua chromium (IV) chloride dihydrate
- D. Tetraaquadichlorido chromium(III) chloride dihydrate

Answer: D

Solution:

$$\mu = \sqrt{n(n+2)} \text{ B.M.} = 3.83 \text{ B.M.}$$

$$n = 3 (n = \text{No. of unpaired } e^-)$$

Therefore, oxidation number of Cr should be +3.

Hence complex is $\text{Cr}(\text{H}_2\text{O})_6\text{Cl}_3$

Complex shows geometrical isomerism therefore formula

of complex is $[\text{Cr}(\text{H}_2\text{O})_4\text{Cl}_2]\text{Cl} \cdot 2\text{H}_2\text{O}$

IUPAC Name: Tetraaquadichlorido chromium(III)

Chloride dihydrate

Question219

The isomer(s) of $[\text{Co}(\text{NH}_3)_4\text{Cl}_2]$ that has/have a $\text{Cl} - \text{Co} - \text{Cl}$ angle of 90° , is/are:

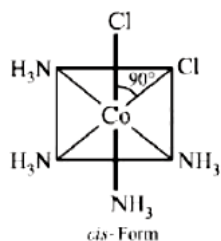
[Jan. 09,2020 (II)]

Options:

- A. meridional and trans
- B. cis and trans
- C. trans only
- D. cis only

Answer: D

Solution:



Question220

The complex that can show fac - and mer-isomers is:

[Jan. 08,2020 (I)]

Options:

- A. $[\text{Co}(\text{NH}_3)_4\text{Cl}_2]^+$
- B. $[\text{Pt}(\text{NH}_3)_2\text{Cl}_2]$
- C. $[\text{CoCl}_2(\text{en})_2]$
- D. $[\text{Co}(\text{NH}_3)_3(\text{NO}_2)_3]$

Answer: D

Solution:

$[\text{Ma}_3\text{b}_3]$ type complex shows facial and meridional isomerism.

So, the complex $[\text{Co}(\text{NH}_3)_3(\text{NO}_2)_3]$ will show fac-and merisomers.

Question221

Among (a) - (d) , the complexes that can display geometrical isomerism are:

- (a) $[\text{Pt}(\text{NH}_3)_3\text{Cl}]^+$
- (b) $[\text{Pt}(\text{NH}_3)_3\text{Cl}]^+$
- (c) $[\text{Pt}(\text{NH}_3)_2\text{Cl}(\text{NO}_2)]$
- (d) $[\text{Pt}(\text{NH}_3)_4\text{ClBr}]^{2+}$

[Jan. 08,2020 (II)]

Options:

- A. (b) and (c)
- B. (d) and (a)
- C. (c) and (d)
- D. (a) and (b)

Answer: C

Solution:

$[\text{Pt}(\text{NH}_3)_2\text{Cl}(\text{NO}_2)]$ and $[\text{Pt}(\text{NH}_3)_4\text{ClBr}]^{2+}$ are the Ma_4bc and Ma_2bc type of complexes.

Each shows two geometrical isomers i.e., cis and trans isomers.

Question222

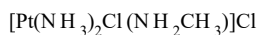
The IUPAC name of the complex $[\text{Pt}(\text{NH}_3)_2\text{Cl}(\text{NH}_2\text{CH}_3)]\text{Cl}$ is:

[Jan. 07,2020 (I)]

Options:

- A. Diamminechlorido(methanamine) platinum (II)chloride
- B. Diammine (methanamine) chloridoplatinum(II)chloride
- C. Diamminechlorido(aminomethane) platinum (II)chloride
- D. Bisammine(methanamine) chlorido platinum(II)chloride

Answer: A

Solution:

Diamine chlorido (methanamine) platinum (II) chloride

Question223

The number of possible optical isomers for the complexes MA_2B_2 with sp^3 and dsp^2 hybridized metal atom, respectively, is:

Note: A and B are unidentate neutral and unidentate monoanionic ligands, respectively.

[Jan. 07,2020 (II)]

Options:

A. 0 and 2

B. 2 and 2

C. 0 and 0

D. 0 and 1

Answer: C

Solution:

$[\text{MA}_2\text{B}_2]$ will not exhibit optical isomerism in both conditions.

Question224

Among the statements (A) – (D), the incorrect ones are:

(A) Octahedral Co(III) complexes with strong field ligands have very high magnetic moments.

(B) When $\Delta_0 < P$, the d -electron configuration of Co(II) in an octahedral complex is $t_{2g}^4 e_g^2$

(C) Wavelength of light absorbed by $[\text{Co(en)}_3]^{3+}$ is lower than that of $[\text{CoF}_6]^{3-}$

(D) If the Δ_0 for an octahedral complex of Co(II) is $18,000 \text{ cm}^{-1}$, the Δ_t for its tetrahedral complex with the same ligand will be $16,000 \text{ cm}^{-1}$.

[Jan. 07,2020 (II)]

Options:

A. (A) and (D) only

B. (C) and (D) only

C. (A) and (B) only

D. (B) and (C) only

Answer: A

Solution:

Co^{3+} with strong field ligand forms complex of low magnetic moment.

$$\Delta_t = \left(\frac{4}{9}\right) \Delta_0$$

$$\Rightarrow \Delta_t = \frac{4}{9} \times 18000 = 8000 \text{ cm}^{-1}$$

Question225

$[\text{Pd}(\text{F})(\text{Cl})(\text{Br})(\text{I})]^{2-}$ has n number of geometrical isomers.

Then, the spin-only magnetic moment and crystal field stabilisation energy [CFSE] of $[\text{Fe}(\text{CN})_6]^{n-6}$, respectively, are:

[Note : Ignore the pairing energy]

[Jan. 09,2020(I)]

Options:

A. 2.84BM and $-1.6\Delta_0$

B. 5.92BM and 0

C. 1.73BM and $-2.0\Delta_0$

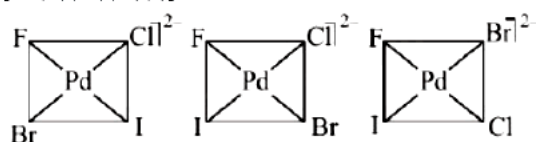
D. 0BM and $-2.4\Delta_0$

Answer: C

Solution:

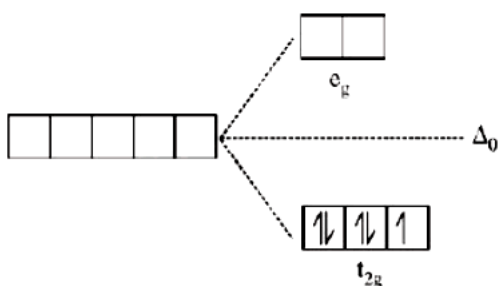
Number of Geometrical isomers (n) in square planar

$$[\text{Pd}(\text{F})(\text{Cl})(\text{Br})(\text{I})]^{2-} = 3$$



$$[\text{Fe}(\text{CN})_6]^{3-6} = [\text{Fe}(\text{CN})_6]^{3-}$$

$$\text{Fe}^{3+} = 3d^5$$



$$\mu = \sqrt{n(n+2)} = 1.73B.M$$

$$CFSE = -0.4 \Delta_0 \times n_{t_{2g}} + 0.6 \Delta_0 \times n_{e_g}$$

$$= -0.4 \Delta_0 \times 5 = -2.0 \Delta_0$$

Question226

The correct order of the spin-only magnetic moments of the following complexes is:

(A) $[\text{Cr}(\text{H}_2\text{O})_6]\text{Br}_2$

(B) $\text{Na}_4[\text{Fe}(\text{CN})_6]$

(C) $\text{Na}_3[\text{Fe}(\text{C}_2\text{O}_4)_3](\Delta_0 > P)$

(D) $(\text{Et}_4\text{N})_2[\text{CoCl}_4]$

[Jan. 09,2020(II)]

Options:

A. (C) > (A) > (D) > (B)

B. $(C) > (A) > (B) > (D)$

C. $(A) > (D) > (C) > (B)$

D. $(B) \approx (A) > (D) > (C)$

Answer: C

Solution:

As, $\mu_s = \sqrt{n(n+2)}$

$\text{Na}_4[\text{Fe}(\text{CN})_6] \rightarrow \text{Fe}^{2+}$

$\Rightarrow \text{S.F.L.} \Rightarrow t_{2g}^6 e_g^0 \Rightarrow \mu_s = 0$

$[\text{Cr}(\text{H}_2\text{O})_6]\text{Br}_2 \rightarrow \text{Cr}^{2+}$

$\Rightarrow \text{W.F.L.} \Rightarrow t_{2g}^3 e_g^1 \Rightarrow \mu_s = \sqrt{24} \text{ B.M.}$

$(\text{Et}_4\text{N})_2[\text{CoCl}_4] \rightarrow \text{Co}^{2+}$

$\Rightarrow \text{W.F.L.} \Rightarrow e^4 t_{2g}^3 \Rightarrow \mu_s = \sqrt{15} \text{ B.M.}$

$\text{Na}_3[\text{Fe}(\text{C}_2\text{O}_4)_3] \rightarrow \text{Fe}^{3+}$

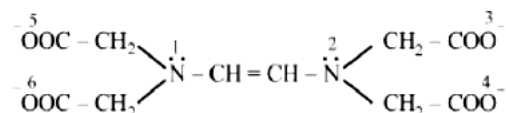
$\Rightarrow \text{S.F.L.} \Rightarrow t_{2g}^5 e_g^0 \Rightarrow \mu_s = \sqrt{3} \text{ B.M.}$

Question 227

The total number of coordination sites in ethylenediaminetetraacetate (EDTA^+) is _____.
[NV, Sep. 05, 2020 (I)]

Answer: 6

Solution:



EDTA^{4-} is a hexadentate ligand, since hexa means six and the ligand attaches six times.

Question 228

Consider the complex ions, $\text{trans-}[\text{Co}(\text{en})_2\text{Cl}_2]^+(A)$ and $\text{cis-}[\text{Co}(\text{en})_2\text{Cl}_2]^+(B)$. The correct statement regarding them is :
[Sep. 05, 2020 (II)]

Options:

A. both (A) and (B) cannot be optically active.

B. (A) can be optically active, but (B) cannot be optically active.

C. both (A) and (B) can be optically active.

D. (A) cannot be optically active, but (B) can be optically active.

Answer: D

Solution:

$\text{trans-[Co(en)}_2\text{Cl}_2\text{)]}^+$ (A) contains a plane of symmetry so (A) cannot be optically active, whereas, (B) $\text{cis-[Co(en)}_2\text{Cl}_2\text{)]}^-$ does not contain any plane of symmetry, so (B) will be optically active.

Question229

The number of isomers possible for $[\text{Pt(en)(NO}_2)_2]$ is

[Sep. 04, 2020 (I)]

Options:

A. 2

B. 4

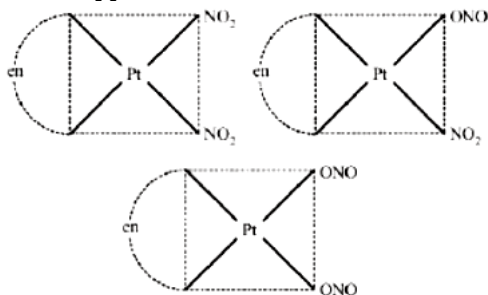
C. 1

D. 3

Answer: D

Solution:

$[\text{Pt(en)(NO}_2)_2]$



So, total possible isomers are 3.

Question230

The complex that can show optical activity is:

[Sep. 03,2020 (I)]

Options:

A. $\text{trans-[CrCl}_2\text{(ox)}_2\text{)]}^3$

B. $\text{trans-[Fe(NH}_3\text{)}_2\text{(CN)}_4\text{]}$

C. $\text{cis-[Fe(NH}_3\text{)}_2\text{(CN)}_4\text{]}$

D. $\text{cis-[CrCl}_2\text{(ox)}_2\text{)]}^{3-}$ (ox = oxalate)

Answer: D

Solution:

Only cis- $[\text{CrCl}_2\text{(ox)}_2]^{3-}$ shows optical isomerism while its trans form do not show optical isomerism due to presence of plane of symmetry.

Question231

Complex A has a composition of $\text{H}_{12}\text{O}_6\text{Cl}_3\text{Cr}$. If the complex on treatment with conc. H_2SO_4 loses 13.5% of its original mass, the correct molecular formula of A is:

[Given : atomic mass of Cr = 52 amu and Cl = 35 amu]

[Sep. 03,2020 (II)]

Options:

- A. $[\text{Cr}(\text{H}_2\text{O})_6]\text{Cl}_3$
- B. $[\text{Cr}(\text{H}_2\text{O})_3\text{Cl}_3] \cdot 3\text{H}_2\text{O}$
- C. $[\text{Cr}(\text{H}_2\text{O})_5\text{Cl}]\text{Cl}_2 \cdot \text{H}_2\text{O}$
- D. $[\text{Cr}(\text{H}_2\text{O})_4\text{Cl}_2]\text{Cl} \cdot 2\text{H}_2\text{O}$

Answer: D

Solution:

Conc. H_2SO_4 acts as dehydrating agent. Molar mass of given complex = 266.5g/mol On treating with conc. H_2SO_4 the mass lost by the complex
 $= \frac{13.5}{100}(266.5) \approx 36\text{g} = 2\text{mol of H}_2\text{O}$

Formula of the complex = $[\text{Cr}(\text{H}_2\text{O})_4\text{Cl}_2]\text{Cl} \cdot 2\text{H}_2\text{O}$

Question232

The one that is not expected to show isomerism is:

[Sep. 02,2020 (II)]

Options:

- A. $[\text{Ni}(\text{NH}_3)_4(\text{H}_2\text{O})_2]^{2+}$
- B. $[\text{Ni}(\text{en})_3]^{2+}$
- C. $[\text{Ni}(\text{NH}_3)_2\text{Cl}_2]$
- D. $[\text{Pt}(\text{NH}_3)_2\text{Cl}_2]$

Answer: C

Solution:

(a) $[\text{Ni}(\text{NH}_3)_4(\text{H}_2\text{O})_2]^{2+}$ shows geometrical isomerism.

(b) $[\text{Ni}(\text{en})_3]^{2+}$ shows optical isomerism.

(c) $[\text{Ni}^{2+}(\text{NH}_3)_2\text{Cl}_2] \Rightarrow \text{Ni}^{2+} \Rightarrow 3d^8 4s^0$

$\Rightarrow sp^3$ hybridisation

$\Rightarrow$ tetrahedral

So, $[\text{Ni}(\text{NH}_3)_2\text{Cl}_2]$ does not show isomerism.

(d) $[\text{Pt}(\text{NH}_3)_2\text{Cl}_2]$ shows geometrical isomerism.

Question233

The species that has a spin-only magnetic moment of 5.9 BM, is : $T_d =$ tetrahedral)

[Sep. 06, 2020 (I)]

Options:

A. $[\text{Ni}(\text{CN})_4]^{2-}$ (square planar)

B. $[\text{NiCl}_4]^{2-}(\text{T}_d)$

C. $\text{Ni}(\text{CO})_4(\text{T}_d)$

D. $[\text{MnBr}_4]^{2-}(\text{T}_d)$

Answer: D

Solution:



[$\because$ Br is weak field ligand]

$d^5(\text{T}_d)$ is high spin complex.

So, $\mu = \sqrt{5(5+2)} = 5.91 \text{ B.M.}$

Question 234

For a d^4 metal ion in an octahedral field, the correct electronic configuration is :
[Sep. 06, 2020 (II)]

Options:

A. $t_{2g}^3 e_g^1$ when $\Delta_0 < P$

B. $t_{2g}^3 e_g^1$ when $\Delta_0 > P$

C. $t_{2g}^4 e_g^0$ when $\Delta_0 < P$

D. $e_g^2 t_{2g}^2$ when $\Delta_0 < P$

Answer: A

Solution:

If $\Delta_0 > P$ then for $d^4 \Rightarrow t_{2g}^4 e_g^0$

If $\Delta_0 < P$ then for $d^4 \Rightarrow t_{2g}^3 e_g^1$

Question 235

The values of the crystal field stabilization energies for a high spin d^6 metal ion in octahedral and tetrahedral fields, respectively, are:
[Sep. 05, 2020 (I)]

Options:

A. $-0.4\Delta_0$ and $-0.6\Delta_t$

B. $-2.4\Delta_0$ and $-0.6\Delta_t$

C. $-1.6\Delta_0$ and $-0.4\Delta_t$

D. $-0.4\Delta_0$ and $-0.27\Delta_t$

Answer: A

Solution:

For d^6 configuration, high spin complex.

(i) In case of octahedral field, $t_{2g}^4 e_g^2$

$$\begin{aligned} \text{CF SE} &= [-0.4p + 0.6q] \Delta_0 + n(P) \\ &= [-0.4 \times 4 + 0.6 \times 2] \Delta_0 + 0 = -0.4\Delta_0 \end{aligned}$$

(ii) In case of tetrahedral field, e^3, t_2^3

$$\begin{aligned} \text{CF SE} &= [-0.6p + 0.4q] \Delta_t \\ &= [-0.6 \times 3 + 0.4 \times 3] \Delta_t = -0.6 \Delta_t. \end{aligned}$$

Question 236

Considering that $\Delta_0 > P$, the magnetic moment (in BM) of $[\text{Ru}(\text{H}_2\text{O})_6]^{2+}$ would be _____.
[NV Sep. 05, 2020 (II)]

Answer: 0

Solution:

$\text{Ru}^{2+} = 4d^6 = t_{2g}^6 e_g^0$ since $\Delta_0 > P$

No. of unpaired electrons = 0

$\therefore$ Magnetic moment = 0 B.M.

Question 237

The pair in which both the species have the same magnetic moment (spin only) is:
[Sep. 04, 2020 (I)]

Options:

A. $[\text{Cr}(\text{H}_2\text{O})_6]^{2+}$ and $[\text{Fe}(\text{H}_2\text{O})_6]^{2+}$

B. $[\text{Co}(\text{OH})_4]^{2-}$ and $[\text{Fe}(\text{NH}_3)_6]^{2+}$

C. $[\text{Mn}(\text{H}_2\text{O})_6]^{2+}$ and $[\text{Cr}(\text{H}_2\text{O})]^{2+}$

D. $[\text{Cr}(\text{H}_2\text{O})_6]^{2+}$ and $[\text{CoCl}_4]^{2-}$

Answer: A

Solution:

$[\text{Cr}(\text{H}_2\text{O})_6]^{2+} \Rightarrow \text{Cr}^{2+} \Rightarrow 3d^4 \Rightarrow t_{2g}^{1,1,1}, e_g^{1,0}$

No. of unpaired e : 4

$[\text{Fe}(\text{H}_2\text{O})_6]^{2+} \Rightarrow \text{Fe}^{2+} \Rightarrow 3d^6 \Rightarrow t_{2g}^{2,1,1}, e_g^{1,1}$

No. of unpaired e : 4

$[\text{Fe}(\text{NH}_3)_6]^{2+} \Rightarrow \text{Fe}^{2+} \Rightarrow 3d^6 \Rightarrow t_{2g}^{2,1,1}, e_g^{1,1}$

No. of unpaired e : 4

$[\text{Co}(\text{OH})_4]^{2-} \Rightarrow \text{Co}^{2+} \Rightarrow 3d^7 \Rightarrow e^{2,2}, t_2^{1,1,1}$

No. of unpaired e : 3

$[\text{CoCl}_4]^{2-} \Rightarrow \text{Co}^{2+} \Rightarrow 3d^7 \Rightarrow e^{2,2}, t_2^{1,1,1}$

No. of unpaired e : 3

$[\text{Mn}(\text{H}_2\text{O})_6]^{2+} \Rightarrow \text{Mn}^{2+} \Rightarrow 3d^5 \Rightarrow t_{2g}^{1,1,1}, e_g^{1,1}$

No. of unpaired e : 5

So $[\text{Cr}(\text{H}_2\text{O})_6]^{2+}$ and $[\text{Fe}(\text{H}_2\text{O})_6]^{2+}$ have same magnetic moment (spin only).

Question238

The Crystal Field Stabilization Energy (CFSE) of $[\text{CoF}_3(\text{H}_2\text{O})_3](\Delta_0 < P)$ is :
[Sep. 04, 2020 (II)]

Options:

- A. $-0.8 \Delta_0 + 2P$
- B. $-0.4\Delta_0$
- C. $-0.8\Delta_0$
- D. $-0.4 \Delta_0 + P$

Answer: B

Solution:

$$\begin{aligned}[\text{CoF}_3(\text{H}_2\text{O})_3] &\Rightarrow \text{Co}^{3+} \Rightarrow d^6 \text{ or } t_{2g}^4 e_g^2 \\ \text{CF SE} &= -[0.4p + 0.6q] \Delta_0 + n(P) \\ &= [-4 \times 0.4 + 2 \times 0.6] \Delta_0 + 0 = -0.4\Delta_0\end{aligned}$$

Question239

The one that can exhibit highest paramagnetic behaviour among the following is :
gly = glycinato; bpy = 2, 2'-bipyridine
[Sep. 04, 2020 (II)]

Options:

- A. $[\text{Pd}(\text{gly})_2]$
- B. $[\text{Fe}(\text{en})(\text{bpy})(\text{NH}_3)_2]^{2+}$
- C. $[\text{Co}(\text{OX})_2(\text{OH})_2](\Delta_0 > P)$
- D. $[\text{Ti}(\text{NH}_3)_6]^{3+}$

Answer: C

Solution:

No, of unpaired electrons	
$[\text{Fe}(\text{en})(\text{bpy})(\text{NH}_3)_2]^{2+}$	$\text{Fe}^{2+} - 3d^6 0$
$[\text{Pd}(\text{gly})_2]$	$\text{Pd}^{2+} - 3d^8 0$
$[\text{Co}(\text{OX})_2(\text{OH})_2]$	$\text{Co}^{5+} - 3d^4 2$
$[\text{Ti}(\text{NH}_3)_6]^{3+}$	$\text{Ti}^{3+} - 3d^1 1$

Thus, $[\text{Co}(\text{OX})_2(\text{OH})_2]$ exhibits highest paramagnetic behaviour due to highest number of unpaired electrons.

Question240

The electronic spectrum of $[\text{Ti}(\text{H}_2\text{O})_6]^{3+}$ shows a single broad peak with a maximum at $20,300\text{cm}^{-1}$.

The crystal field stabilization energy (CFSE) of the complex ion, in kJ mol^{-1} , is:

$$\left(\text{kJ mol}^{-1} = 83.7\text{cm}^{-1} \right)$$

[Sep. 03, 2020 (I)]

Options:

A. 145.5

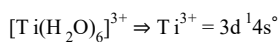
B. 242.5

C. 83.7

D. 97

Answer: D

Solution:



$\therefore$ Electronic configuration is $t_{2g}^1 e_g^0$

$$\text{CFSE} = [-0.4n_{t_{2g}} + 0.6n_{e_g}] \Delta_0 + n(p)$$

$$= [(-0.4) \times 1 + 0] 20300$$

$$= -8120\text{cm}^{-1}$$

$$= \frac{-8120}{83.7} \text{kJ/mol} = -97 \text{kJ/mol}$$

Question241

The d -electron configuration of $[\text{Ru}(\text{en})_3]\text{Cl}_2$ and $[\text{Fe}(\text{H}_2\text{O})_6]\text{Cl}_2$, respectively are:

[Sep. 03, 2020 (II)]

Options:

A. $t_{2g}^6 e_g^0$ and $t_{2g}^6 e_g^0$

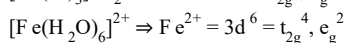
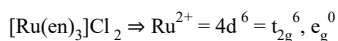
B. $t_{2g}^4 e_g^2$ and $t_{2g}^6 e_g^0$

C. $t_{2g}^6 e_g^0$ and $t_{2g}^4 e_g^2$

D. $t_{2g}^4 e_g^2$ and $t_{2g}^4 e_g^2$

Answer: C

Solution:



So, correct answer is (c).

Question242

Consider that a d^6 metal ion (M^{2+}) forms a complex with aqua ligands, and the spin only magnetic moment of the complex is 4.90BM . The geometry and the crystal field stabilization energy of the complex is:

[Sep. 02, 2020 (I)]

Options:

- A. octahedral and $-2.4 \Delta_0 + 2P$
- B. tetrahedral and $-0.6\Delta_t$
- C. octahedral and $-1.6\Delta_0$
- D. tetrahedral and $-1.6 \Delta_t + 1P$

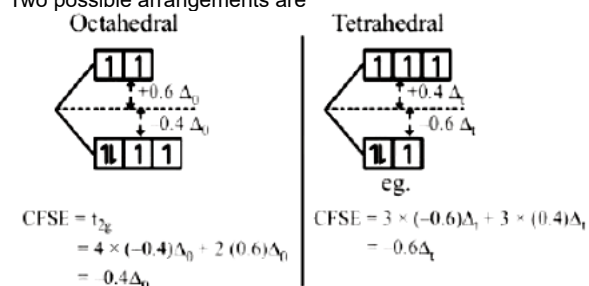
Answer: B

Solution:

Spin only magnetic moment $= 4 \cdot 9 = \sqrt{n(n+2)}$

$\therefore$ number of unpaired electrons $= 4$

Two possible arrangements are



So, option (b) is correct.

Question243

For octahedral $Mn(II)$ and tetrahedral $Ni(II)$ complexes, consider the following statements:

- (I) both the complexes can be high spin.
- (II) $Ni(II)$ complex can very rarely be low spin.
- (III) with strong field ligands, $Mn(II)$ complexes can be low spin.
- (IV) aqueous solution of $Mn(II)$ ions is yellow in color.

The correct statements are:

[Sep. 02, 2020 (I)]

Options:

- A. (I) and (II) only
- B. (I), (III) and (IV) only
- C. (I), (II) and (III) only
- D. (II), (III) and (IV) only

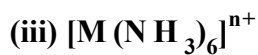
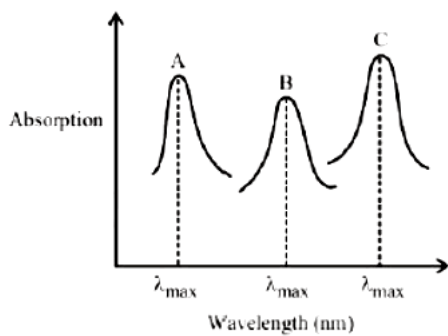
Answer: C

Solution:

With weak field ligands $Mn(II)$ will be of high spin and with strong field ligands it will be of low spin. $Ni(II)$ tetrahedral complexes will be generally of high spin due to sp^3 hybridisation. $Mn(II)$ is of light pink colour in aqueous solution.

Question244

Simplified absorption spectra of three complexes ((i), (ii) and (iii)) of M^{n+} ion are provided below; their λ_{max} values are marked as A, B and C respectively. The correct match between the complexes and their λ_{max} values is :



[Sep. 02, 2020 (II)]

Options:

A. A – (iii), B – (i), C – (ii)

B. A – (ii), B – (i), C – (iii)

C. A – (ii), B – (iii), C – (i)

D. A – (i), B – (ii), C – (iii)

Answer: A

Solution:

Stronger the ligand greater is splitting of d orbitals and smaller will be wavelength of light absorbed. According to spectrochemical series, the splitting power of ligands is $NH_3 > NCS^- > F^-$.

Thus, order of wavelength of light absorbed is

$$\lambda_{NH_3} < \lambda_{NCS^-} < \lambda_{F^-}$$

Question245

The coordination number of Th in $K_4[Th(C_2O_4)_4(H_2O)_2]$ is: ($C_2O_4^{2-}$ = oxalato)

[Jan. 11, 2019 (II)]

Options:

A. 14

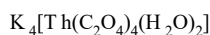
B. 6

C. 8

D. 10

Answer: D

Solution:



$C_2O_4^{2-}$ (oxalato): bidentate ligand

H_2O (aqua): monodentate

$$\therefore \text{Co-ordination no. of Th} = 2 \times 4 + 2 = 10$$

Question246

The total number of isomers for a square planar complex $[M(F)(Cl)(SCN)(NO_2)]$ is:

[Jan. 10, 2019(I)]

Options:

- A. 16
- B. 8
- C. 4
- D. 12

Answer: D

Solution:

Possible complexes No. of isomers
 $[M(F)(Cl)(SCN)(NO_2)] \rightarrow 3$
 $[M(F)(Cl)(SCN)(ONO)] \rightarrow 3$
 $[M(F)(Cl)(NCS)(NO_2)] \rightarrow 3$
 $[M(F)(Cl)(NCS)(ONO)] \rightarrow 3$
Total = 12

Question 247

A reaction of cobalt(III) chloride and ethylenediamine in a 1: 2 mole ratio generates two isomeric products A (violet coloured) and B (green coloured). A can show optical activity, but, B is optically inactive. What type of isomers does A and B represent?

[Jan. 10, 2019(II)]

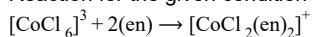
Options:

- A. Geometrical isomers
- B. Coordination isomers
- C. Linkage isomers
- D. Ionisation isomers

Answer: A

Solution:

Reaction for the given condition can be written as:



(1 : 2 mole ratio) (cis-trans-isomer)

A = optically active (cis-isomer), violet

B = optically inactive (trans-isomer), green

Question 248

The metal d -orbitals that are directly facing the ligands in $K_3[Co(CN)_6]$ are :

[Jan.12, 2019(I)]

Options:

- A. d_{xy} and $d_{x^2-y^2}$
- B. $d_{x^2-y^2}$ and d_{z^2}

C. d_{xz} , d_{yz} and d_{z^2}

D. d_{xy} , d_x and d_{yz}

Answer: B

Solution:

$K_3[Co(CN)_6]$ is an octahedral complex.

During splitting of d orbitals in octahedral complexes, $d_{x^2-y^2}$ and d_{z^2} orbitals point towards the direction of ligands (i.e. they experience more repulsion and will be raised in energy by $\frac{3}{5}\Delta_0$).

Question249

The magnetic moment of an octahedral homoleptic Mn(II) complex is 5.9BM . The suitable ligand for this complex is:

[Jan.12, 2019(I)]

Options:

A. Ethylenediamine

B. CN

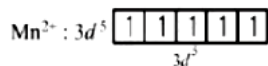
C. NCS

D. CO

Answer: C

Solution:

Electronic configuration of Mn^{2+} is,



Presence of 5 unpaired e^- shows that the complex of Mn^{2+} has only weak field ligand (NCS⁻).

Question250

Match the metals (column I) with the coordination compound(s)/enzyme(s) (column II) :

(column I)	(column II)
Metals	Coordination compound(s)/enzyme(s)
(A) Co	(i) Wilkinson catalyst
(B) Zn	(ii) Chlorophyll
(C) Rh	(iii) Vitamin B ₁₂
(D) Mg	(iv) Carbonic anhydrase

[Jan.11, 2019(I)]

Options:

A. (A) – (iii); (B) – (iv); (C) – (i); (D) – (ii)

B. (A) – (i); (B) – (ii); (C) – (iii); (D) – (iv)

C. (A) – (ii); (B) – (i); (C) – (iv); (D) – (iii)

D. (A) – (iv); (B) – (iii); (C) – (i); (D) – (ii)

Answer: A

Solution:

Wilkinson catalyst: $[\text{Rh}(\text{PPh})_3\text{Cl}]$

Chlorophyll: $\text{C}_{55}\text{H}_{72}\text{O}_5\text{N}_4\text{Mg}$

Vitamin B_{12} contains Co.

Carbonic anhydrase contains a Zn ion.

Question251

The difference in the number of unpaired electrons of a metal ion in its high-spin and low-spin octahedral complexes is two. The metal ion is:
[Jan.10, 2019(II)]

Options:

A. Ni^{2+}

B. Fe^{2+}

C. Co^{2+}

D. Mn^{2+}

Answer: C

Solution:

Metal ion	High spin	Low spin	Difference in the unpaired electrons
$\text{Ni}^{2+}(3d^8)$	2	2	0
$\text{Mn}^{2+}(3d^5)$	5	1	4
$\text{Fe}^{2+}(3d^6)$	4	0	4
$\text{Co}^{2+}(3d^7)$	3	1	2

Question252

Two complexes $[\text{Cr}(\text{H}_2\text{O})_6]\text{Cl}_3$ (A) and $[\text{Cr}(\text{NH}_3)_6]\text{Cl}_3$ (B) are violet and yellow coloured, respectively. The incorrect statement regarding them is:
[Jan.9, 2019(I)]

Options:

A. Δ_0 values of (A) and (B) are calculated from the energies of violet and yellow light, respectively.

B. both are paramagnetic with three unpaired electrons.

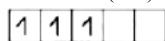
C. both absorb energies corresponding to their complementary colors.

D. Δ_0 value for (A) is less than that of (B).

Answer: A

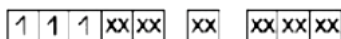
Solution:

E.C. of $\text{Cr}^{3+}(3d^3)$:



$3d$

For complex A $[\text{Cr}(\text{H}_2\text{O})_2]^{3+}$



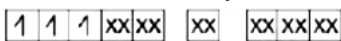
$3d$

$4s$

$4p$

d^2sp^3 hybridisation

For complex B $[\text{Cr}(\text{NH}_3)_6]^{3+}$:



$3d$

$4s$

$4p$

d^2sp^3 hybridisation

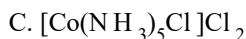
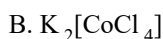
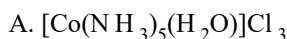
Here, both the complexes (A) and (B) are paramagnetic with 3 unpaired electrons each. H_2O is also a weak field ligand, which causes lesser splitting than NH_3 , which is comparatively stronger field ligand. Hence, the (Δ_0) value of (A) and (B) are calculated from the wavelengths of light absorbed and not from the wavelengths of light emitted.

Question253

The complex that has highest crystal field splitting energy (Δ) is:

[Jan.9, 2019(II)]

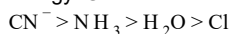
Options:



Answer: D

Solution:

In case of similar metal atom or ion, the value of coordination number and the strength of the ligands determine the value of crystal field splitting energy. Greater the co-ordination number and strength of value of the ligand, greater will be the value of CFSE. Strength of ligands:



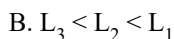
$\therefore \text{K}_3[\text{Co}(\text{CN})_6]$ has the highest crystal field splitting energy.

Question254

Homoleptic octahedral complexes of a metal ion M^{3+} with three monodentate ligands L_1 , L_2 and L_3 absorb wavelengths in the region of green, blue and red respectively. The increasing order of the ligand strength is:

[Jan.9, 2019(II)]

Options:



D. $L_2 < L_1 < L_3$

Answer: A

Solution:

Lesser the wavelength of light absorbed (more energy) greater will be ligand strength.

Energy : Blue > Green > Red

Ligand strength: $L_2 > L_1 > L_3$

Question 255

$Mn_2(CO)_{10}$ is an organometallic compound due to the presence of :

[Jan. 12, 2019(I)]

Options:

A. Mn – C bond

B. Mn-Mn bond

C. Mn – O bond

D. C-O bond

Answer: A

Solution:

Compounds having at least one carbon metal (M – C) bond are known as organometallic compounds. It contains Mn – C bond.

Question 256

The number of bridging CO ligand(s) and Co-Co bond(s) in $Co_2(CO)_8$, respectively are:

[Jan. 11, 2019(I)]

Options:

A. 2 and 1

B. 2 and 0

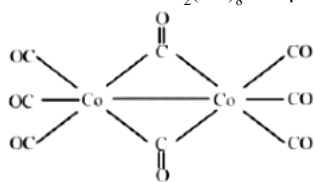
C. 0 and 2

D. 4 and 0

Answer: A

Solution:

The structure of $Co_2(CO)_8$ is represented as



It contains two bridging CO ligands and one metal-metal (Co – Co) bond.

Question 257

Wilkinson catalyst is:
[Jan. 10,2019(I)]

Options:

- A. $[(\text{Ph}_3\text{P})_3\text{IrCl}]$
- B. $[(\text{Et}_3\text{P})_3\text{RhCl}]$
- C. $[(\text{Ph}_3\text{P})_3\text{RhCl}]$
- D. $[(\text{Et}_3\text{P})_3\text{IrCl}]$

Answer: C

Solution:

Wilkinson's catalyst is $[\text{Rh}(\text{PPh}_3)_3\text{Cl}]$

Question258

The coordination numbers of Co and Al in $[\text{Co}(\text{Cl})(\text{en})_2]\text{Cl}$ and $\text{K}_3[\text{Al}(\text{C}_2\text{O}_4)_3]$, respectively, are:
(en = ethane- 1,2 -diamine)
[April 12,2019 (II)]

Options:

- A. 5 and 3
- B. 3 and 3
- C. 6 and 6
- D. 5 and 6

Answer: D

Solution:

$[\text{Co}(\text{en})_2\text{Cl}]\text{Cl}$

Cl^- - monodentate ligand

en - bidentate ligand

$\therefore$ Co-ordination Number of Co = $(2 \times 2) + 1 = 5$

$\text{K}_3[\text{Al}(\text{C}_2\text{O}_4)_3]$

$\text{C}_2\text{O}_4^{2-}$ - bidentate ligand

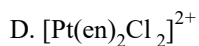
$\therefore$ Co-ordination Number of Al = $2 \times 3 = 6$

Question259

The species that can have a trans-isomer is:
(en = ethane –1, 2– diamine, Ox = oxalate)
[April 10,2019 (I)]

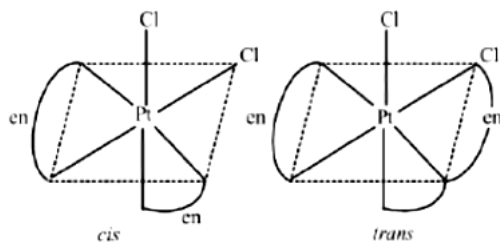
Options:

- A. $[\text{Zn}(\text{en})\text{Cl}_2]$
- B. $[\text{Pt}(\text{en})\text{Cl}_2]$
- C. $[\text{Cr}(\text{en})_2(\text{Ox})]^+$



Answer: D

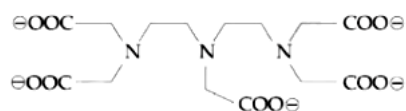
Solution:



cis-trans Isomerism is possible with $[\text{Pt}(\text{en})_2\text{Cl}_2]^{2+}$. $[\text{Cr}(\text{en})_2\text{Ox}]^+$ shows only optical isomerism. The other two complexes, i.e., $[\text{Pt}(\text{en})\text{Cl}_2]$ and $[\text{Zn}(\text{en})\text{Cl}_2]$ do not show stereoisomerism.

Question260

The maximum possible denticities of a ligand given below towards a common transition and inner-transition metal ion, respectively, are:



[April 9,2019 (II)]

Options:

- A. 8 and 6
- B. 6 and 8
- C. 6 and 6
- D. 8 and 8

Answer: B

Solution:

The maximum possible denticities of the given ligand towards transition metal ion is 6 and towards inner transition metal ion (due to greater ionic radii and more atomic orbitals) is 8 .

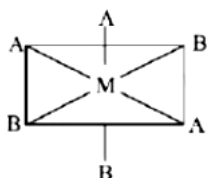
Question261

The one that will show optical activity is:
(en = ethane 1,2 -diamine)

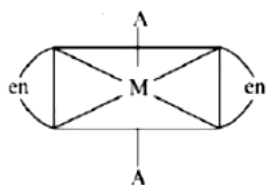
[April 9,2019 (I)]

Options:

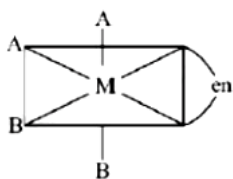
A.



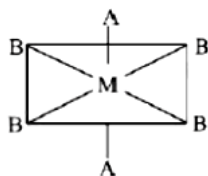
B.



C.

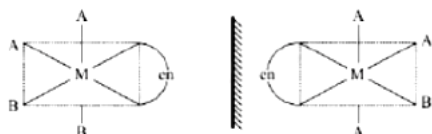


D.



Answer: C

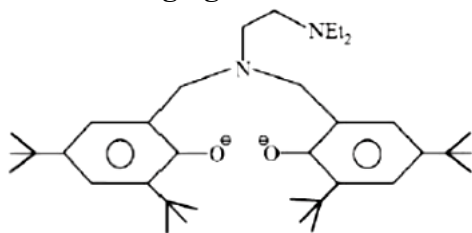
Solution:



No plane of symmetry or centre of symmetry is present. Hence, it will be optically active.

Question262

The following ligand is:



[April 8,2019 (I)]

Options:

- A. hexadentate
- B. tetradentate
- C. bidentate
- D. tridentate

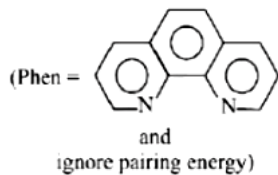
Answer: B

Solution:

It has four atoms containing lone pair of e^- , therefore, it will be able to donate these lone pairs and acts as a tetradentate ligand.

Question263

The complex ion that will lose its crystal field stabilization energy upon oxidation of its metal to +3 state is:



[April 12, 2019 (I)]

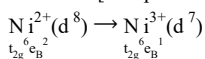
Options:

- A. $[\text{Co}(\text{phen})_3]^{2+}$
- B. $[\text{Ni}(\text{phen})_3]^{2+}$
- C. $[\text{Zn}(\text{phen})_3]^{2+}$
- D. $[\text{Fe}(\text{phen})_3]^{2+}$

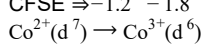
Answer: D

Solution:

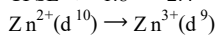
$$\text{CFSE} \Rightarrow [-0.4p + 0.6q]\Delta_0$$



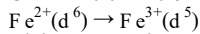
$$\text{CFSE} \Rightarrow -1.2 - 1.8$$



$$\text{CFSE} \Rightarrow -1.8 - 2.4$$



$$\text{CFSE} \Rightarrow 0.0 - 0.6$$



$$\text{CFSE} \Rightarrow -2.4 - 2.0$$

So, only Fe^{2+} will lose crystal field stabilisation energy upon oxidation to +3, others will gain crystal field stabilisation energy.

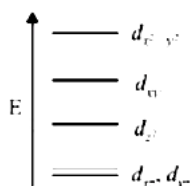
Question264

Complete removal of both the axial ligands (along the zaxis) from an octahedral complex leads to which of the following splitting patterns? (relative orbital energies not on scale)

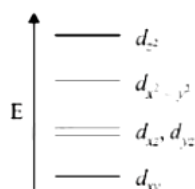
[April 12, 2019 (I)]

Options:

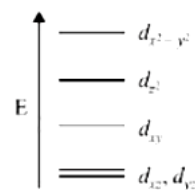
A.



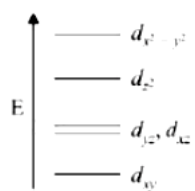
B.



C.



D.



Answer: A

Solution:

(a) After removal of both axial ligands from octahedral complex the field becomes square planar and leads to following energy patterns.
 $d_{x^2-y^2} > d_{xy} > d_{z^2} > d_{xz} = d_{yz}$

Question265

Three complexes, $[\text{CoCl}(\text{NH}_3)_5]^{2+}$ (I), $[\text{Co}(\text{NH}_3)_5\text{H}_2\text{O}]^{3+}$ (II) and $[\text{Co}(\text{NH}_3)_6]^{3+}$ (III) absorb light in the visible region. The correct order of the wavelength of light absorbed by them is :
 [April 10, 2019 (I)]

Options:

- A. (III) > (I) > (II)
- B. (III) > (II) > (I)
- C. (II) > (D) > (III)
- D. (I) > (II) > (III)

Answer: D

Solution:

Wavelength of the energy absorbed by the coordination compound is inversely proportional to ligand field strength of the given co-ordination compound. The strong field ligand causes higher splitting of the d-orbitals. The decreasing order of ligand field strength is $\text{NH}_3 > \text{H}_2\text{O} > \text{Cl}$. Therefore decreasing order of absorbed wavelength is (I) > (II) > (III)

Question266

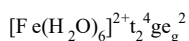
The crystal field stabilization energy (CFSE) of $[\text{Fe}(\text{H}_2\text{O})_6]\text{Cl}_2$ and $\text{K}_2[\text{NiCl}_4]$, respectively are:
 [April 10, 2019 (II)]

Options:

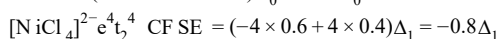
- A. $-0.6\Delta_0$ and $-0.8\Delta_z$
- B. $-0.4\Delta_0$ and $-0.8\Delta_t$
- C. $-2.4\Delta_0$ and $-1.2\Delta_e$
- D. $-0.4\Delta_0$ and $-1.2\Delta_r$

Answer: B

Solution:



$$\text{CFSE} = (-4 \times 0.4 + 2 \times 0.6)\Delta_0 = -0.4\Delta_0$$



Question267

The degenerate orbitals of $[\text{Cr}(\text{H}_2\text{O})_6]^{3+}$ are:

[April 9, 2019 (I)]

Options:

- A. d_{xz} and d_{y^2}
- B. d_{yz} and d_{z^2}
- C. d_{z^2} and d_{xz}
- D. $d_{x^2-y^2}$ and d_{xy}

Answer: A

Solution:

Cr^{3+} has d^3 configuration and forms an octahedral inner orbitals complex, therefore the set of degenerate orbitals are (d_{xy} , d_{yz} and d_{xz}) and ($d_{x^2-y^2}$ and d_{z^2}).

Question268

The correct statements among I to III are:

(I) Valence bond theory cannot explain the color exhibited by transition metal complexes.

(II) Valence bond theory can predict quantitatively the magnetic properties of transition metal complexes.

(III) Valence bond theory cannot distinguish ligands as weak and strong field ones.

[April 9, 2019 (I)]

Options:

- A. (II) and (III) only
- B. (I), (II) and (III)
- C. (I) and (III) only
- D. (I) and (II) only

Answer: C

Solution:

Valence bond theory cannot distinguish between weak field ligands and strong field ligands. Therefore, it cannot predict quantitatively the magnetic properties of transition metal complexes.

Question269

The correct order of the spin-only magnetic moment of metal ions in the following low-spin complexes, $[V(CN)_6]^{4-}$, $[Fe(CN)_6]^{4-}$, $[Ru(NH_3)_6]^{3+}$, and $[Cr(NH_3)_6]^{2+}$, is
[April 8, 2019 (I)]

Options:

- A. $Cr^{2+} > Ru^{3+} > Fe^{2+} > V^{2+}$
- B. $V^{2+} > Cr^{2+} > Ru^{3+} > Fe^{2+}$
- C. $V^{2+} > Ru^{3+} > Cr^{2+} > Fe^{2+}$
- D. $Cr^{2+} > V^{2+} > Ru^{3+} > Fe^{2+}$

Answer: B

Solution:

Complex	Metal ion electrons	No. of unpaired
$[V(CN)_6]^{4-}$	V^{2+}	3
$[Ru(NH_3)_6]^{3+}$	Ru^{3+}	1
$[Fe(CN)_6]^{4-}$	Fe^{2+}	0
$[Cr(NH_3)_6]^{2+}$	Cr^{2+}	2

∴ Spin magnetic moment $\propto$ no. of unpaired electrons. So the order of spin magnetic moment is:
 $V^{2+} > Cr^{2+} > Ru^{3+} > Fe^{2+}$

Question270

The calculated spin-only magnetic moments (BM) of the anionic and cationic species of $[Fe(H_2O)_6]^{2-}$ and $[Fe(CN)_6]^{3-}$, respectively, are:
[April 8, 2019 (II)]

Options:

- A. 0 and 4.9
- B. 2.84 and 5.92
- C. 4.9 and 0
- D. 0 and 5.92

Answer: C

Solution:

$[\text{Fe}(\text{H}_2\text{O})_6]^{2+}$: cationic species of $[\text{Fe}(\text{H}_2\text{O})_6]_2$

$[\text{Fe}(\text{CN})_6]^{4-}$: anionic species of $[\text{Fe}(\text{CN})_6]$

Magnetic Moment (μ) = $\sqrt{n(n+2)}$

Where n = no. of unpaired electrons

Now, $[\text{Fe}(\text{H}_2\text{O})_6]^{2+}$ $[\text{Fe}(\text{CN})_6]^{4-}$

$n = 4$ $n = 0$

$\therefore \mu = \sqrt{4(4+2)}$ $\mu = \sqrt{0(0+2)}$

$\mu = 4.9$ $\mu = 0$

Question 271

The compound used in the treatment of lead poisoning is:
[April 12, 2019 (II)]

Options:

- A. D-penicillamine
- B. desferrioxime B
- C. Cis-platin
- D. EDTA

Answer: D

Solution:

EDTA is used in the treatment of lead poisoning. Deferrioxime B is used in treatment of iron poisoning and D-penicillamine is used in treatment of heavy metal poisoning, while cis-platin is used for treating cancer.

Question 272

The compound that inhibits the growth of tumors is :
[April 8, 2019 (II)]

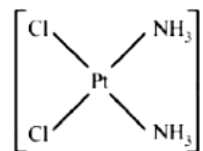
Options:

- A. trans- $[\text{Pt}(\text{Cl})_2(\text{NH}_3)_2]$
- B. cis- $[\text{Pd}(\text{Cl})_2(\text{NH}_3)_2]$
- C. cis- $[\text{Pt}(\text{Cl})_2(\text{NH}_3)_2]$
- D. trans- $[\text{Pd}(\text{Cl})_2(\text{NH}_3)_2]$

Answer: C

Solution:

cis-Platin is used as an anti-cancer drug.



cis- $[\text{Pt}(\text{Cl})_2(\text{NH}_3)_2]$

Question273

Consider the following reaction and statements: $[\text{Co}(\text{NH}_3)_4\text{Br}_2]^+ + \text{Br}^- \rightarrow [\text{Co}(\text{NH}_3)_3\text{Br}_3] + \text{NH}_3$

(I) Two isomers are produced if the reactant complex ion is a cis-isomer.

(II) Two isomers are produced if the reactant complex ion is a trans -isomer

(III) Only one isomer is produced if the reactant complex ion is a trans-isomer

(IV) Only one isomer is produced if the reactant complex ion is a cis-isomer.

The correct statements are:

[2018]

Options:

A. (I) and (II)

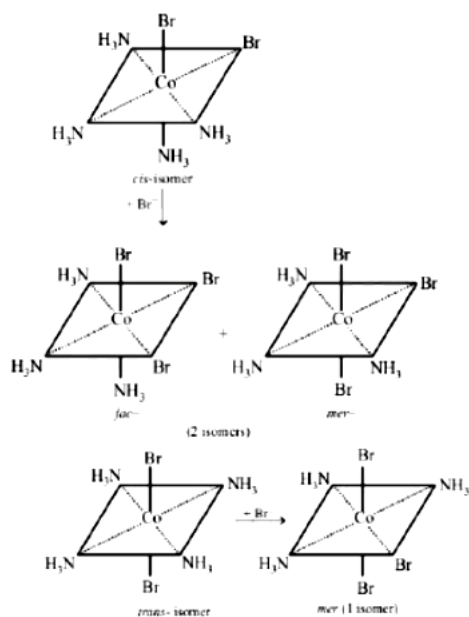
B. (I) and (III)

C. (III) and(IV)

D. (II) and (IV)

Answer: B

Solution:



Question274

Which of the following complexes will show geometrical isomerism?

[Online April 16, 2018]

Options:

A. Potassium tris(oxalato) chromate (III)

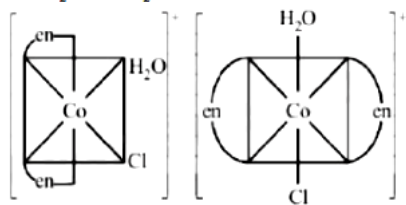
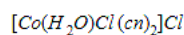
B. Pentaquachlorochromium (III) chloride

C. Aquachlorobis (ethylenediamine) cobalt(II) chloride

D. Potassium amminetrichloroplatinate (II)

Answer: C

Solution:



(Geometrical isomers)

Question275

The total number of possible isomers for square-planar $[Pt(Cl)(NO_2)(NO_3)(SCN)]^{2-}$ is:

[Online April 15, 2018(II)]

Options:

- A. 16
- B. 12
- C. 8
- D. 24

Answer: B

Solution:

The square planar complex of the type $[Mabcd]^{n+}$, where all four ligands are different, has 3 geometrical isomers. But if one of the ligands is ambidentate, then $2 \times 3 = 6$ geometrical isomers are possible. But if two ligands are ambidentate, then $4 \times 3 = 12$ geometrical isomers are possible. In the given example, NO_2^- and SCN^- are ambidentate ligands

Question276

In Wilkinson's catalyst, the hybridisation of central metal ion and its shape are respectively:

[Online April 16, 2018]

Options:

- A. sp^3d , trigonal bipyramidal
- B. d^2sp^3 , octahedral
- C. dsp^2 , square planar
- D. sp^3 , tetrahedral

Answer: C

Solution:

Wilkinson catalyst is $[RhCl(PPh_3)_3]$.

Rh belongs to 4d series, so it forms square planar complexes and contains dsp^2 hybridisation.

Question277

The correct combination is:
[Online April 15, 2018(I)]

Options:

- A. $[\text{NiCl}_4]^{2-}$ – square-planar; $[\text{Ni}(\text{CN})_4]^{2-}$ -paramagnetic
- B. $[\text{Ni}(\text{CN})_4]^{2-}$ -tetrahedral; $[\text{Ni}(\text{CO})_4]$ -paramagnetic
- C. $[\text{NiCl}_4]^{2-}$ -paramagnetic; $[\text{Ni}(\text{CO})_4]$ -tetrahedral
- D. $[\text{NiCl}_4]^{2-}$ – diamagnetic; $[\text{Ni}(\text{CO})_4]$ -square-planar

Answer: C

Solution:

$[\text{Ni}(\text{CN})_4]^{2-}$ is square planar, diamagnetic (0 unpaired electrons) with $d\text{sp}^2$ hybridisation.

$[\text{Ni}(\text{CO})_4]$ is tetrahedral, diamagnetic (0 unpaired electrons) with sp^3 hybridisation.

$[\text{NiCl}_4]^{2-}$ is tetrahedral, paramagnetic (2 unpaired electrons) with sp^3 hybridisation.

Hence, the option (c) is the correct answer.

Question278

The correct order of spin-only magnetic moments among the following is:
(Atomic number: Mn = 25, Co = 27, Ni = 28, Zn = 30)
[Online April 15, 2018(II)]

Options:

- A. $[\text{ZnCl}_4]^{2-} > [\text{NiCl}_4]^{2-} > [\text{CoCl}_4]^{2-} > [\text{MnCl}_4]^{2-}$
- B. $[\text{CoCl}_4]^{2-} > [\text{MnCl}_4]^{2-} > [\text{NiCl}_4]^{2-} > [\text{ZnCl}_4]^{2-}$
- C. $[\text{NiCl}_4]^{2-} > [\text{CoCl}_4]^{2-} > [\text{MnCl}_4]^{2-} > [\text{ZnCl}_4]^{2-}$
- D. $[\text{MnCl}_4]^{2-} > [\text{CoCl}_4]^{2-} > [\text{NiCl}_4]^{2-} > [\text{ZnCl}_4]^{2-}$

Answer: D

Solution:

The complex having highernumber of unpaired electrons will have higher value of spin only magnetic moment.

In all these complexes, the central metal ion is in +2 oxidation state.

Zn^{2+} has $3d^{10}$ outer electronic configuration with 0 unpaired electron.

Ni^{2+} has $3d^8$ outer electronic configuration with 2 unpaired electrons.

Co^{2+} has $3d^7$ outer electronic configuration with 3 unpaired electrons.

Mn^{2+} has $3d^5$ outer electronic configuration with 5 unpaired electrons

Hence the correct order of spin-only magnetic moments is $[\text{MnCl}_4]^{2-} > [\text{CoCl}_4]^{2-} > [\text{NiCl}_4]^{2-} > [\text{ZnCl}_4]^{2-}$

Question279

On treatment of 100mL of 0.1M solution of $\text{CoCl}_3 \cdot 6\text{H}_2\text{O}$ with excess AgNO_3 ; 1.2×10^{22} ions are precipitated. The complex is:
[2017]

Options:

- A. $[\text{Co}(\text{H}_2\text{O})_4\text{Cl}_2]\text{Cl} \cdot 2\text{H}_2\text{O}$
- B. $[\text{Co}(\text{H}_2\text{O})_3\text{Cl}_3] \cdot 3\text{H}_2\text{O}$
- C. $[\text{Co}(\text{H}_2\text{O})_6]\text{Cl}_3$
- D. $[\text{Co}(\text{H}_2\text{O})_5\text{Cl}]\text{Cl}_2, \text{H}_2\text{O}$

Answer: D

Solution:

$$\text{Moles of complex} = \frac{\text{Molarity} \times \text{Volume(mL)}}{1000}$$

$$= \frac{100 \times 0.1}{1000} = 0.01 \text{ mole}$$

Moles of ions precipitated with excess of

$$\text{AgNO}_3 = \frac{1.2 \times 10^{22}}{6.02 \times 10^{23}} = 0.02 \text{ moles}$$

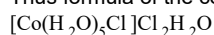
$\therefore$ Number of Cl^- present in ionisation sphere

$$= \frac{\text{Number of moles of ions precipitated}}{\text{Number of moles of complex}}$$

$$= \frac{0.02}{0.01} = 2$$

It means 2Cl^- ions present in ionization sphere.

Thus formula of the complex is



Question 280

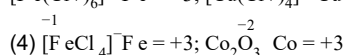
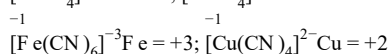
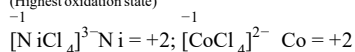
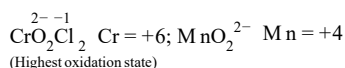
The pair of compounds having metals in their highest oxidation state is :
[Online April 8, 2017]

Options:

- A. MnO_2 and CrO_2Cl_2
- B. $[\text{NiCl}_4]^{2-}$ and $[\text{CoCl}_4]^{2-}$
- C. $[\text{Fe}(\text{CN})_6]^{3-}$ and $[\text{Cu}(\text{CN})_2]^{2-}$
- D. $[\text{FeCl}_4]^-$ and Co_2O_3

Answer: A

Solution:



Question 281

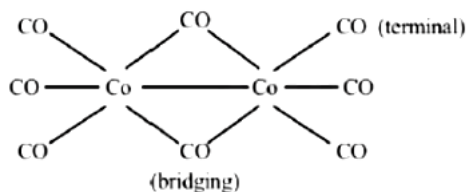
$[\text{Co}_2(\text{CO})_8]$ displays:
[Online April 9, 2017]

Options:

- A. one Co-Co bond, six terminal CO and two bridging CO
- B. one Co-Co bond, four terminal CO and four bridging CO
- C. no Co-Co bond, six terminal CO and two bridging CO
- D. no Co-Co bond, four terminal CO and four bridging CO

Answer: A

Solution:



Question 282

Which one of the following complexes shows optical isomerism?

[2016]

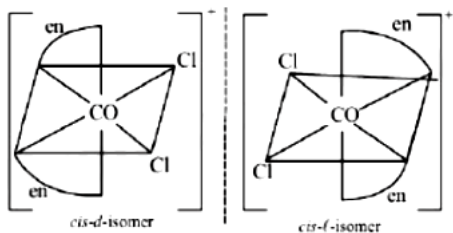
Options:

- A. $\text{trans}[\text{Co}(\text{en})_2\text{Cl}_2]\text{Cl}$
- B. $[\text{Co}(\text{NH}_3)_4\text{Cl}_2]\text{Cl}$
- C. $[\text{Co}(\text{NH}_3)_3\text{Cl}_3]$
- D. $\text{cis}[\text{Co}(\text{en})_2\text{Cl}_2]\text{Cl}$ (en = ethylenediamine)

Answer: D

Solution:

Optical isomerism occurs when a molecule is nonsuperimposable with its mirror image hence the complex $\text{cis}[\text{Co}(\text{en})_2\text{Cl}_2]\text{Cl}$ is optically active.



Question 283

Which one of the following complexes will consume more equivalents of aqueous solution of AgNO_3 ?

[Online April 9, 2016]

Options:

- A. $\text{Na}_2[\text{CrCl}_5(\text{H}_2\text{O})]$
- B. $\text{Na}_3[\text{CrCl}_6]$
- C. $[\text{Cr}(\text{H}_2\text{O})_5\text{Cl}]\text{Cl}_2$

D. $[\text{Cr}(\text{H}_2\text{O})_6]\text{Cl}_3$

Answer: D

Solution:

More equivalents of AgNO_3 aqueous solution will be consumed if complex will furnish more Cl^- ions in solution. Hence complex $[\text{Cr}(\text{H}_2\text{O})_6]\text{Cl}_3$ will consume more equivalents of aqueous solution of AgNO_3 .

Question284

Which of the following is an example of homoleptic complex?

[Online April 10, 2016]

Options:

A. $[\text{Co}(\text{NH}_3)_6]\text{Cl}_3$

B. $[\text{Pt}(\text{NH}_3)_2\text{Cl}_2]$

C. $[\text{Co}(\text{NH}_3)_4\text{Cl}_2]$

D. $[\text{Co}(\text{NH}_3)_5\text{Cl}]\text{Cl}_2$

Answer: A

Solution:

Complexes having only one type of ligands are examples of homoleptic complex.

Question285

The pair having the same magnetic moment is:

[At. No.: Cr = 24, Mn = 25, Fe = 26, Co = 27]

[2016]

Options:

A. $[\text{Mn}(\text{H}_2\text{O})_6]^{2+}$ and $[\text{Cr}(\text{H}_2\text{O})_6]^{2+}$

B. $[\text{CoCl}_4]^{2-}$ and $[\text{Fe}(\text{H}_2\text{O})_6]^{2+}$

C. $[\text{Cr}(\text{H}_2\text{O})_6]^{2+}$ and $[\text{CoCl}_4]^{2-}$

D. $[\text{Cr}(\text{H}_2\text{O})_6]^{2+}$ and $[\text{Fe}(\text{H}_2\text{O})_6]^{2+}$

Answer: D

Solution:

	Metal ion electron	Unpaired moment	Magnetic
(i)	Cr^{2+}	4	$\sqrt{24}$ B.M.
(ii)	Fe^{2+}	4	$\sqrt{24}$ B.M.
(iii)	Co^{2+}	3	$\sqrt{15}$ B.M.
(iv)	Mn^{2+}	5	$\sqrt{35}$ B.M.

Since (i) and (ii), each has 4 unpaired electron, so they will exhibit same magnetic moment. Thus option (d) is correct.

Question 286

Identify the correct trend given below:

(Atomic No. = Ti : 22, Cr : 24 and Mo : 42)

[Online April 9, 2016]

Options:

- A. Δ_o of $[Cr(H_2O)_6]^{2+} > [Mo(H_2O)_6]^{2+}$ and Δ_o of $[Ti(H_2O)_6]^{3+} > [Ti(H_2O)_6]^{2+}$
- B. Δ_o of $[Cr(H_2O)_6]^{2+} > [Mo(H_2O)_6]^{2+}$ and Δ_o of $[Ti(H_2O)_6]^{3+} < [Ti(H_2O)_6]^{2+}$
- C. Δ_o of $[Cr(H_2O)_6]^{2+} < [Mo(H_2O)_6]^{2+}$ and Δ_o of $[Ti(H_2O)_6]^{3+} > [Ti(H_2O)_6]^{2+}$
- D. Δ_o of $[Cr(H_2O)_6]^{2+} < [Mo(H_2O)_6]^{2+}$ and Δ_o of $[Ti(H_2O)_6]^{3+} < [Ti(H_2O)_6]^{2+}$

Answer: D

Solution:

The splitting is affected by the oxidation state of the central metal ion. A higher oxidation state leads to larger splitting, hence

$$\Delta_o \text{ of } [Ti(H_2O)_6]^{3+} < \Delta_o \text{ of } [Ti(H_2O)_6]^{2+}$$

Further Δ_o also depends of Z_{eff} and Z_{eff} of 4d series is more than 3d series. Hence

$$\Delta_o \text{ of } [Cr(H_2O)_6]^{2+} < \Delta_o \text{ of } [Mo(H_2O)_6]^{2+}$$

Question 287

The number of geometric isomers that can exist for square planar complex

$[Pt(Cl)(py)(NH_3)(NH_2OH)]^+$ is (py = pyridine):

[2015]

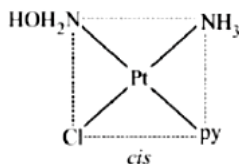
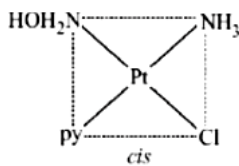
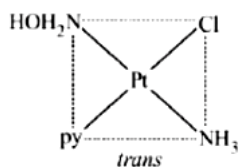
Options:

- A. 4
- B. 6
- C. 2
- D. 3

Answer: D

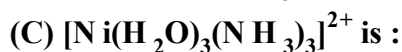
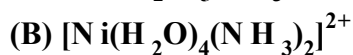
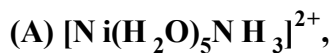
Solution:

Square planar complexes of type $M[ABCD]$ form three isomers. Their position may be obtained by fixing the position of one ligand and placing at the trans position any one of the remaining three ligands one by one.



Question288

The correct statement on the isomerism associated with the following complex ions,



[Online April 10,2015]

Options:

- A. (A) and (B) show only geometrical isomerism
- B. (A) and(B) show geometrical and optical isomerism
- C. (B) and (C) show geometrical and optical isomerism
- D. (B) and (C) show only geometrical isomerism

Answer: D

Solution:

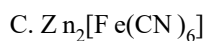
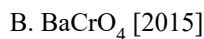
It does not show optical and geometrical isomerism (B) and (C) shows only geometrical isomerism.

Question289

Which of the following compounds is not colored yellow?

[2015]

Options:



Answer: C

Question290

The color of KMnO_4 is due to :
[2015]

Options:

- A. $\text{L} \rightarrow \text{M}$ charge transfer transition
- B. $\sigma - \sigma^*$ transition
- C. $\text{M} \rightarrow \text{L}$ charge transfer transition
- D. $\text{d} - \text{d}$ transition

Answer: A

Solution:

$\text{L} \rightarrow \text{M}$ charge transfer spectra. KMnO_4 is colored because it absorbs light in the visible range of electromagnetic radiation. The permanganate ion is the source of color, as a ligand to metal ($\text{L} \rightarrow \text{M}$) charge transfer takes place between oxygen's p orbitals and the empty d orbitals on the metal. This charge transfer takes place when a photon of light is absorbed which leads to the purple color of the compound.

Question291

Which of the following complex ions has electrons that are symmetrically filled in both t_{2g} and e_g orbitals?
[Online April 11, 2015]

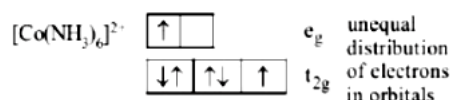
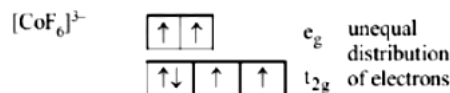
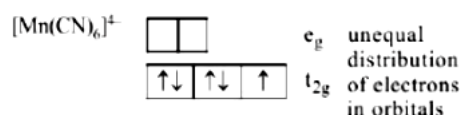
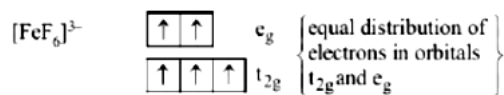
Options:

- A. $[\text{FeF}_6]^{3-}$
- B. $[\text{Mn}(\text{CN})_6]^{4-}$
- C. $[\text{CoF}_6]^{3-}$
- D. $[\text{Co}(\text{NH}_3)_6]^{2+}$

Answer: A

Solution:

Symmetrically filled t_2 and e_g are those, which contain equal distribution of electrons.



Question292

Which molecule/ion among the following cannot act as a ligand in complex compounds?
[Online April 10,2015]

Options:

- A. CH_4
- B. CO
- C. CN
- D. Br

Answer: A

Solution:

The donor atoms, molecules or anions which donate a pair of electrons to the metal atom or ion and form a coordinate bond with it are called ligands. In methane there is no electrons for donation to central metal atom/ ion, it is stable with complete octet configuration.

Question293

Which of the following name formula combinations is not correct?

	Formula	
(1)	$K_2[\text{Pt}(\text{CN})_4]$	Potassium tetracyanoplatinate (II)
(2)	$[\text{Mn}(\text{CN})_5]^{2-}$	Pentacyanomagnate (II) ion
(3)	$K[\text{Cr}(\text{NH}_3)_2\text{Cl}_4]$	Potassium diammine tetrachlorochromate (III)
(4)	$[\text{Co}(\text{NH}_3)_4(\text{H}_2\text{O})\text{I}]\text{SO}_4$	Tetraammine aquaioda cobalt (III) sulphate.

[Online April 11,2014]

Answer: 2

Solution:

Pentacyanomanganate (III) is the correct IUPAC name of $[\text{Mn}(\text{CN})_5]^{2-}$.

Question 294

An octahedral complex with molecular composition $\text{M} \cdot 5 \text{NH}_3 \cdot \text{Cl} \cdot \text{SO}_4$ has two isomers, A and B. The solution of A gives a white precipitate with AgNO_3 solution and the solution of B gives white precipitate with BaCl_2 solution. The type of isomerism exhibited by the complex is:

[Online April 19, 2014]

Options:

- A. Linkage isomerism
- B. Ionisation isomerism
- C. Coordinate isomerism
- D. Geometrical isomerism

Answer: B

Solution:

The two possible isomers for the given octahedral complex are $[\text{M}(\text{NH}_3)_5\text{SO}_4]\text{Cl}$ and $[\text{M}(\text{NH}_3)_5\text{Cl}]\text{SO}_4$. They respectively give chloride ion (indicated by precipitation with BaCl_2) and SO_4 ion (indicated by precipitation with AgNO_3). Hence the type of isomerism exhibited by the complex is ionisation isomerism.

Question 295

The octahedral complex of a metal ion M^{3+} with four monodentate ligands L_1 , L_2 , L_3 and L_4 absorb wavelengths in the region of red, green, yellow and blue, respectively. The increasing order of ligand strength of the four ligands is:

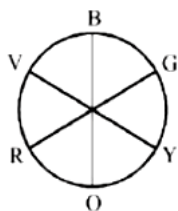
[2014]

Options:

- A. $\text{L}_4 < \text{L}_3 < \text{L}_2 < \text{L}_1$
- B. $\text{L}_1 < \text{L}_3 < \text{L}_2 < \text{L}_4$
- C. $\text{L}_3 < \text{L}_2 < \text{L}_4 < \text{L}_1$
- D. $\text{L}_1 < \text{L}_2 < \text{L}_4 < \text{L}_3$

Answer: B

Solution:



For a given metal ion, weak field ligands create a complex with smaller Δ , which will absorb light of longer λ and thus lower frequency. Conversely, stronger field ligands create a larger Δ , absorb light of shorter λ and thus higher frequency. So order of ligand strength is

$$L_1 < L_3 < L_2 < L_4$$

Question 296

An octahedral complex of Co^{3+} is diamagnetic. The hybridisation involved in the formation of the complex is:

[Online April 9, 2014]

Options:

A. sp^3d^2

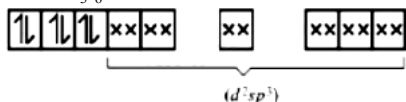
B. d^2sp^3

C. d^2sp^3

D. sp^3d

Answer: C

Solution:



Octahedral and diamagnetic.

Question 297

The correct statement about the magnetic properties of $[\text{Fe}(\text{CN})_6]^{3-}$ and $[\text{FeF}_6]^{3-}$ is ($Z = 26$):

[Online April 9, 2014]

Options:

A. both are paramagnetic.

B. both are diamagnetic.

C. $[\text{Fe}(\text{CN})_6]^{3-}$ is diamagnetic, $[\text{FeF}_6]^{3-}$ is paramagnetic.

D. $[\text{Fe}(\text{CN})_6]^{3-}$ is paramagnetic, $[\text{FeF}_6]^{3-}$ is diamagnetic.

Answer: A

Solution:

Both are paramagnetic, the only difference is that CN is a strong field ligand whereas F^- is a weak field ligand.

Question298

Consider the coordination compound, $[\text{Co}(\text{N H}_3)_6]\text{Cl}_3$. In the formation of this complex, the species which acts as the Lewis acid is:

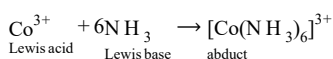
[Online April 11, 2014]

Options:



Answer: C

Solution:



Question299

Among the following species the one which causes the highest CFSE, Δ_0 as a ligand is:

[Online April 12, 2014]

Options:



Answer: D

Solution:

Following is the order of increasing value of Δ_0 for the ligands $\text{I}^- < \text{Br}^- < \text{Cl}^- < \text{F}^- < \text{H}_2\text{O} < \text{Py} = \text{N H}_3 < \text{en} < \text{N O}_2^- < \text{CN}^- < \text{CO}$

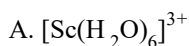
Question300

Which one of the following complexes will most likely absorb visible light?

(At nos. Sc = 21, Ti = 22, V = 23, Zn = 30)

[Online April 12, 2014]

Options:





Answer: C

Solution:

The absorption of visible light is responsible for coloured nature of the transition metal cation due to the promotion of one or more unpaired d -electron from a lower to higher level within same d -subshell. Hence higher the number of unpaired electron higher will be the absorption in visible light.

The electronic configuration of the given elements is $Sc^{3+}(18) = 1s^2 2s^2 2p^6 3s^2 3p^6 3d^0 4s^0$ – no unpaired e^-

$Ti^{4+}(18) = 1s^2 2s^2 2p^6 3s^2 3p^6 3d^0 4s^0$ – no unpaired e^-

$V^{3+}(20) = 1s^2 2s^2 2p^6 3s^2 3p^6 3d^2 4s^0$ – Two unpaired e^-

$Zn^{2+}(28) = 1s^2 2s^2 2p^6 3s^2 3p^6 3d^{10} 4s^0$ – no unpaired e^- .

hence $[V(NH_3)_6]^{3+}$ will most likely absorb visible light.

Question301

Nickel ($Z = 28$) combines with a uninegative monodentate ligand to form a diamagnetic complex $[NiL_4]^2$. The hybridisation involved and the number of unpaired electrons present in the complex are respectively:

[Online April 19, 2014]

Options:

A. sp^3 , two

B. $d sp^2$, zero

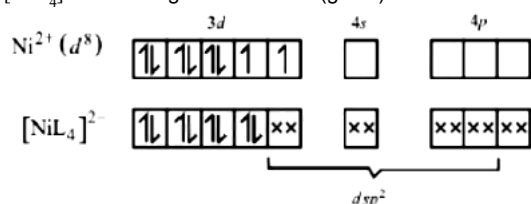
C. $d sp^2$, one

D. sp^3 , zero

Answer: B

Solution:

$[NiL_4]^{2-} \Rightarrow$ Diamagnetic in nature (given)



So, no. of unpaired electron = 0

hybridisation – $d sp^2$

Question302

Which of the following complex species is not expected to exhibit optical isomerism? [2013]

Options:

A. $[Co(en)_3]^{3+}$

B. $[Co(en)_2Cl_2]^+$

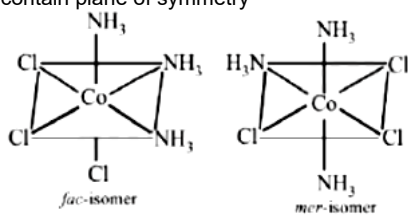
C. $[Co(NH_3)_3Cl_3]$



Answer: C

Solution:

Octahedral coordination entities of the type $\text{M a}_3\text{b}_3$ exhibit geometrical isomerism. The compound exists both as facial and meridional isomers, both contain plane of symmetry



Question303

Type of isomerism which exists between $[\text{Pd}(\text{C}_6\text{H}_5)_2(\text{SCN})_2]$ and $[\text{Pd}(\text{C}_6\text{H}_5)_2(\text{NCS})_2]$ is:
[Online April 9,2013]

Options:

- A. Linkage isomerism
- B. Coordination isomerism
- C. Ionisation isomerism
- D. Solvate isomerism

Answer: A

Solution:

The compound shows linkage isomerism because the ligand in the compound is an ambidentate ligand that can bond at more than one atomic site. i.e., $:\text{NCS}^-$ and $:\text{SCN}^-$

Question304

Which of the following is diamagnetic?
[Online April 22, 2013]

Options:

- A. $[\text{Fe}(\text{CN})_6]^{3-}$
- B. $[\text{Co}(\text{ox})]^{3-}$
- C. $[\text{FeF}_6]^{3-}$
- D. $[\text{Co}(\text{F}_6)]^{3-}$

Answer: B

Solution:

Among four ligands CN^- and OX^{2-} are strong field ligands. So pairing takes places in $[\text{Fe}(\text{CN})_6]^{3-}$ and $[\text{Co}(\text{OX})_3]^{3-}$ complexes. Fe^{3+} having one unpaired electron in its complex while Co^{3+} does not have unpaired electron in its complex. So option b is correct.

Question305

The magnetic moment of the complex anion $[\text{Cr}(\text{NO})(\text{NH}_3)(\text{CN})_4]^{2-}$ is:

[Online April 23, 2013]

Options:

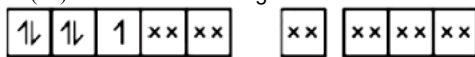
- A. 5.91BM
- B. 3.87BM
- C. 1.73BM
- D. 2.82BM

Answer: C

Solution:

In $[\text{Cr}(\text{NO})(\text{NH}_3)(\text{CN})_4]^{2-}$,

$\text{Cr}^{+}(\text{d}^5)$ has electronic configuration as:



So, 1 unpaired electron is present.

Thus, $\mu = \sqrt{1(1+2)} = \sqrt{3} = 1.73\text{BM}$

Question306

In which of the following octahedral complex species the magnitude of Δ_0 will be maximum?

[Online April 25, 2013]

Options:

- A. $[\text{Co}(\text{H}_2\text{O})_6]^{2+}$
- B. $[\text{Co}(\text{CN})_6]^{3-}$
- C. $[\text{Co}(\text{C}_2\text{O}_4)_3]^{3-}$
- D. $[\text{Co}(\text{NH}_3)_6]^{3+}$

Answer: B

Solution:

Crystal field splitting depends upon the nature of ligand. The nature of ligand decreases as shown below

$\text{C}_2\text{O}_4^{2-} < \text{H}_2\text{O} < \text{NH}_3 < \text{CN}^-$

Hence the crystal field splitting will be maximum for $[\text{Co}(\text{CN})_6]^{3-}$

Question307

The structure of which of the following chloro species can be explained on the basis of d sp^2 hybridisation?

[Online April 25, 2013]

Options:



Answer: A

Solution:

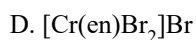
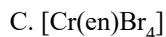
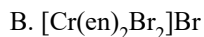
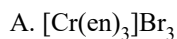
$[\text{PdCl}_4]^{2-}$ is dsp^2 hybridised and square planar in shape.

Question308

Which among the following will be named as dibromido bis (ethylenediamine) chromium (III) bromide?

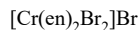
[2012]

Options:



Answer: B

Solution:



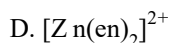
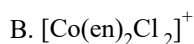
dibromido bis(ethylenediamine) chromium (III) Bromide.

Question309

Which of the following complex ions will exhibit optical isomerism? (en = 1, 2 -diamine ethane).

[Online May 12,2012]

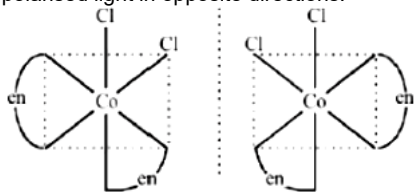
Options:



Answer: B

Solution:

Non superimposable mirror images are called optical isomers and may be described as "chiral". They are also called enantiomers and rotate plane polarised light in opposite directions.



Question310

The complex ion $[\text{Pt}(\text{NO}_2)(\text{Py})(\text{NH}_3)(\text{NH}_2\text{OH})]^+$ will give
[Online May 26,2012]

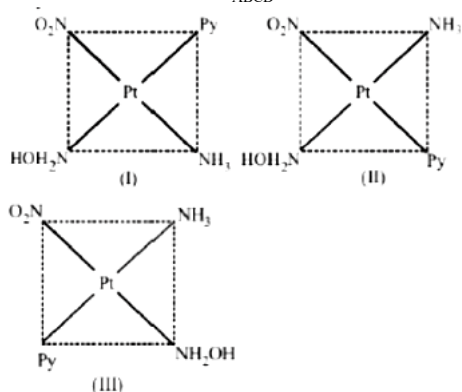
Options:

- A. 2 isomers (Geometrical)
- B. 3 isomers (Geometrical)
- C. 6 isomers (Geometrical)
- D. 4 isomers (Geometrical)

Answer: B

Solution:

Complexes of the type M_{ABCD} may exist in three isomeric form. Similarly $[\text{Pt}(\text{NO}_2)(\text{Py})(\text{NH}_3)(\text{NH}_2\text{OH})]^+$ may exist in three isomeric form.



Question311

The d -electron configurations of Cr^{2+} , Mn^{2+} , Fe^{2+} and Co^{2+} are d^4 , d^5 , d^6 and d^7 respectively. Which one of the following will exhibit the lowest paramagnetic behaviour?

(Atomic no. Cr = 24, Mn = 25, Fe = 26, Co = 27)

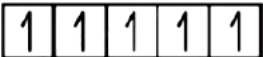
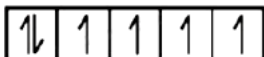
[Online May 7,2012]

Options:

- A. $[\text{Co}(\text{H}_2\text{O})_6]^{2+}$
- B. $[\text{Cr}(\text{H}_2\text{O})_6]^{2+}$
- C. $[\text{Mn}(\text{H}_2\text{O})_6]^{2+}$
- D. $[\text{Fe}(\text{H}_2\text{O})_6]^{2+}$

Answer: A

Solution:

Electronic Configuration		No. of unpaired electrons
Co^{2+}	d^7 	3
Cr^{2+}	d^4 	4
Mn^{2+}	d^5 	5
Fe^{2+}	d^6 	4

$\therefore$ Since Co^{3+} has lowest no. of unpaired electrons hence lowest paramagnetic behaviour is shown by $[\text{Co}(\text{H}_2\text{O})_6]^{2+}$

Question312

The correct order of ligands in the spectrochemical series is
[Online May 19,2012]

Options:

- A. $\text{Cl}^- > \text{en} > \text{CN}^- > \text{NCS}^-$
- B. $\text{CN}^- > \text{en} > \text{NCS}^- > \text{Cl}^-$
- C. $\text{en} > \text{CN}^- > \text{Cl}^- > \text{NCS}^-$
- D. $\text{NCS}^- > \text{CN}^- > \text{Cl}^- > \text{en}$

Answer: B

Solution:

Correct order is : $\text{CN}^- > \text{en} > \text{NCS}^- > \text{Cl}^-$

Question313

Which of the following paramagnetic ions would exhibit a magnetic moment (spin only) of the order of 5 BM?

(At. Nos. Mn = 25, Cr = 24, V = 23, Ti = 22)

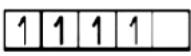
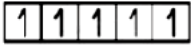
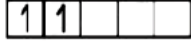
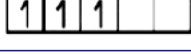
[Online May 19,2012]

Options:

- A. Mn^{2+}
- B. Ti^{2+}
- C. V^{2+}
- D. Cr^{2+}

Answer: D

Solution:

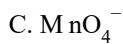
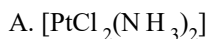
Ion	Electronic Configuration	Magnetic moment ($\sqrt{n(n+2)}$)
$Cr^{2+}d^4$		$\sqrt{4(4+2)} = 4.9$
$Mn^{2+}d^5$		$\sqrt{5(5+2)} = 5.8$
$Ti^{2+}d^2$		$\sqrt{2(2+2)} = 2.4$
$V^{2+}d^3$		$\sqrt{3(3+2)} = 3.8$

Hence Cr^{2+} has magnetic moment of the order of 5B . M .

Question314

**Square-planar geometry is shown by
[Online May 19,2012]**

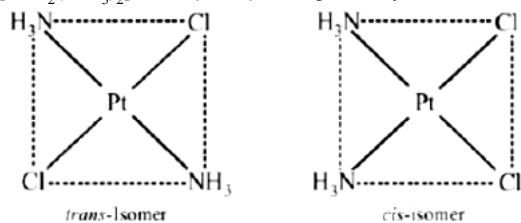
Options:



Answer: A

Solution:

$[PtCl_2(NH_3)_2]$ has square planar geometry.

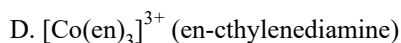
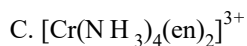


Question315

**Which one of the following complex ions has geometrical isomers?
[2011]**

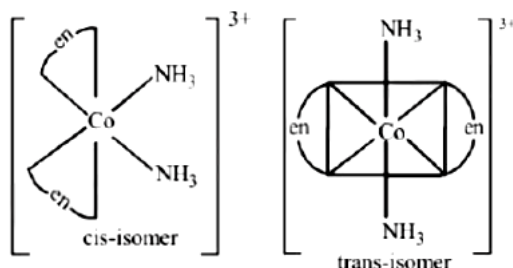
Options:





Answer: B

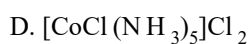
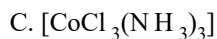
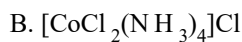
Solution:



Question316

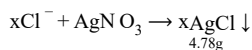
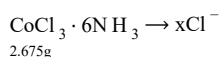
A solution containing 2.675g of $\text{CoCl}_3 \cdot 6\text{NH}_3$ (molar mass = 267.5g mol^{-1}) is passed through a cation exchanger. The chloride ions obtained in solution were treated with excess of AgNO_3 to give 4.78g of AgCl (molar mass = 143.5g mol^{-1}). The formula of the complex is (At. mass of $\text{Ag} = 108\text{u}$) [2010]

Options:



Answer: A

Solution:



Number of moles of the complex

$$= \frac{2.675}{267.5} = 0.01\text{mol}$$

Number of moles of AgCl obtained

$$= \frac{4.78}{143.5} = 0.03\text{mol}$$

$\therefore$ No. of moles of AgCl obtained

$$= 3 \times \text{No. of moles of complex}$$

$$\therefore n = \frac{0.03}{0.01} = 3$$

Question317

Which one of the following has an optical isomer?

[2010]

Options:

- A. $[\text{Zn}(\text{en})(\text{NH}_3)_2]^{2+}$
- B. $[\text{Co}(\text{en})_3]^{3+}$
- C. $[\text{Co}(\text{H}_2\text{O})_4(\text{en})]^{3+}$
- D. $[\text{Zn}(\text{en})_2]^{2+}$ (en = ethylenediamine)

Answer: B

Solution:

For a substance to be optical isomer following conditions should be fulfilled

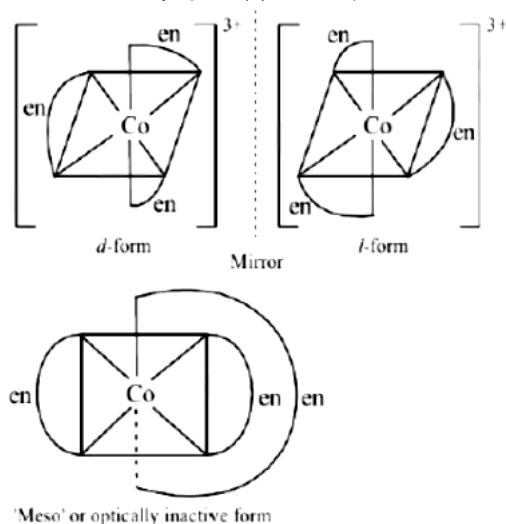
(a) A coordination compound which can rotate the plane of polarised light is said to be optically active.

(b) When the coordination compounds have same formula but differ in their abilities to rotate directions of the plane of polarised light are said to exhibit optical isomerism and the molecules are optical isomers. The optical isomers are a pair of molecules which are non-superimposable mirror images of each other.

(c) This is due to the absence of elements of symmetry in the complex.

(d) Optical isomerism is expected in tetrahedral complex of the type Mabod .

Based on this only option (b) shows optical isomerism $[\text{Co}(\text{en})_3]^{3+}$



Complexes of Zn^{2+} cannot show optical isomerism as they are tetrahedral complexes with plane of symmetry.

$[\text{Co}(\text{H}_2\text{O})_4(\text{en})]^{3+}$ has two planes of symmetry hence it is also optically inactive.

Question318

Which of the following shows optical isomerism

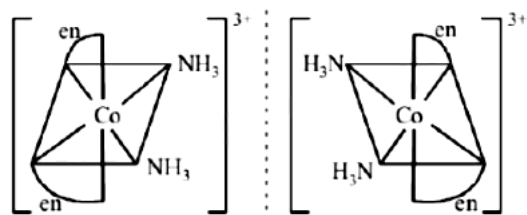
[2009]

Options:

- A. $[\text{Co}(\text{en})(\text{NH}_3)_2]^{2+}$
- B. $[\text{Co}(\text{H}_2\text{O})_4(\text{en})]^{3+}$
- C. $[\text{Co}(\text{en})_2(\text{NH}_3)_2]^{3+}$
- D. $[\text{Co}(\text{NH}_3)_3\text{Cl}]^+$

Answer: C

Solution:



Enantiomers of cis- $[Co(en)_2(NH_3)_2]^{3+}$

Question319

Which of the following pairs represent linkage isomers?

[2009]

Options:

- A. $[Pd(PPh_3)_2(NCS)_2]$ and $[Pd(PPh_3)_2(SCN)_2]$
- B. $[Co(NH_3)_5NO_3]SO_4$ and $[Co(NH_3)_5SO_4]NO_3$
- C. $[PtCl_2(NH_3)_4]Br_2$ and $[PtBr_2(NH_3)_4]Cl_2$
- D. $[Cu(NH_3)_4][PtCl_4]$ and $[Pt(NH_3)_4][CuCl_4]$

Answer: A

Solution:

The SCN⁻ ion can coordinate through S or N atom giving rise to linkage isomerism

M ← SCN thiocyanato

M ← NCS isothiocyanato.

Question320

The coordination number and the oxidation state of the element ' E ' in the complex

$[E(en)_2(C_2O_4)]NO_2$ (where (en) is ethylene diamine) are, respectively,

[2008]

Options:

- A. 6 and 2
- B. 4 and 2
- C. 4 and 3
- D. 6 and 3

Answer: D

Solution:

In the given complex we have two bidentate ligands (i.e en and C_2O_4), so coordination number of E is 6

$(2 \times 2 + 1 \times 2 = 6)$

Let the oxidation state of E in complex be x, then

$[x + (-2) = 1]$ or $x - 2 = 1$ or $x = +3$, so its oxidation state is +3

Thus option (d) is correct.

Question 321

In which of the following complexes of the Co(at. no. 27), will the magnitude of Δ_0 be the highest?
[2008]

Options:

- A. $[\text{Co}(\text{CN})_6]^{3-}$
- B. $[\text{Co}(\text{C}_2\text{O}_4)_3]^{3-}$
- C. $[\text{Co}(\text{H}_2\text{O})_6]^{3+}$
- D. $[\text{Co}(\text{NH}_3)_6]^{3+}$

Answer: A

Solution:

In octahedral complex, the magnitude of Δ_0 will be highest in a complex having strongest ligand. Of the given ligands, CN^- is strongest, so Δ_0 will be highest for $[\text{Co}(\text{CN})_6]^{3-}$. Thus option (a) is correct.

Question 322

Which of the following has a square planar geometry?
(At. nos: Fe = 26, Co = 27, Ni = 28, Pt = 78)
[2007]

Options:

- A. $[\text{PtCl}_4]^{2-}$
- B. $[\text{CoCl}_4]^{2-}$
- C. $[\text{FeCl}_4]^{2-}$
- D. $[\text{NiCl}_4]^{2-}$

Answer: A

Solution:

Complexes with d_{sp^2} hybridisation are square planar. All the complexes of Pt^{2+} are square planar including those with weak field ligand such as halide ions. Thus option (a) is correct.

Question 323

The IUPAC name for the complex $[\text{Co}(\text{NO}_2)(\text{NH}_3)_5]\text{Cl}_2$ is :
[2006]

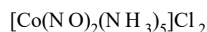
Options:

- A. pentaammine nitrito-N-cobalt(II) chloride [2006]
- B. pentaammine nitrito-N-cobalt(III) chloride
- C. nitrito-N-pentaamminecobalt(III) chloride

D. nitrito-N-pentaamminecobalt(II) chloride

Answer: B

Solution:



Pentaamminenitrito-N-cobalt (III) chloride

Question324

Nickel ($Z = 28$) combines with a uninegative monodentate ligand X^- to form a paramagnetic complex $[\text{NiX}_4]^{2-}$. The number of unpaired electron(s) in the nickel and geometry of this complex ion are, respectively:
[2006]

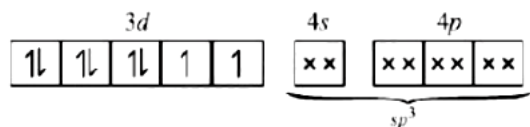
Options:

- A. one, square planar
- B. two, square planar
- C. one, tetrahedral
- D. two, tetrahedral

Answer: D

Solution:

$[\text{NiX}_4]^{2-}$, the electronic configuration of Ni^{2+} is



It contains two unpaired electrons and the hybridisation is sp^3 (tetrahedral).

Question325

How many EDTA (ethylenediaminetetraacetic acid) molecules are required to make an octahedral complex with a Ca^{2+} ion?
[2006]

Options:

- A. One
- B. Two
- C. Six
- D. Three

Answer: A

Solution:

EDTA is hexadentate, four donor O atoms and 2 donor N atoms, and for the formation of octahedral complex one molecule is required.

Question 326

In $\text{Fe}(\text{CO})_5$, the $\text{Fe}-\text{C}$ bond possesses
[2006]

Options:

- A. ionic character
- B. σ -character only
- C. π -character
- D. both σ and π characters

Answer: D

Solution:

Due to some backbonding by sideways overlapping between d-orbitals of metal and p-orbital of carbon, the $\text{Fe}-\text{C}$ bond in $\text{Fe}(\text{CO})_5$ has both σ and π character.

Question 327

The oxidation state Cr in $[\text{Cr}(\text{NH}_3)_4\text{Cl}_2]^+$ is
[2005]

Options:

- A. 0
- B. +1
- C. +2
- D. +3

Answer: D

Solution:

Oxidation state of Cr in $[\text{Cr}(\text{NH}_3)_4\text{Cl}_2]^+$.
Let it be x, $1 \times x + 4 \times 0 + 2(-1) = 1$
Therefore x = 3.

Question 328

The IUPAC name of the coordination compound $\text{K}_3[\text{Fe}(\text{CN})_6]$ is
[2005]

Options:

- A. Tripotassium hexacyanoiron (II)
- B. Potassium hexacyanoiron (II)
- C. Potassium hexacyanoferrate(III)
- D. Potassium hexacyanoferrate (II)

Answer: C

Solution:

$K_3[Fe(CN)_6]$ is potassium hexacyanoferrate (III).

Question329

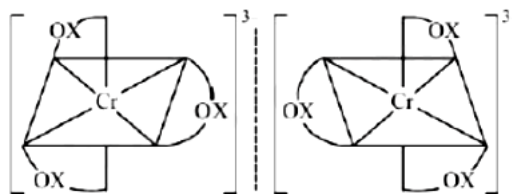
Which of the following compounds shows optical isomerism?

[2005]

Options:

- A. $[Co(CN)_6]^{3-}$
- B. $[Cr(C_2O_4)_3]^{3-}$
- C. $[ZnCl_4]^{2-}$
- D. $[Cu(NH_3)_4]^{2+}$

Answer: B

Solution:

Non-superimposable mirror images, hence optical isomers.

Question330

Which one of the following cyano complexes would exhibit the lowest value of paramagnetic behaviour?

(At. Nos: Cr = 24, Mn = 25, Fe = 26, Co = 27)

[2005]

Options:

- A. $[Co(CN)_6]^{3-}$
- B. $[Fe(CN)_6]^{3-}$
- C. $[Mn(CN)_6]^{3-}$
- D. $[Cr(CN)_6]^{3-}$

Answer: B

Solution:

No. of unpaired electrons		
(a)	Co^{3+}	4
(b)	Fe^{3+}	1
(c)	Mn^{3+}	4
(d)	Cr^{3+}	3

The effective magnetic moment is given by the number of unpaired electrons in a substance, the lesser the number of unpaired electrons lower is its magnetic moment in Bohr - Magneton and lower shall be its paramagnetic

Question331

The coordination number of a central metal atom in a complex is determined by [2004]

Options:

- A. the number of ligands around a metal ion bonded by sigma and pi-bonds both
- B. the number of ligands around a metal ion bonded by pi-bonds
- C. the number of ligands around a metal ion bonded by sigma bonds
- D. the number of only anionic ligands bonded to the metal ion.

Answer: C

Solution:

The coordination number of central metal atom in a complex is equal to number of monovalent ligands, twice the number of bidentate ligands and so on, around the metal ion bonded by coordinate bonds.
Hence coordination number = No. of σ bonds formed by metals with ligands.

Question332

Coordination compounds have great importance in biological systems. In this context which of the following statements is incorrect? [2004]

Options:

- A. Cyanocobalamin is B_{12} and contains cobalt
- B. Haemoglobin is the red pigment of blood and contains iron
- C. Chlorophylls are green pigments in plants and contain calcium
- D. Carboxypeptidase - A is an enzyme and contains zinc.

Answer: C

Solution:

The chlorophyll molecule plays an important role in photosynthesis. It has porphyrin rings and the metal Mg, not Ca.

Question333

Which one of the following has largest number of isomers? [2004]

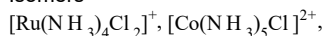
Options:

- A. $[\text{Ir}(\text{PR}_3)_2\text{H}(\text{CO})]^{2+}$
- B. $[\text{Co}(\text{NH}_3)_5\text{Cl}]^{2+}$ [2004]
- C. $[\text{Ru}(\text{NH}_3)_4\text{Cl}_2]^+$ (R = alkyl group, en = ethylenediamine)
- D. $[\text{Co}(\text{en})_2\text{Cl}_2]^+$

Answer: D

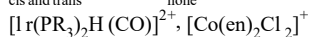
Solution:

Isomers



cis and trans

none



cis and trans

cis and trans and optical isomers

Question 334

Among the properties (a) reducing (b) oxidising (c) complexing, the set of properties shown by CN ion towards metal species is [2004]

Options:

- A. c, a
- B. b, c
- C. a, b
- D. a, b, c

Answer: A

Solution:

CN ion acts good complexing as well as reducing agent.

Question 335

Which one of the following complexes is an outer orbital complex? (Atomic nos. : Mn = 25; Fe = 26; Co = 27, Ni = 28) [2004]

Options:

- A. $[\text{Co}(\text{NH}_3)_6]^{3+}$
- B. $[\text{Mn}(\text{CN})_6]^{4-}$
- C. $[\text{Fe}(\text{CN})_6]^{4-}$
- D. $[\text{Ni}(\text{NH}_3)_6]^{2+}$

Answer: D

Solution:

$[\text{Fe}(\text{CN})_6]^{4-}$, $[\text{Mn}(\text{CN})_6]^{4-}$, $[\text{Co}(\text{NH}_3)_6]^{3+}$, $[\text{Ni}(\text{NH}_3)_6]^{2+}$
 d^2sp^3 d^2sp^3 d^2sp^3 sp^3d^2
 Hence $[\text{Ni}(\text{NH}_3)_6]^{2+}$ is outer orbital complex.

Question336

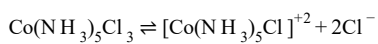
One mole of the complex compound $\text{Co}(\text{NH}_3)_5\text{Cl}_y$ gives 3 moles of ions on dissolution in water. One mole of the same complex reacts with two moles of AgNO_3 solution to yield two moles of $\text{AgCl}(\text{s})$. The structure of the complex is [2003]

Options:

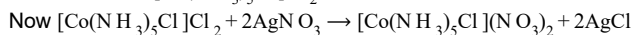
- A. $[\text{Co}(\text{NH}_3)_3\text{Cl}_3] \cdot 2\text{NH}_3$
- B. $[\text{Co}(\text{NH}_3)_4\text{Cl}_2]\text{Cl} \cdot \text{NH}_3$
- C. $[\text{Co}(\text{NH}_3)_5\text{Cl}]\text{Cl}_2$
- D. $[\text{Co}(\text{NH}_3)_4\text{Cl}]\text{Cl}_2, \text{NH}_3$

Answer: C

Solution:



$\therefore$ Structure is $[\text{Co}(\text{NH}_3)_5\text{Cl}]\text{Cl}_2$



Question337

In the coordination compound, $\text{K}_4[\text{Ni}(\text{CN})_4]$, the oxidation state of nickel is [2003]

Options:

- A. 0
- B. +1
- C. +2
- D. -1

Answer: A

Solution:

Let the O.N., of N in $\text{K}_4[\text{Ni}(\text{CN})_4]$ be = x then

$$4(+1) + x + (-1) \times 4 = 0$$

$$\Rightarrow 4 + x - 4 = 0$$

$$x = 0$$

Question338

The type of isomerism present in nitropentammine chromium (III) chloride is [2002]

Options:

- A. optical
- B. linkage
- C. ionization
- D. polymerisation.

Answer: B

Solution:

The chemical formula of nitropentammine chromium (III) chloride is $[\text{Cr}(\text{NH}_3)_5\text{NO}_2]\text{Cl}_2$

It can exist in following two structures

$[\text{Cr}(\text{NH}_3)_5\text{NO}_2]\text{Cl}_2$ and

Nitropentamminechromium (III) chloride

$[\text{Cr}(\text{NH}_3)_5\text{ONO}]\text{Cl}_2$

Nitritopentamminechromium (III) chloride

Therefore the type of isomerism found in this compound is linkage isomerism as NO_2 group is linked through N as NO_2 , or through O as $-\text{ONO}$.

Question339

A square planar complex is formed by hybridisation of which atomic orbitals? [2002]

Options:

- A. s, p_x, p_y, d_{yz}
- B. $s, p_x, p_y, d_{x^2-y^2}$
- C. s, p_x, p_y, d_{z^2}
- D. s, p_y, p_z, d_{xy}

Answer: B

Solution:

A square planar complex is formed by hybridisation of s, p_x, p_y and $d_{x^2-y^2}$ atomic orbitals

Question340

$\text{CH}_3 - \text{Mg} - \text{Br}$ is an organo metallic compound due to [2002]

Options:

A. $\text{Mg} - \text{Br}$ bond

B. $\text{C} - \text{Mg}$ bond

C. $\text{C} - \text{Br}$ bond

D. $\text{C} - \text{H}$ bond.

Answer: B

Solution:

Compounds that contain carbon-metal bond are known as organometallic compounds.

In $\text{CH}_3 - \text{Mg} - \text{Br}$ (Grignard's reagent), a bond is present between carbon and Mg (metal), hence it is an organometallic compound.
